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mathmatics

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* Properties of Determinants :-

i) If two rows or columns of a matrix are identical then the determinant = 0

$$\Delta = \begin{vmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 0 & 1 & 2 \end{vmatrix} = 0$$

ii) If two rows or columns of a matrix are interchanged, then the sign of determinant changes

$$-\Delta = \begin{vmatrix} 1 & 0 & 2 \\ 2 & 1 & 3 \\ 1 & 0 & 2 \end{vmatrix}$$

iii) If three rows or columns of matrix are interchanged, then the sign of determinant is unaltered.

$$\Delta = \begin{vmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 3 \\ 3 & 1 & 1 \\ 0 & 1 & 2 \end{vmatrix}$$

iv) In the determinant of a matrix, if any column containing the sum or difference of two elements then it can be split into sum or difference of two determinants.

$$\begin{vmatrix} a & a^2 & a^3+1 \\ b & b^2 & b^3+1 \\ c & c^2 & c^3+1 \end{vmatrix} = \begin{vmatrix} a & a^2 & a^3 \\ b & b^2 & b^3 \\ c & c^2 & c^3 \end{vmatrix} + \begin{vmatrix} a & a^2 & 1 \\ b & b^2 & 1 \\ c & c^2 & 1 \end{vmatrix}$$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\Delta = ad - bc$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\Delta = a_{11}\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12}\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13}\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

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* Lower Triangular Matrix :- If all elements above the principal digonals are zeroes.

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 1 & 2 & 5 \end{bmatrix} \qquad \Delta = 15$$

* Upper Triangular Matrix :- If all elements below the principal digonal are zeroes, then it is said to be an upper triangular Matrix.

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -4 & 2 \\ 0 & 0 & 7 \end{bmatrix} \qquad \Delta = -28$$

NOTE :-
If a matrix is either lower triangular or upper triangular then the determinant is the product of the principle digonal elements.

Q. Find the determinant of Matrix

$$\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}$$

$$\rightarrow \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} \qquad R_1 \rightarrow R_1 - R_2 \qquad R_2 \rightarrow R_2 - R_3$$

$$= \begin{vmatrix} 0 & a-b & a^2-b^2 \\ 0 & b-c & b^2-c^2 \\ 1 & c & c^2 \end{vmatrix}$$

$$= (a-b)(b-c) \begin{vmatrix} 0 & 1 & a+b \\ 0 & 1 & b+c \\ 1 & c & c^2 \end{vmatrix} = (a-b)(b-c)\,[bc - a\cancel{b}]$$

$$= (a-b)(b-c)(c-a)$$

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* $\boxed{|A| = |A^T|}$

$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}$$

$$|A| = |A^T| = (a-b)(b-c)(c-a)$$

Q. $\begin{vmatrix} 1 & 1 & 1 \\ 1 & 1+a & 1 \\ 1 & 1 & 1+b \end{vmatrix}$ $R_2 - R_1$, $R_3 - R_1$

Imp Result

$$= \begin{vmatrix} 1 & 1 & 1 \\ 0 & a & 0 \\ 0 & 0 & b \end{vmatrix} = ab$$

Q. $\begin{vmatrix} 1 & 1 & 1 \\ 1 & 5 & 1 \\ 1 & 1 & 6 \end{vmatrix} = ?$

(a) 20 (b) 30 (c) 40 (d) 0

Q. $\begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & 1+a & 1 & 1 \\ 1 & 1 & 1+b & 1 \\ 1 & 1 & 1 & 1+c \end{vmatrix} = abc$

Imp Result

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Q. $\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix}$

Imp Result

$= abc \begin{vmatrix} 1+\frac{1}{a} & \frac{1}{a} & \frac{1}{a} \\ \frac{1}{b} & 1+\frac{1}{b} & \frac{1}{b} \\ \frac{1}{c} & \frac{1}{c} & 1+\frac{1}{c} \end{vmatrix}$ $R_1 \to R_1 + (R_2+R_3)$

$= abc\left(1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right) \begin{vmatrix} 1 & 1 & 1 \\ \frac{1}{b} & 1+\frac{1}{b} & \frac{1}{b} \\ \frac{1}{c} & \frac{1}{c} & 1+\frac{1}{c} \end{vmatrix}$ $C_2-C_1,\ C_3-C_1$

$= abc\left(1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right) \begin{vmatrix} 1 & 0 & 0 \\ \frac{1}{b} & 1 & 0 \\ \frac{1}{c} & 0 & 1 \end{vmatrix}$

$= abc\left(1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)$

Q. $\begin{vmatrix} 1+a & 1 & 1 & 1 \\ 1 & 1+b & 1 & 1 \\ 1 & 1 & 1+c & 1 \\ 1 & 1 & 1 & 1+d \end{vmatrix}$

Imp Result

$= abcd\left(1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c}+\frac{1}{d}\right)$

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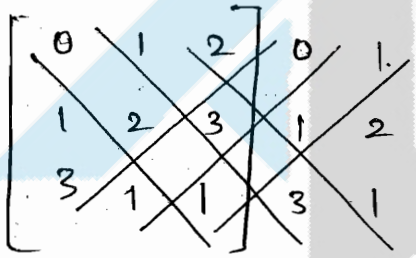
Q. $\begin{vmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{vmatrix} = ?$

(a) 4 (b) 5 (c) 6 (d) 7

Q. $\begin{vmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{vmatrix} = 0(\ ) - 1(1-9) + 2(1-6) = 8-10 = -2.$

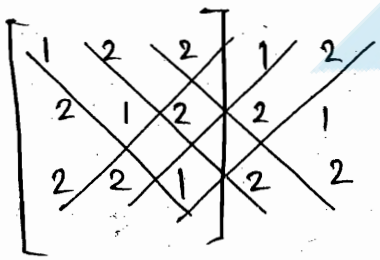
* SHORTCUT to determinant of Matrix :-

[Applicable only for 3×3 Matrix].

Q. 

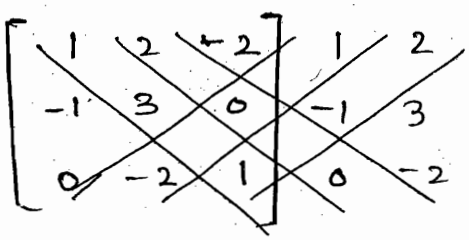
$0 + 9 + 2 - 12 - 0 - 1$

$= -2.$

Q. 

$\Delta = 1 + 8 + 8 - 4 - 4 - 4$

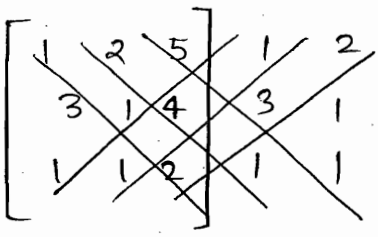
$\Delta = 17 - 12 = 5$

Q. 

$\Delta = 3 + 0 - 4 - 0 - 0 + 2$

$\Delta = 1$

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Q. 

$$\Delta = 2 + 8 + 15 - 5 - 4 - 12$$

$$\Delta = 25 - 21$$

$$\Delta = 4$$

INVERSE OF MATRIX :-

$$\boxed{A^{-1} = \frac{\text{adj } A}{\Delta}}$$

If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

Principle

→ Interchange diagonal element

→ & sign change in second diagonal element.

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Q. $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

$$A^{-1} = \frac{1}{4 - 6} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$$

$$= \frac{1}{-2} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$$

Q. $A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$

$$A^{-1} = \frac{1}{10 - 4} \begin{bmatrix} 2 & -4 \\ -1 & 5 \end{bmatrix}$$

$$= \frac{1}{6} \begin{bmatrix} 2 & -4 \\ -1 & 5 \end{bmatrix}$$

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Q. $A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$

$$A^{-1} = \frac{1}{\cos^2\theta + \sin^2\theta} \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$$

$$= \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$$

For 3×3 inverse matrix :-

$$A^{-1} = \frac{\text{adj } A}{\Delta}$$

adj A = Transpose of cofactors Matrix

Cofactor of elements = $(-1)^{i+j}$ minor

↳ Just to check sign for even or odd (row+column) no.

Minor of an Element = determinant of sq. sub matrix in which the row & column of the particular element lines to be deleted.

ex:-

$$\begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$$

Minor of 1 → $\begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} = 1 - 2 = -1$

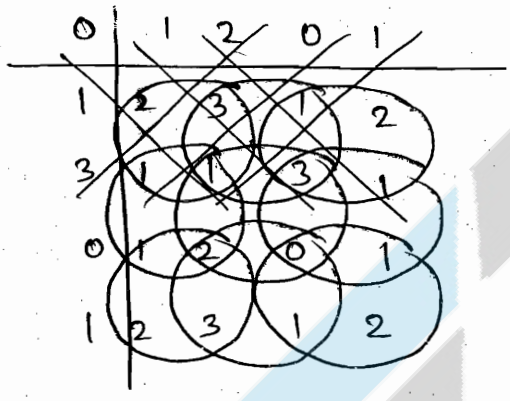
Cofactor of 1 = $(-1)^{2+1}(-1) = 1$

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* Shortcut to find Cofactor of 3×3 Matrix :-

Q. $\begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$.

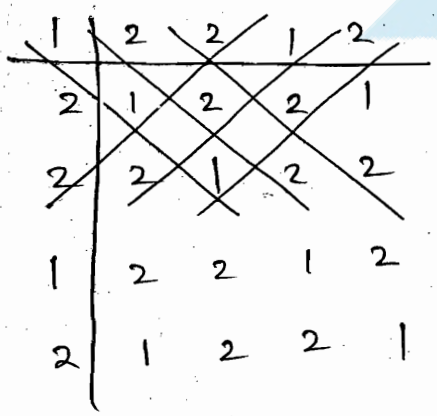
$\Delta = 0+9+2-12-0-1 = -2$



$$A^{-1} = \frac{1}{-2}\begin{bmatrix} -1 & 1 & -1 \\ 8 & -6 & 2 \\ -5 & 3 & -1 \end{bmatrix}$$

Q. $\begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$

$\Delta = 1+8+8-4-4-4 = 17-12 = 5$

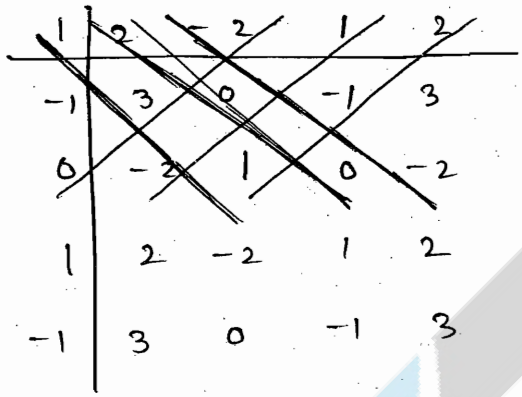


$$A^{-1} = \frac{1}{5}\begin{bmatrix} -3 & 2 & 2 \\ 2 & -3 & 2 \\ 2 & 2 & -3 \end{bmatrix}$$

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Q. $\begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix}$

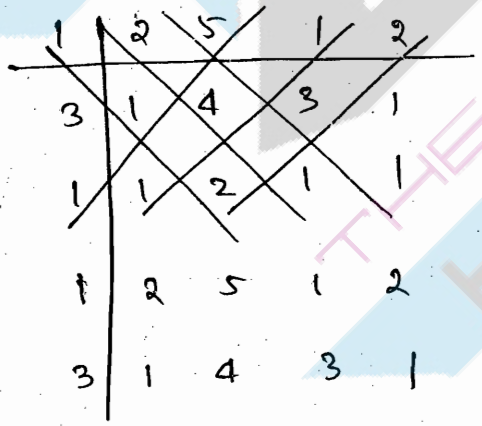
$\Delta = 0+0+(-2)-0 \cdot 3+0-4-0-0+2 = 1.$



$A^{-1} = \begin{bmatrix} 3 & 2 & 6 \\ 1 & 1 & 2 \\ 2 & 2 & 5 \end{bmatrix}$

Q. $\begin{bmatrix} 1 & 2 & 5 \\ 3 & 1 & 4 \\ 1 & 1 & 2 \end{bmatrix}$

$\Delta = 2+8+15-5-4-12 = 25-21 = 4$



$A^{-1} = \frac{1}{4}\begin{bmatrix} -2 & 1 & 3 \\ -2 & -3 & 11 \\ 2 & 1 & -5 \end{bmatrix}$

Q.

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Q/ $\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ - & . & \end{bmatrix}$

Q $\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$\Delta = \cos^2\theta + \sin^2\theta = 1$

$A^{-1} =$

$\cos\theta$	$-\sin\theta$	0	$\cos\theta$	$-\sin\theta$
$\sin\theta$	$\cos\theta$	0	$\sin\theta$	$\cos\theta$
0	0	1	0	0
$\cos\theta$	$-\sin\theta$	0	$\cos\theta$	$-\sin\theta$
$\sin\theta$	$\cos\theta$	0	$\sin\theta$	$\cos\theta$

Q. $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

0	1	0	0	1
1	0	0	1	0
0	0	1	0	0
0	1	0	0	1
1	0	0	1	0

$A^{-1} = \frac{1}{-1}\begin{bmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$

$= \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

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RANK OF MATRIX!-

If all minors of order $(r+1)$ are zeroes, but there is atleast one non zero minor of order r. if exists, It is called the rank of matrix and is denoted by $\boxed{\rho(A) = r}$

* Properties of Rank!-

i) If A is a null matrix or zero matrix then rank of A = 0

ex!- $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, $\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$, $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ etc.

ii) If A is a non zero matrix, then rank of $A \geq 1$

iii) If 'I' be the unit matrix or identity matrix of order $n \times n$, then rank of $I_n = n$

ex!- $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ rank = 2

$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ rank = 3

$I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ rank = 4.

iv) If A is a matrix of order $m \times n$, then rank of $A \leq \min\{m, n\}$

ex!- $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}_{2\times 3}$

rank ≤ 2.

* If all determinants of $2 \times 2 = 0$ rank < 2
* If at least one determinant of 2×2 is non zero, then rank $= 2$.

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Q. Find rank of $\begin{bmatrix} 1 & 2 & 3 & 4 \\ -2 & 0 & 5 & 7 \end{bmatrix}_{2\times4}$

$\rho(A) \leq 2$

By checking determinant of any one of $2\times2 \neq 0$ here

$\rho(A) = 2$

** Determinant checking
1st D check = 0
2nd D check = 0
3rd D check = 0
4th D check (no need)
of course There must be rows identical with some common factors.
∴ rank will $< \min\{m,n\}$
$\neq \min\{m,n\}$.

Q. $\begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 2 & 6 & 5 \end{bmatrix}$

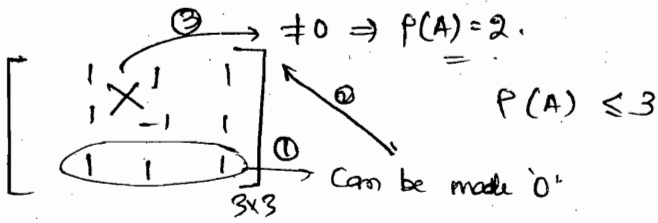
* purpose → make at least one row = 0

$\begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 2 & 6 & 5 \end{bmatrix}$ $\rho(A) \leq 3$.

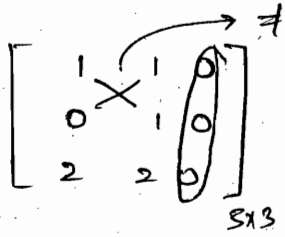
$R_3 - (R_1 + R_2)$

$\begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 0 & 0 & 0 \end{bmatrix}$ $\neq 0 \Rightarrow \rho(A) = 2$.

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Q. 

$\rho(A) \leq 3$

Q. 

$\rho(A) \leq 3$

Q. $\begin{bmatrix} 1 & 2 & 3 \\ -1 & 2 & 2 \\ 0 & 5 & 6 \end{bmatrix}_{3\times3}$

$\rho(A) \leq 3$

$\rho(A) = 3$.

Q. $\begin{bmatrix} 2 & 3 & 4 & -1 \\ 5 & 2 & 0 & -1 \\ -4 & 5 & 12 & -1 \end{bmatrix}_{3\times4}$

$\rho(A) \leq 3$.

$R_3 - 3R_1$

$$\left[\begin{array}{cc|c|c} 2 & 3 & 4 & -1 \\ \hline 5 & 2 & 0 & -1 \\ -10 & -4 & 0 & 2 \end{array}\right]$$

$= 0 \Rightarrow \rho(A) < 3$

$\rho(A) = 2$

Q. $\begin{bmatrix} 1 & -1 & 3 & 6 \\ 1 & 8 & -3 & -4 \\ 5 & 3 & 3 & 11 \end{bmatrix}_{3\times4}$

$\rho(A) \leq 3$.

$R_2 + R_1$

$R_3 - R_1$

$$\left[\begin{array}{cc|c|c} 1 & -1 & 3 & 6 \\ \hline 2 & 7 & 0 & 2 \\ 4 & 4 & 0 & 5 \end{array}\right]$$

$\neq 0 \Rightarrow \rho(A) = 3$

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Q. $\begin{bmatrix} 1 & 3 & 4 & 3 \\ 3 & 9 & 12 & 3 \\ \hline 1 & 3 & 4 & 1 \end{bmatrix}$ $\to \rho(A) = 2$

$\Rightarrow \rho(A) \leq 3$

$\rho(A) < 3$ $\{\because R_2, R_1 \text{ identical}\}$

Q $\begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 4 & 3 & 2 \\ 3 & 2 & 1 & 3 \\ 6 & 8 & 7 & 5 \end{bmatrix}_{4\times4}$ $\rho(A) \leq 4$

$R_4 - (R_1 + R_2 + R_3)$

$\begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 4 & 3 & 2 \\ 3 & 2 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ $\rho(A) < 4$

$R_2 - 2R_1$

$\begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & 0 & -3 & 2 \\ 3 & 2 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$\neq 0 \Rightarrow \rho(A) = 3.$

Q. $\begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$ $\rho(A) \leq 4$

R_{12}

$\begin{bmatrix} 1 & -1 & -2 & -4 \\ 2 & 3 & -1 & -1 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$ $R_2 - 2R_1$, $R_3 - 3R_1$

$R_4 - (R_1 + R_2 + R_3)$

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$$= \begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 5 & 3 & 7 \\ 0 & 4 & 9 & 10 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\rho(A) < 4$

$\neq 0 \Rightarrow \rho(A) = 3.$

(b)

$$\begin{bmatrix} 6 & 1 & 3 & 8 \\ 4 & 2 & 6 & -1 \\ 10 & 3 & 9 & 7 \\ 16 & 4 & 12 & 15 \end{bmatrix}$$

$\rho(A) \leq 4$

$R_4 - (R_1 + R_3)$

$R_3 - (R_1 + R_2)$

$$\begin{bmatrix} 6 & 1 & 3 & 8 \\ 4 & 2 & 6 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\rho(A) < 4$

$\rho(A) < 3$

$\rho(A) = 2$

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The consistency (→ have soln) & Inconsistency (→ have no soln) of the system of eqn:-

$$a_1x + b_1y + c_1z = d_1$$
$$a_2x + b_2y + c_2z = d_2$$
$$a_3x + b_3y + c_3z = d_3$$

$$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$$ Coefficient Matrix

$$B = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$ Const. Matrix.

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$ Variable Matrix.

$$AB = \begin{bmatrix} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \end{bmatrix}$$

Augmented Matrix.

Eqn form → $\boxed{AX = B}$

* Any system is said to be a consistent system, if it has a solution.

* Inconsistent system has no solution.

→ $\boxed{\text{If } P(AB) = P(A) = \text{No. of Unknowns}}$, then the system is said to be consistent and having **unique solution**.

→ $\boxed{\text{If } P(AB) = P(A) < \text{No. of Unknowns}}$ then the system is said to be consistent and having more than one solution, or **∞ no. of solutions**.

* Remember → either 1 sol^n or ∞ sol^n exist for system
To confuse, in exam option will be 2 sol^n or 3 sol^n ✗

→ $\boxed{\text{If } P(AB) \neq P(A)}$, then the System is said to Inconsistent and have **no solution**.

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Q. $x + y + z = 3$

$x + 2y + 3z = 4$

$x + 4y + 9z = 6$

(a) 0 (b) 1 (c) ~~2~~ (d) ∞

NOTE!-

→ If AB is rectangle matrix, then last column must be excluded.

→ If AB is a square matrix, then last column must be included

→ If last column is excluded, then it represents both the matrice's rank.

→ If last column included, then it represents only AB matrix rank.

Now,

$$[AB] = \begin{bmatrix} 1 & 1 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 1 & 4 & 9 & 6 \end{bmatrix}_{3\times 4} \quad R_2 - R_1 \,,\; R_3 - R_1$$

$$= \left[\begin{array}{c|ccc} 1 & 1 & 1 & 3 \\ \hline 0 & 1 & 2 & 1 \\ 0 & 3 & 8 & 3 \end{array}\right]$$

$\neq 0 \Rightarrow \rho(AB) = \rho(A) = 3 = 3$ (no. of unknown)

∴ System has unique soln.

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Q. Check the consistency of system –

$$x - 2y + 3z = 2$$
$$2x - 3z = 3$$
$$x + y + z = 0$$

→

$$\begin{bmatrix} 1 & -2 & 3 & 2 \\ 2 & 0 & -3 & 3 \\ 1 & 1 & 1 & 0 \end{bmatrix}_{3\times 4} \quad R_1 + 2R_3$$

$$\begin{bmatrix} 3 & 0 & 5 & 2 \\ 2 & 0 & -3 & 3 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

→ $\neq 0 \Rightarrow \rho(AB) = \rho(A) = 3 =$ no. of unknowns $\therefore$

$\therefore$ System having consistent and having unique soln.

Q. $3x + 3y + 2z = 1$

$x + 2y = 4$

$10y + 3z = -2$

$2x - 3y - z = 5$

$$[AB] = \begin{bmatrix} 1 & 2 & 0 & 4 \\ 3 & 3 & 2 & 1 \\ 0 & 10 & 3 & -2 \\ 2 & -3 & -1 & 5 \end{bmatrix}_{4\times 4} \quad R_2 - 3R_1,\ R_4 - 2R_1$$

$$= \begin{bmatrix} 1 & 2 & 0 & 4 \\ 0 & -3 & 2 & -11 \\ 0 & 10 & 3 & -2 \\ 0 & -7 & -1 & 3 \end{bmatrix} \quad R_2 + 2R_4,\ R_3 + 3R_4$$

Note:-

If 0 present → make adjacent '0'

If no '0' present, then check if 1 or −1 → make adjacent of 1 or −1 = 0

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$$\begin{bmatrix} 1 & 2 & 0 & 4 \\ 0 & -17 & 0 & -17 \\ 0 & -11 & 0 & -11 \\ 0 & -7 & -1 & -3 \end{bmatrix}$$

take common

take common

then $R_3 - R_2$ and interchange R_4 & R_3

$$\left[\begin{array}{c|ccc} 1 & 2 & 0 & 4 \\ \hline 0 & 1 & 0 & 1 \\ 0 & -7 & -1 & -3 \\ \hline 0 & 0 & 0 & 0 \end{array}\right]$$

$\neq 0 \Rightarrow \rho(A) = \rho(AB) = 3 =$ no. of unknowns

$\therefore$ System is consistent & having unique solution.

Q. $4x - 2y + 6z = 8$

$x + y - 3z = -1$

$15x - 3y + 9z = 21$

$$[AB] = \begin{bmatrix} 1 & 1 & -3 & -1 \\ 2 & -1 & 3 & 4 \\ 5 & -1 & 3 & 7 \end{bmatrix}_{3\times 4} \quad \begin{matrix} R_2 + R_1 \\ R_3 + R_1 \end{matrix}$$

$$\left[\begin{array}{c|ccc} 1 & 1 & -3 & -1 \\ \hline 3 & 0 & 0 & 3 \\ \hline 6 & 0 & 0 & 6 \end{array}\right]$$

$= 0 \Rightarrow$

$\therefore \rho(AB) = \rho(A) = 2 < 3.$

System – Consistent

But ∞ Solution.

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We can write system again as—

$x + y - 3z = -1$

$3x = 3$

$\Rightarrow x = 1.$

$1 + y - 3z = -1$

$\Rightarrow y - 3z = -2$ —— ①

Since, we are discussing the consistency of only real system of equations, therefore solutions always real numbers.

Let $z = k$ where $k \in R$

$y - 3k = -2$

$$\Rightarrow \boxed{\begin{aligned} y &= 3k - 2 \\ z &= k \\ x &= 1. \end{aligned}}$$

for different values of K, we will have diff. solutions and these solutions are infinite.

Q. $2x - y + z = 4$

$3x - y + z = 6$

$4x - y + 2z = 7$

$-x + y - z = 9$

Check the consistency of System.

→ $[AB] = \begin{bmatrix} -1 & 1 & -1 & 9 \\ 2 & -1 & 1 & 4 \\ 3 & -1 & 1 & 6 \\ 4 & -1 & 2 & 7 \end{bmatrix}_{4\times4}$

$R_2 + R_1$

$R_3 + R_1$

$R_4 + R_1$

$\sim \begin{bmatrix} -1 & 1 & -1 & 9 \\ 1 & 0 & 0 & 13 \\ 2 & 0 & 0 & 15 \\ 3 & 0 & 1 & 16 \end{bmatrix}$

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C_{12}

1	-1	-1	9
0	1	0	13
0	2	0	15
0	3	1	16

$\neq 0 \Rightarrow \rho(AB) = 4$

But $\rho(A) = 3$

$\therefore \rho(AB) \neq \rho(A)$

Hence, system is inconsistent and having no solution.

Q. for what values of λ and μ, does the system of equations

$x + y + z = 6$

$x + 2y + 3z = 10$

$x + 2y + \lambda z = \mu$

has —

(A) no solution

(B) Unique sol^n

(C) More than one solution.

→

$$[AB] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & \lambda & \mu \end{array}\right]_{3\times 4} \quad R_3 - R_2$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 0 & 0 & \lambda-3 & \mu-10 \end{array}\right]$$

$\neq 0 \Rightarrow \rho(AB) = \rho(A) = 3$ if $\lambda - 3 \neq 0$

< 3 if $\lambda - 3 = 0$

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Case - I

Case - II

Case - III.

Q. $2x + 3y + 5z = 9$
$7x + 3y - 2z = 8$
$2x + 3y + \lambda z = \mu$

For what values of λ & μ System has —

(A) no soln
(B) unique soln.
(C) more than one soln.

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Eigen Values and Eigen Vectors :-

Characteristic equation :- Let A be the sq. matrix of order $n \times n$ and I be the unit matrix of order $n \times n$ then $|A - \lambda I| = 0$ is called the characteristic eqn where λ is a parameter.

The roots of characteristic eqn are called Characterestic roots / latent / eigen / proper values.

$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ or $[x_1\ x_2 \dots x_n]^T$ which satisfies the matrix eqn $[A - \lambda I]X = 0$ is called the coresponding eigen vector of the matrix.

Note :-

i) The sum of eigen values of any matrix is equal to sum of the elements of its principal digonals.

Ex :- $\begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$ (A) 1 (B) 3 (C) 5 (D) 7.

Trace of A = 1+5+1 = 7.

ii) The Product of eigen values of any matrix is equal to its determinant.

Ex :- The characteristic roots of matrix $\begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$

(A) 2, 5 ✓
(B) 3, 4 ✗
(C) 1, 6 ✓
(D) 2, 3 ✗

$\therefore$ { → Sum = sum of principle digonal 6+1 = 7
→ Product = |determinant value| }

iii) The eigen values of a symmetric matrix ($A^T = A$) are purely real.

Ex :- $A = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix} = A^T$ ∴ Eigen values are 1 & 5 which are purely real

Cross check → 1+5 = 6 equal to Sum of Principle digonal
1×5 = 5 equal to determinant value

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iv) The eigen value of skew symmetric matrix are ($A^T = -A$) either purely imaginary or zeroes

Ex:- $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

Eigen values ?

(a) 1, -1 ✗ (b) i, -i ✓ (c) -1, 0 ✗ (d) 0, 1 ✗

$$A^T = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = -\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = -A$$

v) If a matrix is either lower triangular or upper triangular, then principle digonal elements are called eigen values.

* If Given Matrix only is L·T·M or U·T·M (without operation only original) → Then only follow this rule.

Ex:-

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -4 & 2 \\ 0 & 0 & 7 \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 4 & 5 & 6 \end{bmatrix}$$

$\lambda = 1, -4, 7$ $\qquad$ $\lambda = 1, 3, 6$

Don't Use this Property to find eigen value when L·T·M or U·T·M is made by some operation.

vi) If λ is an eigen value of A

then λ^2 ——— A^2

λ^3 ——— A^3

λ^n ——— A^n

$\frac{1}{\lambda}$ ——— A^{-1}

Ex:- if $S = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}$ and $\lambda = 1, 5$

then for S^2 what will be eigen value → After getting result λ, verify it by Sum & product rule.

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

then eigen values of A^3 are —

a) 5, 4

b) 9, 1

c) 16, -1

d) 27, -1

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Q. Find the eigen values and corresponding eigen vectors of Matrix $A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$

$\rightarrow \quad |A - \lambda I| = 0$

$$\left| \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right| = 0$$

$$\begin{vmatrix} 5-\lambda & 4 \\ 1 & 2-\lambda \end{vmatrix} = 0.$$

$\Rightarrow \quad 10 - 7\lambda + \lambda^2 - 4 = 0$

$\Rightarrow \quad \lambda^2 - 7\lambda + 6 = 0$

$\Rightarrow \quad (\lambda - 1)(\lambda - 6) = 0$

$\lambda = 1, 6.$

Eigen Vector —

$[A - \lambda I] X = 0.$

Substract λ from matrix and multiply ~~vector~~ X Matrix

for $\lambda = 1$ $\quad \begin{bmatrix} 4 & 4 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$

$\Rightarrow \quad 4x_1 + 4x_2 = 0$

$x_1 + x_2 = 0$

$\Rightarrow x_1 = -x_2$

$\Rightarrow \frac{x_1}{1} = \frac{x_2}{-1}$ eigen vector $= \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ or $[1 \quad -1]^T$.

for $\lambda = 6$ —

$$\begin{bmatrix} -1 & 4 \\ 1 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0.$$

$-x_2 + 4x_1 = 0$

$\frac{x_1}{4} = \frac{x_2}{4}$ eigen vector $= \begin{bmatrix} 4 \\ 4 \end{bmatrix}$ or $[4, 4]^T$.

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Q. find the ch. roots of matrix or Eigen Value.

$$\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

1st step

(a) 3, 7, 8 ✓
(b) 2, 2, 14 ✓
(c) 0, 3, 15 ✓
(d) 1, 4, 9 ✗

} By sum of digonal rule = Sum of eigen value.

2nd step

check determinant.

$\Delta = 0$.

So, one root must be 0 so that to make product rule of eigen values correct

Q. find the latent root of matrix —

$$\begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$$

(a) 1, 2, 3 — 1st step ✓ ✓ — 2nd step $4 + 2 = 6$.
(b) 0, 2, 4 ✓ ✗
(c) 1, 1, 4 ✓ ✗
(d) 2, 2, 2. ✓ ✗

Q. Find Eigen Values and Corresponding eigen vectors of

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -4 & 2 \\ 0 & 0 & 7 \end{bmatrix}$$

$\lambda = 1, -4, 7$. [∵ It is U.T.M]

for $\lambda = 1$

$$\begin{bmatrix} 0 & 2 & 3 \\ 0 & -5 & 2 \\ 0 & 0 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0.$$

$2x_2 + 3x_3 = 0$.
$-5x_2 + 2x_3 = 0$

$6x_3 = 0. \Rightarrow x_3 = 0 \Rightarrow x_2 = 0$ → not able to identify any info abt x_1.

Remember
* for any eigen values if in eigen vectors set, two values are = 0 then third must be only non zero

∴ eigen vector set = $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

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for $\lambda = -4$

$$\begin{bmatrix} 5 & 2 & 3 \\ 0 & 0 & 2 \\ 0 & 0 & 11 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0.$$

$5x_1 + 2x_2 + 3x_3 = 0. \Rightarrow 5x_1 + 2x_2 = 0.$

$2x_3 = 0 \Rightarrow x_3 = 0 \Rightarrow 5x_1 = -2x_2$

$11x_3 = 0. \Rightarrow x_3 = 0. \Rightarrow \frac{x_1}{2} = \frac{x_2}{-5}$

Eigen Vector $\therefore$ $\begin{bmatrix} 2 \\ -5 \\ 0 \end{bmatrix}$.

for $\lambda = 7$.

$$\begin{bmatrix} -6 & 2 & 3 \\ 0 & -11 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0.$$

$-6x_1 + 2x_2 + 3x_3 = 0.$

$-11x_2 + 2x_3 = 0 \Rightarrow \frac{x_2}{2} = \frac{x_3}{11}$ $\Rightarrow 6x_1 = 37 \Rightarrow x_1 = \frac{37}{6}$.

eigen vector : $\begin{bmatrix} \frac{37}{6} \\ 2 \\ 11 \end{bmatrix}$ or $\begin{bmatrix} 37 \\ 12 \\ 66 \end{bmatrix}$

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Q. for the matrix $P = \begin{bmatrix} 5 & -2 & 2 \\ 0 & -2 & 1 \\ 0 & 0 & 3 \end{bmatrix}$ one of the eigen value is -2. which of the following is an eigen vector.

(A) $\begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$ (B) $\begin{bmatrix} -3 \\ 2 \\ -1 \end{bmatrix}$ (C) $\begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$ (D) ✓ $\begin{bmatrix} 2 \\ 5 \\ 0 \end{bmatrix}$

→ for $\lambda = -2$ $\begin{bmatrix} 7 & -2 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0.$

$7x_1 - 2x_2 + 2x_3 = 0. \Rightarrow 5x_1 - 2x_2 = 0.$

$\frac{x_1}{2} = \frac{x_2}{5}.$

$x_3 = 0.$
$3x_3 = 0$ $\Big\}$ $x_3 = 0$

eigen vector $= \begin{bmatrix} 2 \\ 5 \\ 0 \end{bmatrix}$ ✓

Q. for the matrix $P = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}$ which one of the following is an eigen vector.

a) $[1 \quad -1 \quad 1]^T$

b) $[1 \quad 2 \quad 1]^T$

c) $[2 \quad -1 \quad 1]^T$

d) $[1 \quad 1 \quad -1]^T$

Give 1st pref to least eigen value.
Check only by substracting $-\lambda$ in 1st row
If option not found check eigen value.
Again 2nd pref to 2nd least eigen value
Again if option not found: Same substract λ from 2nd row check eigen value
Again 3rd pref to 3rd least eigen value but do not substract to 3rd row substract from either 1st or 2nd row (no effect) ∴ check eigen value ✓

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Cayley - Hamilton Theorem :-

Every sq. matrix satisfies its own characteristic eqn.

Ex:- $A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$.

$$\begin{vmatrix} 5-\lambda & 4 \\ 1 & 2-\lambda \end{vmatrix} = 0.$$

$\Rightarrow 10 - 7\lambda + \lambda^2 - 4 = 0$

$\Rightarrow \lambda^2 - 7\lambda + 6 = 0$

$\Rightarrow \lambda =$

Means —

$A^2 - 7A + 6I = 0 \rightarrow$ A/c to Cayle Hamilton Theorem

↓

For Inverse finding without knowing adj of A & determinant.

$\Rightarrow 6I = 7A - A^2$

$\Rightarrow 6IA^{-1} = 7AA^{-1} - A^2A^{-1}$

$\Rightarrow 6A^{-1} = 7I - A.$

$$= \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix} - \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$$

$$= \frac{1}{6} \begin{bmatrix} 2 & -4 \\ -1 & 5 \end{bmatrix}.$$

Q. Let M be 3×3 matrix with ch. eqn $x^3+1=0$ then the inverse of M = —

→ A/c to Cayle Hamilton theorem —

$M^3 + I = 0$

$I = -M^3$

$IM^{-1} = -M^3M^{-1}$

$\boxed{M^{-1} = -M^2}$

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Q. Let P be a 3×3 matrix with ch. eqⁿ $\lambda^3+\lambda^2+2\lambda+1=0$ then inverse of P = ___

(a) P^2+P+I

(b) P^2+P+2I

(c) $-(P^2+P+2I)$

(d) $-(P^2+P+I)$.

→ $P^3+P^2+2P+1=0$

$\Rightarrow I = -(P^3+P^2+2P)$

$\Rightarrow \boxed{P^{-1} = -(P^2+P+2I)}$

Q. Find the value of λ for which the following system of eqⁿs

$\lambda x + 3y + 5z = 0$

$2x - 4\lambda y + \lambda z = 0$

$-4x + 18y + 7z = 0$

have a non trivial solution.

Note:-

$\boxed{AX=0}$ is called Homogeneous.

$X=0$ $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ will always be a solution.

⇓

This is called Trivial equation.

Note:-

for the system of Eqn $AX=0$

$X=0$ will always a solution

and this soln is called a trivial soln.

for a non trivial solution determinant of coefficient matrix $=0$ (always)

$$\begin{bmatrix} \lambda & 3 & 5 \\ 2 & -4\lambda & \lambda \\ -4 & 18 & 7 \end{bmatrix} \begin{matrix} \lambda & 3 \\ 2 & -4\lambda \\ -4 & 18 \end{matrix}$$

$-28\lambda^2 - 12\lambda + 180 - 80\lambda - 18\lambda^2 - 42 = 0$

$\Rightarrow -46\lambda^2 - 92\lambda + 138 = 0$

$\Rightarrow \lambda^2 + 2\lambda - 3 = 0$

$\therefore \lambda = -3$ or 1.

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Q. Rank of $\begin{bmatrix} 4 & 2 & 1 & 3 \\ 6 & 3 & 4 & 7 \\ 2 & 1 & 0 & 1 \end{bmatrix}$

(a) 1 (b) 2 (c) 3 (d) 4.

→ $\begin{bmatrix} 4 & 2 & 1 & 3 \\ 6 & 3 & 4 & 7 \\ 2 & 1 & 0 & 1 \end{bmatrix}_{3\times 4}$ $\rho(A) \leq 3$

$R_2 - 4R_1$

4	2	1	3
-10	-5	0	+5
2	1	0	1

$= 0 \Rightarrow \rho(A) < 3$

$\rho(A) = 2$

Q. Rank and nullity of matrix.

$\begin{bmatrix} 6 & 1 & 8 & 3 \\ 2 & 3 & 0 & 2 \\ 4 & -1 & -8 & -3 \end{bmatrix}$

Nullity of A = no. of columns − Rank

10	0	0	0
2	3	0	2
4	-1	-8	-3

$\rho(A) \leq 3$

$\neq 0 \Rightarrow \rho(A) = 3$

∴ nullity of A = 4 − 3

= 1 Ans

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Q. The rank of $\begin{bmatrix} {}^{1}C_3 & {}^{2}C_1 & {}^{3}C_2 \\ 4 & 1 & K \end{bmatrix}$ is 2

then $K = \underline{5}$

Q. The inverse of $\begin{bmatrix} 0.5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0.2 \end{bmatrix}$

$\rightarrow \begin{bmatrix} \frac{1}{0.5} & 0 & 0 \\ 0 & \frac{1}{1} & 0 \\ 0 & 0 & \frac{1}{0.2} \end{bmatrix}$

**** Inverse of Digonal matrix is inverse of each element in that digonal.

Q. The rank of $\begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & -3 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & -4 \end{bmatrix}$ 4×4

rank = 4

**** Rank of digonal matrix = no. of non zero elements in that digonal.

Q. Let A, B, C, D be n×n matrices each with non zero determinant and if ABCD = I then $B^{-1} = $ ____

$(ABCD)^{-1} = I^{-1}$

$D^{-1}C^{-1}B^{-1}A^{-1} = I$

~~$B^{-1} = I$~~

~~$BB^{-1}CC^{-1}B^{-1}AA^{-1}$~~

$DD^{-1}C^{-1}B^{-1}A^{-1} = ID$

$CC^{-1}B^{-1}A^{-1} = IDC$

$B^{-1}AA^{-1} = IDCA = DCA$

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Q. The eigen vector pair of matrix $\begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$

(a) $\begin{bmatrix} 2 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

(b) $\begin{bmatrix} 2 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \end{bmatrix}$

(c) $\begin{bmatrix} -2 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

(d) $\begin{bmatrix} -2 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \end{bmatrix}$.

→ $\begin{bmatrix} 3-\lambda & \\ & \end{bmatrix}$ ⇒ $-9 - 3\lambda + 3\lambda + \lambda^2 - 16 = 0$

⇒ $\lambda^2 = 25$

$\lambda = 5, -5$.

for $\lambda = 5$

$$\begin{bmatrix} -2 & 4 \\ 4 & -8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0.$$

$-2x_1 + 4x_2 = 0$

$4x_1 - 8x_2 = 0.$

$\frac{x_1}{8} = \frac{x_2}{4}$

$2 \quad 1$

$\begin{bmatrix} 2 \\ 1 \end{bmatrix}$

for $\lambda = -5$.

$$\begin{bmatrix} 8 & 4 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0.$$

$8x_1 + 4x_2 = 0$

$\frac{x_1}{4} = \frac{x_2}{-8}$

$1 \quad -2$

$\begin{bmatrix} 1 \\ -2 \end{bmatrix}$

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Q. How many of the following matrices have an eigen value = 1

$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ $\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$ $\begin{bmatrix} -1 & 0 \\ 1 & -1 \end{bmatrix}$

$\lambda = 1, 0$ $\lambda = 0, 0$ $(1-\lambda)^2 + 1 = 0$ $(-1-\lambda)^2 = 0$

(a) 1 (b) 2 (c) 3 (d) 4.

Q. If the following represents eqn of line then line passes through the point

$$\begin{bmatrix} x & 2 & 4 \\ y & 8 & 0 \\ 1 & 1 & 1 \end{bmatrix} = 0$$

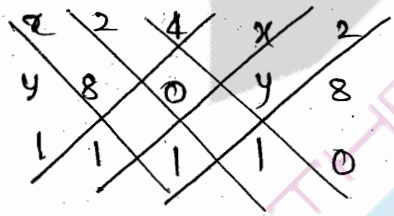
a) (0, 0)

b) (3, 4)

c) (4, 3)

d) (4, 4)

→



$8x + 4y - 32 - 2y = 0$

$\Rightarrow 8x + 2y = 32$

$-32 + 0 + 2y - 8x - 0 + 0 = 0$

$\Rightarrow -8x + 2y + 32 = 0$

$\Rightarrow 8x + 2y = 32.$

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Q. A is 3×4 real matrix and $Ax = B$ is an inconsistent system of eq^{n}s then the highest possible rank of A —

(a) 1 (b) 2 (c) 3 (d) 4 ..

$A = 3 \times 4$

$[AB] = 3 \times 5 \Rightarrow \rho(A) \leq 3$

for inconsistent system $\rho(A) \neq \rho[AB]$ { $\because \rho(A)$ can't be greater than $\rho(AB)$ }

$\therefore \rho(A) < 3$

$\therefore \rho(A) = 2$

Q. Given an orthogonal matrix $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix}$ then $(AA^T)^{-1} =$ I

NOTE!-

A matrix is said to be orthogonal matrix if $AA^T = I$.

Q. $A = \begin{bmatrix} 2 & -0.1 \\ 0 & 3 \end{bmatrix}$, $A^{-1} = \begin{bmatrix} \frac{1}{2} & a \\ 0 & b \end{bmatrix}$

then $a+b =$ ____

$A^{-1} = \frac{1}{6} \begin{bmatrix} 3 & 0.1 \\ 0 & 2 \end{bmatrix}$

$\therefore a = \frac{0.1}{6}$

$b = \frac{2}{6}$

$\therefore a+b = \frac{2.1}{6} = \frac{21}{60} = \frac{7}{20}$ Ans

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Q. Let $A = \begin{bmatrix} 2 & 0 & 0 \\ -1 & -2 & 0 \\ 4 & 2 & -1 \end{bmatrix}$ and $B = 2A^2$ then $|B| =$ ____

(a) 16 (b) 32 (c) 64 (d) 128.

$B = 2A^2$

$|B| = |2A^2|$ → order = 3×3

$= 2^3 |A^2|$

$= 8(16)$

$= 128.$

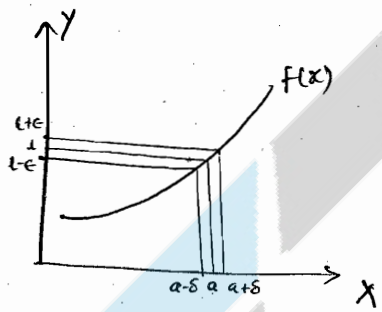
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DIFFERENTIAL CALCULUS

Limit :- A no l is said to be the limit of the fn $f(x)$ as $x \to a$ if $\forall \epsilon > 0$ (however small) $\exists$ a $\delta > 0$ such that $|f(x) - l| < \epsilon$ $\forall |x-a| < \delta$

$$\lim_{x \to a} f(x) = l$$



$\therefore \quad l - \epsilon < f(x) < l + \epsilon$

$\forall \quad a - \delta < x < a + \delta$

* Left limit :- $\lim_{x \to a^-} f(x) = \lim_{h \to 0} f(a-h), \quad (a-\delta < x < a)$

* Right limit :- $\lim_{x \to a^+} f(x) = \lim_{h \to 0} f(a+h), \quad (a < x < a+\delta)$

* Existence of Limit :- The limit of a function exists if both the left and right limits are existed and are equal.

ex :- $\lim_{x \to a} \frac{1}{x-a}$.

$$\lim_{x \to a^-} f(x) = \lim_{h \to 0} f(a-h) = \frac{1}{a-h-a} = \frac{-1}{h}$$

$$\lim_{x \to a^+} f(x) = \lim_{h \to 0} f(a+h) = \frac{1}{a+h-a} = \frac{1}{h}$$

$\therefore$ LH limit $\neq$ R.H limit

$\therefore$ Limit not exists

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Formulae :-

$$\lim_{x\to a}\frac{x^n-a^n}{x-a} = n a^{n-1}$$

$$\lim_{x\to a}\frac{x^m-a^m}{x^n-a^n} = \frac{m}{n}a^{m-n}$$

$$\lim_{x\to 0}\frac{e^x-1}{x} = 1$$

$$\lim_{x\to 0}\frac{e^{mx}-1}{x} = m$$

$$\lim_{x\to 0}\frac{a^x-1}{x} = \log a$$

$$\lim_{x\to 0}\frac{a^x-b^x}{x} = \log\left(\frac{a}{b}\right)$$

- $\lim_{x\to 0}(1+x)^{1/x} = e$
- $\lim_{x\to 0}(1+ax)^{1/x} = e^a$

$$\lim_{x\to 0}(1+ax)^{b/x} = e^{ab}$$

$$\lim_{x\to 0}(1+ax)^{1/bx} = e^{a/b}$$

$$\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^x = e$$

$$\lim_{x\to\infty}\left(1-\frac{1}{x}\right)^x = e^{-1}$$

$$\lim_{x\to 0}\frac{\sin x}{x} = 1$$

$$\lim_{x\to 0}\frac{\tan x}{x} = 1$$

$$\lim_{x\to 0}\frac{\sin^{-1}x}{x} = 1$$

$$\lim_{x\to 0}\frac{\tan^{-1}x}{x} = 1$$

$$\lim_{x\to 0}\frac{1-\cos mx}{1-\cos nx} = \frac{m^2}{n^2}$$

$$\lim_{x\to 0}\frac{\cos ax-\cos bx}{x^2} = \frac{b^2-a^2}{2}$$

$\frac{0}{0}, \frac{\infty}{\infty}, \frac{\infty}{0}, 0\times\infty, 1^\infty$ etc are Indeterminate form.

If $\frac{0}{0}, \frac{\infty}{\infty}$ then apply L·H·rule

$$\lim_{x\to a}\frac{f(x)}{g(x)} = \lim_{x\to a}\frac{f'(x)}{g'(x)} = \lim_{x\to a}\frac{f''(x)}{g''(x)} = \cdots\cdots$$

$$\lim_{x\to a}[f(x)]^{g(x)} = e^{\lim_{x\to a} g(x)[f(x)-1]}$$

$\lim_{x\to\infty}\frac{f(x)}{g(x)}$, if $f(x)$ & $g(x)$ are algebric fn.

Case-1 if degree of $f(x) > g(x)$ degree, then result $= \infty$

ex:- $\lim_{x\to\infty}\frac{2x^3+3x+5}{5x^2+6x+7} = \infty$

Case-2 if degree of $f(x) < g(x)$ degree then result $= 0$

ex:- $\lim_{x\to\infty}\frac{2x^2+3x+4}{x^3+8} = 0$

Case-3 If $f(x)$ and $g(x)$ have equal degrees then —

result $= \frac{\text{co·eff of } N^r}{\text{coeff of } D^r}$

ex:- $\lim_{x\to\infty}\frac{2x^2+3x+4}{5x^2+6x+8} = \frac{2}{5}$

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* $|x| = x$ if $x > 0$

* $|x| = -x$ if $x < 0$.

* $[x]$ = greatest integer not greater than x.

* **Examples :-**

① $\lim_{x \to 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$

② $\lim_{x \to 0} \frac{1 - \cos x}{x \sin x} = \frac{1 - \cos x}{x^2} \cdot \frac{\sin x}{2x} = \frac{\cos x}{2} = \frac{1}{2}$

In Trigonometric fn., avoid LH rule Try alternates

③ $\lim_{x \to 0} \frac{\tan x - \sin x}{\sin^3 x} = \frac{1 - \cos x}{\sin x \cdot \sin^2 x} \cdot \frac{1}{(1-\cos x)(1+\cos x)} = \frac{1}{\cos x (1 + \cos x)} = \frac{1}{1 \cdot (1+1)} = \frac{1}{2}$

④ $\lim_{x \to 1} \frac{\cos\left(\frac{\pi x}{2}\right)}{1 - \sqrt{x}} = \frac{-\frac{\pi}{2} \sin\left(\frac{\pi x}{2}\right)}{-\frac{1}{2\sqrt{x}}} = \frac{\pi/2}{1/2} = \pi$

⑤ $\lim_{x \to 0} \frac{2^{8 \cos x}}{8} \cdot \frac{\left(\sin^8\left(\frac{\pi}{6} + x\right) - \sin^8 \frac{\pi}{6}\right)}{x}$

$= \frac{2^{8 \cdot 1}}{8} \left[\frac{8 \sin^7\left(\frac{\pi}{6} + x\right) \cos\left(\frac{\pi}{6} + x\right) - 0}{1} \right]$

$= \frac{2^8}{8} \left[8 \times \frac{1}{2^8} \cdot \frac{\sqrt{3}}{2} \right]$

$= \sqrt{3}$ Ans

⑥ $\lim_{x \to 0} \frac{\sqrt{1 - x^2} - \sqrt{1 + x^2}}{x^2} = -1$

⑦ $\lim_{x \to 0} \frac{\sqrt[3]{1+x} - \sqrt[3]{1-x}}{x} = \frac{2}{3}$

⑧ $\lim_{x \to 0} \frac{\log(1+x)}{x} = 1.$

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(9) $\lim_{x\to 0} (1+\sin x)^{\operatorname{cosec} x} = e^{\lim_{x\to 0} \operatorname{cosec} x [1+\sin x - 1]}$ $\to 0$

$= 1$

(10) $\lim_{x\to 0} (1+2x)^{1/3x} = e^{\lim_{x\to 0} \frac{1}{3x}[1+2x-1]} = e^{\frac{2}{3}}$

(11) $\lim_{n\to\infty} \left(1-\frac{1}{n}\right)^{2n} = e^{\lim_{n\to\infty} 2n\left[1-\frac{1}{n}-1\right]} = e^{\lim_{n\to\infty}[-2]} = e^{-2}$

(12) $\lim_{x\to\infty} \left(\frac{x-1}{x-2}\right)^{x} = e^{\lim_{x\to\infty} x\left[\frac{x-1-x+2}{x-2}\right]} = e$

(13) $\lim_{n\to\infty} \frac{1+2+3+\cdots+n}{n^2} = \frac{n(n+1)}{2n^2} = \frac{1}{2}$

(14) $\lim_{n\to\infty} \frac{1+4+9+\cdots+n^2}{n^3} = \frac{n(n+1)(2n+1)}{6n^3} = \frac{2n^2+3n+1}{6n^2} = \frac{\infty}{\infty} = \frac{1}{3}$

(15) $\lim_{n\to\infty} \frac{n[1^3+2^3+\cdots+n^3]^2}{(1^2+2^2+\cdots+n^2)^3} = \frac{n\left[\frac{n^2(n+1)^2}{4}\right]^2 \times 6^3}{\left[\frac{n(n+1)(2n+1)}{6}\right]^3}$

$= \frac{6^3}{4^2}\cdot\frac{n^3(n+1)^2}{(2n+1)^3} = \frac{6^3}{4^2\times 8} = \frac{27}{16}$

Trick for partial fraction for two terms. दो भाग - दोनों भाई

(16) $\lim_{n\to\infty} \left[\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\cdots+\frac{1}{n(n+1)}\right] = \sum \frac{1}{n(n+1)} = \sum \frac{1}{n} - \frac{1}{n+1}$

$= 1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \cdots - \frac{1}{n+1} = 1$

(17) $\lim_{x\to 0} \frac{1-\cos 2x}{1-\cos 3x} = \frac{+2\sin 2x}{3\sin 3x} = \frac{4\cos 2x}{9\cos 3x} = \frac{4}{9}$

(18) $\lim_{x\to 0} \frac{\cos 2x - \cos 3x}{x^2} = \frac{-2\sin 2x + 3\sin 3x}{2x}$

$= \frac{-4\cos 2x + 9\cos 3x}{2} = \frac{5}{2}$

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Continuity of function :-

A function f is said to be continuous at $x = a$, if $\lim_{x \to a} f(x) = f(a)$, otherwise it is said to be a discontinuous function.

* Types of Discontinuous fn :-

i) Discontinuity of 1st Type (Jumped Discontinuity)

A fn. f is said to have discontinuity of 1st type if

$$\boxed{\lim_{x \to a^-} f(x) \neq \lim_{x \to a^+} f(x)}$$

ii) Discontinuity of 2nd Type

A fn. f is said to have discontinuity of 2nd Type if either the left limit or the right limit are both does not exist

iii) Discontinuity of 3rd Type (Removal Discontinuity)

A fn 'f' is said to have discontinuity of 3rd type

$$\boxed{\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) \neq f(a)}$$

* ~~Certain Standard functions~~ :-

* Standard Continuous function :-

① x^n is a continuous fn $\forall x$ when $n > 0$

$(x, x^2, x^3, x^{2/3}, x^{3/2}, \ldots)$

② x^n is a continuous fn $\forall x$ except $x = 0$ when $n < 0$

$(x^{-2} = \frac{1}{x^2}, x^{-3} = \frac{1}{x^3}, \ldots)$

③ $|x|$ is a continuous fn $\forall x$.

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iv) $\log x$ is a continuous fn $\forall x > 0$.

v) Every exponential function (e^x or a^x) is continuous.

vi) $\sin x$ and $\cos x$ are continuous fn $\forall x$

vii) $\tan x$ and $\sec x$ are continuous fn $\forall x$ except $x = (2n+1)\frac{\pi}{2}$

viii) $\cot x$, $\operatorname{cosec} x$ are continuous fn $\forall x$ except $x = n\pi$

Q. $f(x) = \dfrac{1}{1+2^{1/x}}$ at $x = 0$

$$\lim_{h\to 0^-} f(0-h) = \frac{1}{1+2^{\frac{1}{0-h}}} = 1$$

$$\lim_{h\to 0^+} f(0+h) = \frac{1}{1+2^{\frac{1}{0+h}}} = 0.$$

$\because$ L.H limit $\neq$ R.H limit

$\therefore$ discontinuous at $x = 0$.

Q $f(x) = \dfrac{x\left(e^{1/x}-1\right)}{e^{\frac{1}{x}}+1}$, $x \neq 0$

$= 0$, $x = 0$.

$$\lim_{h\to 0^-} f(0-h) = \frac{-h\left(e^{\frac{1}{-h}}-1\right)}{e^{\frac{1}{-h}}+1}$$

$$= \frac{-h \times -1}{1} = 0$$

$$\lim_{h\to 0^+} f(0+h) = \frac{h\left(e^{\frac{1}{h}}-1\right)}{e^{\frac{1}{h}}+1} = \frac{h\left(1-\frac{1}{e^{1/h}}\right)}{\left(1+\frac{1}{e^{1/h}}\right)} = h = 0$$

$f(0) = 0$

$\therefore$ LH limit = RH limit = $f(0)$ $\therefore$ Continuous at $x = 0$

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Q. A fn $f(x)$ is defined by —

$$f(x) = \begin{cases} 0, & x \le 0 \\ \frac{1}{2} - x, & 0 < x < \frac{1}{2} \\ \frac{1}{2}, & x = \frac{1}{2} \\ \frac{3}{2} - x, & \frac{1}{2} < x < 1 \\ 1, & x \ge 1 \end{cases}$$

Which of the following is true —

(a) f is continuous at $x = 0$ ✗

(b) f is discontinuous at $x = \frac{1}{2}$ ✓

(c) Continuous at $x = 1$ ✗

(d) All are true.

Q.

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25.9.14:

Q. A fn $f(x)$ is given by —

$$f(x) = \begin{cases} 0, & x \le 0 \\ 5x-4, & 0 < x \le 1 \\ 4x^2-3x, & 1 \le x \le 2 \\ 3x+4, & x > 2 \end{cases}$$

Check continuity @ $x = 0, 1, 2$ —

→ At $x=0$ → discontinuous

At $x=1$ → continuous.

At $x=2$ → continuous.

DIFFERENTIABILITY

— A fn f is said to be differentiable at $x=a$ if both ~~limit~~

$$L.H.D = \lim_{h \to 0} \frac{f(a-h) - f(a)}{-h}$$

$$R.H.D = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

exists and are equal. They are equal to $f'(a)$.

$d(x^n) = n x^{n-1}$

$d\sqrt{x} = \frac{1}{2\sqrt{x}}$

$d\left(\frac{1}{\sqrt{x}}\right) = \frac{-1}{2x\sqrt{x}}$

$d\left(\frac{1}{x}\right) = \frac{-1}{x^2}$

$d\left(\frac{1}{x^2}\right) = \frac{-2}{x^3}$

$d(e^{ax}) = ae^{ax}$

$d(a^x) = a^x \log a$

$d(\sin^{-1}x) = \frac{+1}{\sqrt{1-x^2}}$

$d(\sinh^{-1}x) = \frac{1}{\sqrt{1+x^2}}$

$d(\tan^{-1}x) = \frac{1}{1+x^2}$

*** $\frac{d}{dx}(x^x) = x^x(1+\log x)$

*** $\frac{d}{dx}(x^{\frac{1}{x}}) = x^{\frac{1}{x}} \cdot \frac{1}{x^2}(1-\log x)$

$\frac{d}{dx}(u \pm v) = \frac{du}{dx} \pm \frac{dv}{dx}$

$\frac{d}{dx}(u \cdot v) = u\frac{dv}{dx} + v\frac{du}{dx}$

$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$

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If $y = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$

$= \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$

$= \tan^{-1}\left(\frac{2x}{1-x^2}\right)$

then $\frac{dy}{dx} = \frac{2}{1+x^2}$

Q. If $y = x^{x^{x\cdots}}$ then $\frac{dy}{dx} = ?$

$y = x^y$

$\log y = y \log x.$

$\frac{dy}{y} = y\left(\frac{1}{x}dx\right) + \log x\, dy$

$dy\left(\frac{1}{y} - \log x\right) = \frac{y}{x}dx$

$\Rightarrow \frac{dy}{dx} = \frac{(y/x)}{(1/y - \log x)} = \frac{y^2}{x - xy\log x}$

Q. $x^y = e^{(x-y)}$ in terms of x $\frac{dy}{dx} = ?$

2 marks

$y \log x = x - y$

$y(\log x + 1) = x$

$dy = \frac{(\log x + 1) - x\left(\frac{1}{x}\right)}{(\log x + 1)^2} dx.$

$\frac{dy}{dx} = \frac{\log x}{(\log x + 1)^2}$

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Q. If $x^m \cdot y^n = a^{m+n}$ then $\frac{dy}{dx}$

$$m \log x + n \log y = (m+n) \log a$$

$$\frac{m}{x} dx + \frac{n}{y} dy = 0$$

$$\frac{dx}{dy} = x$$

$$\Rightarrow \frac{dy}{dx} = -\frac{m}{n} \cdot \frac{y}{x}$$

Q. $x^m \cdot y^n = (x+y)^{m+n} \Rightarrow \frac{dy}{dx} = ?$

$$m \log x + n \log y = (m+n) \log(x+y)$$

$$\frac{m}{x} dx + \frac{n}{y} dy = \frac{m+n}{x+y} \cdot (dx + dy)$$

$$\Rightarrow \left(\frac{m}{x} - \frac{m+n}{x+y}\right) dx = \left(\frac{m+n}{x+y} - \frac{n}{y}\right) dy$$

$$\Rightarrow \boxed{\frac{dy}{dx} = \frac{y}{x}}$$

Imp Result.

Q. If $y = \sqrt{x + \sqrt{x + \sqrt{x + \ldots}}}$ then $\frac{dy}{dx} = ?$

$$y = \sqrt{x+y}$$

$$\Rightarrow y^2 = x+y$$

$$\Rightarrow 2y\, dy = dx + dy$$

$$\Rightarrow dy(2y-1) = dx$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{2y-1}$$

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Imp Result

If $y = \sqrt{f(x) + \sqrt{f(x) + \sqrt{f(x) + \dots}}}$ then $\frac{dy}{dx} = \frac{f'(x)}{2y-1}$

Q. $y = \sqrt{\tan x + \sqrt{\tan x + \sqrt{\tan x + \dots}}}$ then $\frac{dy}{dx} = ?$ $\frac{\sec^2 x}{2y-1}$

Q. $y = \sqrt{\log x + \sqrt{\log x + \sqrt{\log x + \dots}}}$ then $\frac{dy}{dx} = ?$ $\frac{1/x}{2y-1}$

Q. $y = \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \dots}}}$ then $\frac{dy}{dx} = ?$ $\frac{\cos x}{2y-1}$

Q. $y = \frac{\sin^{-1} x}{\sqrt{1-x^2}}$ then $(1-x^2)y_1 - xy = $ ____

(a) 0 (b) 1 (c) y (d) $-y$.

To save time → uv form is easy than $\frac{u}{v}$

$(y\sqrt{1-x^2})^2 = (\sin^{-1} x)^2$

$\Rightarrow y^2(1-x^2) = (\sin^{-1} x)^2$

~~$\Rightarrow 2y(dy)x$~~

$\Rightarrow (1-x^2)\,2y\,dy + y^2(-2x\,dx) = 2\sin^{-1}x \cdot \frac{1}{\sqrt{1-x^2}}\,dx$

$\Rightarrow (1-x^2)\,2y\,dy = dx\left(\frac{2\sin^{-1}x}{\sqrt{1-x^2}} + 2y^2 x\right)$

$\Rightarrow 2y\left[(1-x^2)y_1 - xy\right] = 2y$

$\Rightarrow (1-x^2)y_1 - xy = 1.$

Q. $y = a\cos(\log x) + b\sin(\log x)$ then $x^2 y_2 + xy_1 =$ ____

$y_1 = \frac{-a\sin(\log x) + b\cos(\log x)}{x}$

$xy_1 = -a\sin(\log x) + b\cos(\log x)$..

$xy_2 + y_1 \cdot 1 = \frac{-a\cos(\log x) - b\sin(\log x)}{x}$

$x^2 y_2 + xy_1 = y.$

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Q. If $y = \tan^{-1}\left[\frac{(3-x)\sqrt{x}}{1-3x}\right] \Rightarrow \left.\frac{dy}{dx}\right|_{x=1} = ?$

General differentiation consumes time ∴ replacement substitute.

Let $x = \tan^2\theta \Rightarrow \theta = \tan^{-1}\sqrt{x}$.

$\therefore y = \tan^{-1}\left(\frac{(3-\tan^2\theta)\tan\theta}{1-3\tan^2\theta}\right)$

$= \tan^{-1}\left(\frac{3-\tan^3\theta}{1-3\tan^2\theta}\right)$

$= \tan^{-1}(\tan 3\theta)$

$= 3\theta$

$y = 3\tan^{-1}\sqrt{x}$.

$\Rightarrow \frac{dy}{dx} = 3 \cdot \frac{1}{1+(\sqrt{x})^2} \times \frac{1}{2\sqrt{x}}$

$\Rightarrow \left.\frac{dy}{dx}\right|_{x=1} = \frac{3}{1+1} \times \frac{1}{2} = \frac{3}{4}$ Ans.

Q. $y = \tan^{-1}\left(\frac{\cos x}{1+\sin x}\right) \Rightarrow \frac{dy}{dx} = ?$

1 + sin x we don't have formula. But 1 + cos x we have formula. That's why this approach

$= \tan^{-1}\left(\frac{\sin(90-x)}{1+\cos(90-x)}\right)$

$= \tan^{-1}\left(\frac{\sin(90-x)}{2\cos^2(45-\frac{x}{2})}\right)$

$= \tan^{-1}\left(\frac{2\sin(45-\frac{x}{2})\cdot\cos(45-\frac{x}{2})}{2\cos^2(45-x/2)}\right)$

$= 45 - \frac{x}{2}$.

$y = \frac{\pi}{4} - \frac{x}{2}$.

$\Rightarrow \frac{dy}{dx} = -\frac{1}{2}$ Ans

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Q. If $x = a\cos^3\theta$, $y = a\sin^3\theta$ $\Rightarrow \frac{dy}{dx} = ?$

$dx = -3a\cos^2\theta \times \cos\theta \times \sin\theta$

$dy = 3a\sin^2\theta \times \cos\theta \times \sin\theta$

$\Rightarrow \frac{dy}{dx} = \frac{-\sin^2\theta}{\cos^2\theta} = -\tan^2\theta$

$$\frac{dy}{dx} = \frac{3a\sin^2\theta \times \cos\theta}{-3a\cos^2\theta \times \sin\theta} = -\tan\theta.$$

Q. $x = 3\cos\theta - \cos^3\theta$

$y = 3\sin\theta - \sin^3\theta$

$dy = 3\cos\theta - 3\sin^2\theta\cos\theta = 3\cos\theta(1-\sin^2\theta)$

$dx = -3\sin\theta + 3\cos^2\theta \cdot \sin\theta = -3\sin\theta(1-\cos^2\theta)$

$$\therefore \frac{dy}{dx} = \frac{3\cos\theta \cdot \cos^2\theta}{-3\sin\theta \cdot \sin^2\theta} = -\cot^3\theta \quad \text{Ans}$$

Q. $x = a(\theta + \sin\theta)$

$y = a(1-\cos\theta)$

$$\frac{dy}{dx} = \frac{a(0+\sin\theta)}{a(1+\cos\theta)} = \frac{2\sin\frac{\theta}{2}\cdot\cos\frac{\theta}{2}}{2\cos^2\theta/2} = \tan\frac{\theta}{2} \quad \text{Ans}$$

Q. $x = a[\theta\sin\theta + \cos\theta]$

$y = a[\sin\theta - \theta\cos\theta]$

$$\frac{dy}{dx} = \tan\theta \quad \text{Ans}$$

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MEAN VALUE THEOREM :-

(1). Rolle's mean value theorem :-

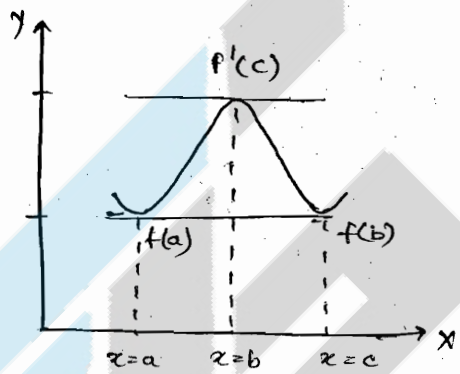
Let $f(x)$ be a function defined such that —

i) $f(x)$ is continuous in $[a, b]$

ii) $f(x)$ is differentiable in (a, b)

iii) $f(a) = f(b)$

then $\exists$ at least one value $c \in (a, b)$ such that $f'(c) = 0$



Proof :-

$f'(c)$ = slope of tangent line.

= slope of $\overline{AB}$ {$\because$ two parallel line have equal slopes}

$= \frac{y_2 - y_1}{x_2 - x_1}$

$= \frac{f(b) - f(a)}{b - a}$

$= \frac{0}{b-a}$ {$\because f(a) = f(b)$}.

$$\boxed{f'(c) = 0}$$

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Q. find c of the Rolle's mean value theorem for -

$f(x) = x^3 - 4x$ in $[-2, 2]$. (a) $\frac{2}{\sqrt{3}}$ (b) $-\frac{2}{\sqrt{3}}$ (c) $\pm\frac{2}{\sqrt{3}}$ (d) none

→ $f(-2) = -8 + 8 = 0$

$f(2) = 8 - 8 = 0$

$\therefore f'(x) = 3x^2 - 4$

$f'(c) = 3c^2 - 4 = 0$

$\Rightarrow c^2 = \pm\sqrt{\frac{4}{3}} = \pm\frac{2}{\sqrt{3}}$ Ans.

But Problem is not over.

Cross-verify the points in case of $\pm\sqrt{}$

Q.

→ are points inside interval or not ?

Here $\pm\frac{2}{\sqrt{3}}$ both are inside so ans is $\pm\frac{2}{\sqrt{3}}$ Ans.

Q. $f(x) = \frac{\sin x}{e^x}$ in $[0, \pi]$.

$f(0) = \frac{0}{1} = 0$

$f(\pi) = \frac{0}{e^\pi} = 0$

$\therefore f'(x) = \frac{e^x \cos x - \sin x\, e^x}{e^{2x}}$

$f'(c) = e^c \cos c - \sin c \cdot e^c = 0$

$\Rightarrow e^c \cos c = e^c \sin c$

$\Rightarrow \tan c = 1$

$\therefore c = \frac{\pi}{4}$ Ans

$c \in [0, \pi]$ must satisfy this

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Q. $f(x) = \dfrac{x^2+ab}{2(a+b)}$, $[a,b]$, $a>0$, $b>0$.

(a) $-\sqrt{ab}$ (b) $\sqrt{ab}$ (c) $\pm\sqrt{ab}$ (d) none.

$$f'(x) = x(a+b)[2x] - [x^2+ab][a+b] = 0$$

$$\Rightarrow 2x^2(a+b) = (x^2+ab)(a+b)$$

$$\Rightarrow x^2 = ab$$

$$\Rightarrow x = \pm\sqrt{ab}$$

But $x = -\sqrt{ab}$ should not $\in [a,b]$

When $a>0$, $b>0$

$$\therefore x = \sqrt{ab}$$

$$\Rightarrow \boxed{c = \sqrt{ab}}$$

Lagrange's mean value theorem :-

Let $f(x)$ be a function defined such that –

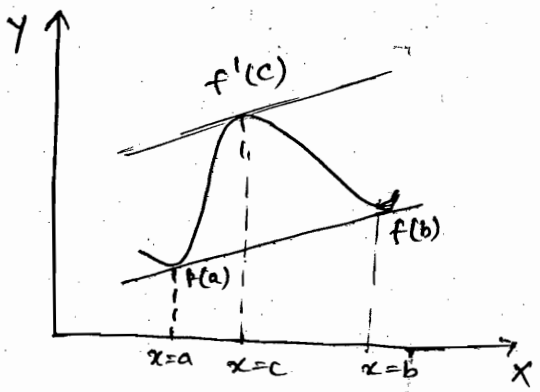
i) $f(x)$ is continuous in $[a,b]$

ii) $f(x)$ is differentiable in (a,b)

iii) $f(a) \neq f(b)$

Then $\exists$ at least one value $c \in (a,b)$ such that

$$\boxed{f'(c) = \frac{f(b)-f(a)}{b-a}}$$



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Q. $f(x) = \log x$ in $[1, e]$.

$f(e) = \log e$

$f(1) = \log 1 = 0.$

$\therefore f'(c) = \dfrac{\log e - \log 1}{e-1} = \dfrac{\log e}{e-1}$

$\Rightarrow \dfrac{1}{c} = \dfrac{\log e}{e-1}$

$\Rightarrow c = \dfrac{e-1}{\log e}$ Ans.

$\Rightarrow c = e-1 \in (1, e)$ $\{\because e \cong 2.71\}$.

Q. $f(x) = x^3 - 6x^2 + 11x - 6$ in $[0, 4]$.

$3x^2 - 12x + 11 = \dfrac{4^3 - 6(4)^2 + 11(4) - 6 - (-6)}{4 - 0}$

$\Rightarrow 3x^2 - 12x + 11 = \dfrac{64 - 96 + 44}{4} = \dfrac{108 - 96}{4} = \dfrac{12}{4} = 3$

$\Rightarrow 3x^2 - 12x + 9 = 0$

$\Rightarrow 3c^2 - 12c + 9 = 0$

$\Rightarrow c = \dfrac{+12 \pm \sqrt{144 - 96}}{6}$

$= \dfrac{12 \pm \sqrt{48}}{6}$

$= 2 \pm \dfrac{4\sqrt{3}}{3}$

$= 2 \pm \dfrac{2}{\sqrt{3}} \in (0, 4)$

$\therefore c = 2 \pm \dfrac{2}{\sqrt{3}}$ Ans

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CAUCHY'S MEAN VALUE THEOREM :-

Let $f(x)$ and $g(x)$ be two functions defined such that

i) f and g are continuous in the closed interval $[a,b]$

ii) f and g are differentiable in (a,b)

iii) $g'(x) \neq 0 \quad \forall x \in (a,b)$ then there exists at least one value $c \in (a,b)$ such that —

$$\boxed{\frac{f'(c)}{g'(c)} = \frac{f(b)-f(a)}{g(b)-g(a)}}$$

Q. $f(x) = \sqrt{x}$, $g(x) = \frac{1}{\sqrt{x}}$ in $[a,b]$

(A) $\frac{ab}{2}$ (B) $\frac{a+b}{2}$ (C) $\sqrt{ab}$ (D) $\frac{2ab}{a+b}$

$$\frac{\frac{1}{2}x^{-\frac{1}{2}}}{-\frac{1}{2}x^{-\frac{3}{2}}} = \frac{\sqrt{b}-\sqrt{a}}{\frac{1}{\sqrt{b}}-\frac{1}{\sqrt{a}}}$$

$$\Rightarrow -x^{-\frac{1}{2}+\frac{3}{2}} = \frac{\sqrt{b}-\sqrt{a}}{\frac{\sqrt{a}-\sqrt{b}}{\sqrt{ab}}}$$

$$\Rightarrow -c = -\sqrt{ab}$$

$$\therefore \boxed{c = \sqrt{ab}} \quad \text{(A) or}$$

Which $\in (a,b)$

Q. $f(x) = e^x$, $g(x) = e^{-x}$ in $[a,b]$.

$$\frac{e^c}{-e^{-c}} = \frac{e^b - a^a}{e^{-b} - e^{-a}}$$

$$\Rightarrow -e^{2c} = \frac{e^b - e^a}{\frac{e^a - e^b}{e^{b+a}}} \Rightarrow e^{2c} = e^{b+a}$$

$$\Rightarrow c = \frac{b+a}{2} \quad \text{(D)}$$

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Q. $f(x) = \frac{1}{x^2}$ $g(x) = \frac{1}{x}$ in $[a, b]$.

$$\rightarrow \frac{-2c^{-3}}{-c^{-2}} = \frac{\frac{1}{b^2} - \frac{1}{a^2}}{\frac{1}{b} - \frac{1}{a}}$$

$$\Rightarrow 2c^{-1} = b^{a+b} \frac{a^2 - b^2}{a+b} \times \frac{ab}{a^2 b^2}$$

$$\Rightarrow c = \frac{2ab}{a+b}.$$

Q.. $f(x) = \sin x$, $g(x) = \cos x$ in $[0, \frac{\pi}{2}]$

$$\Rightarrow \frac{\cos x}{-\sin x} = \frac{1-0}{0-1}$$

$$\Rightarrow -\cot x = -1$$

$$\Rightarrow \tan x = 1$$

$$\therefore c = \frac{\pi}{4}.$$

#

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Taylor's mean value Theorem :-

Let $f(x)$ be a function defined such that —

i) $f(x)$ is continuous in $[a, a+h]$

ii) $f^{n-1}(x)$ is differentiable in $[a, a+h]$

then there exists at least one value $\theta \in (0,1)$ such that —

$$f(a+h) = f(a) + h f'(a) + \frac{h^2}{2!} f''(a) + \frac{h^3}{3!} f'''(a) + \cdots$$

$$\cdots + \frac{h^{n-1}}{n-1!} f^{n-1}(a) + \frac{h^n}{n!} f^n(a)$$

Let $a+h = x \Rightarrow h = x-a$.

*** $\therefore$ $$f(x) = f(a) + (x-a) f'(a) + \frac{(x-a)^2}{2!} f''(a) + \cdots$$

Maclerin's formula :- $a=0$

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \cdots$$

① $f(x) = \sin x$ about $x = 0$

$f(0) = 0$

$f'(0) = \cos 0 = 1$

$f''(0) = -\sin 0 = 0$

$f'''(0) = -\cos 0 = -1$

*** $\therefore$ $$f(x) = \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$$

Imp Result

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$$

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Q. $f(x) = e^x$ about $x = 0$

$f(0) = 1$

$f'(0) = e^0 = 1$

$f''(0) = e^0 = 1$

$f'''(0) = e^0 = 1$

Imp Result ∴ $\boxed{f(x) = e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots\cdots}$

Q. $f(x) = \log(1+x)$ about $x = 0$.

$f(0) = 0$

$f'(0) = \frac{1}{1+x} = 1$

$f''(0) = -(1+x)^{-2} = -1$

$f'''(0) = +2(1+x)^{-3} = 2$

$f''''(0) = -6(1+x)^{-4} = -6$

$\therefore f(x) = \log(1+x) = x - \frac{x^2}{2!} + \frac{x^3}{3!}2 + \frac{x^4}{4!}(-6)$

Imp Result ∴ $\boxed{\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots}$

Q. $f(x) = \frac{x}{1+x}$ about $x = 0$.

$f(x) = \frac{x}{1+x} = 1 - \frac{1}{1+x} = 0$

$f'(x) = \frac{1}{(1+x)^2} = 1$

$f''(x) = \frac{-2}{(1+x)^3} = -2$

$f'''(x) = \frac{6}{(1+x)^4} = 6$

$x+1 \overline{)\, x \,(\, 1 - \frac{1}{x+1}}$

$\frac{x+1}{-1}$

$\frac{-1}{x}$

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$$\therefore \quad \frac{x}{1+x} = 0 + x(1) + \frac{x^2}{2!}(-2) + \frac{x^3}{3!}(6) + \cdots\cdots$$

Imp Result

$$\boxed{\frac{x}{1+x} = x - x^2 + x^3 - x^4 + \cdots\cdots}$$

Alternate Method

$$f(x) = \frac{x}{1+x}$$

$$= x(1+x)^{-1}$$

$$= x(1 - x + x^2 - x^3 + \cdots) \quad \{\text{Binomial Expansion}\}.$$

$$= x - x^2 + x^3 - x^4 + \cdots$$

Q. find the coeff. of x^2 in the expansion of $f(x) = \cos^2 x$ about $x = 0$.

→ $f(x) = \cos^2 x$

$= \cos x \cdot \cos x$

$= \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots\right)\left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots\right)$

$= -\frac{x^2}{2!} - \frac{x^2}{2!}$

$= -\frac{2x^2}{2!}$

$= -x^2 \cdot = -1$

Alternate method.

∵ coeff. of x^2 asked.

↳ $= \frac{f''(0)}{2!}$

$\therefore f(x) = \cos^2 x =$

$f'(x) = 2\cos x(-\sin x) = -\sin 2x$

$f''(x) = -2\cos 2x$

$\therefore \frac{f''(0)}{2!} = \frac{-2}{2!} = -1$

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Q. find the coeff of $(x-2)^4$ in the expansion of e^x about $x=2$

→ coeff of $(x-2)^4 = \frac{f^{iv}(a)}{4!}$

$f(x) = e^x$

$f^{iv}(x) = e^{x^2} = e^2$

$\therefore \frac{f^{iv}(a)}{4!} = \frac{e^2}{4!}$ Ans

Q. find expansion of x^2 about $x=1$

(a) $1-x+x^2$

(b) $1+x^2$

(c) x^2 { $\because f(x) = x^2$ is pure algebric fn
$\therefore$ expansion not possible }

(d) $1+x+\frac{x^2}{2}+\frac{x^3}{3}$

Q. $f(x) = \log\left(\frac{1+x}{1-x}\right)$ about $x=0$.

$f(x) = \log(1+x) - \log(1-x)$

$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} - \cdots$

$\log(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \cdots$

$+ \quad + \quad + \quad +$

$\log\left(\frac{1+x}{1-x}\right) = 2\left(x + \frac{x^3}{3} + \frac{x^5}{5} + \cdots\right)$

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PARTIAL AND TOTAL DERIVATIVES :-

O·D	P·D
$y = f(x)$ $x = g(y)$	$z = f(x,y)$ $u = f(x,y,z)$

Q. $z = x^2 - xy + y^2$

$$\frac{\partial z}{\partial x} = 2x - y \qquad \frac{\partial z}{\partial y} = -x + 2y$$

$$\frac{\partial^2 z}{\partial y\,\partial x} = -1 \qquad \frac{\partial^2 z}{\partial x \cdot \partial y} = -1$$

$$\boxed{\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}}$$

* HOMOGENEOUS FUNCTION :-

A function $f(x,y)$ is said to be homogeneous of degree n in x and y if —

$$\boxed{f(kx, ky) = k^n f(x,y)}$$

or else check overall degree of each term must be same.

ex:- $f(x,y) = x^2 - x^{1+1}y^1 + y^2$ → Homogeneous (Directly by degree)

$f(kx, ky) = k^2x^2 - k^2xy + k^2y^2$

$= k^2(x^2 - xy + y^2)$

Note :-

The product of two homogeneous fn is again a homogeneous function.

ex:- $f(x,y) = (x^3 + y^3)(x^2 - y^2)$

$n = 3 + 2 = 5$

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If a fn is in rational form, if both numerator and Denominator are homogeneous function, therefore the given function is also homogeneous.

ex:- $f(x,y) = \dfrac{x^3+y^3}{x^2-y^2}$

$n = 3-2 = 1$

$f(x,y) = \dfrac{x^{1/4}+y^{1/4}}{x^{1/5}-y^{1/5}}$

$n = \frac{1}{4} - \frac{1}{5} = \frac{1}{20}$

Euler's Theorem :-

If z is a homogeneous fn of degree n in x and y then $x\dfrac{\partial z}{\partial x} + y\dfrac{\partial z}{\partial y} = n \cdot z$

Note:-

This formula is applicable directly if z is an algebric fn

ex:- Q. $z = (x^2+y^2)^{1/3}$ then $x\dfrac{\partial z}{\partial x} + y\dfrac{\partial z}{\partial y} = \dfrac{2}{3}z$

Q. $z = \dfrac{xy}{x+y} \Rightarrow xz_x + yz_y = 1 \cdot z$

Q. If $z = a_0x^n + a_1x^{n-1}y + a_2x^{n-2}y^2 + \cdots + a_ny^n$ then $x\dfrac{\partial z}{\partial x} + y\dfrac{\partial z}{\partial y} = nz$

Q. If $z = \log(x^2+y^2)$ then $x\dfrac{\partial z}{\partial x} + y\dfrac{\partial z}{\partial y} =$ ___

Shortcut

Note:-

If z is not algebric and let $\phi(z)$ is algebric and homogeneous of degree n in x and y then $\boxed{x\dfrac{\partial z}{\partial x} + y\dfrac{\partial z}{\partial y} = \dfrac{n\,\phi(z)}{\phi'(z)}}$

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So, given $z = \log(x^2+y^2)$

$\Rightarrow e^z = x^2+y^2 \rightarrow$ homogeneous of degree 2

$\therefore x\frac{\partial x}{\partial x} + y\frac{\partial y}{\partial y}$

$x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = 2\cdot\frac{e^z}{e^z} = 2$ Ans

Q. If $z = \sin^{-1}\left(\frac{x^2+y^2}{x-y}\right)$

$\Rightarrow x z_x + y z_y = 1\cdot\frac{\sin z}{\cos z} = \tan z$ Ans

Q. If $z = \tan^{-1}\left(\frac{x^3+y^3}{x-y}\right)$ then $x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = ?$

$\Rightarrow \tan z = \frac{x^3+y^3}{x-y}$ homogeneous of order 2.

$\therefore x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = 2\cdot\frac{\tan z}{\sec^2 z} = \sin 2z$ Ans

Q. If $z = x^2\sin^{-1}\left(\frac{y}{x}\right) - y^2\tan^{-1}\left(\frac{x}{y}\right)$

then $x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = 2z$

{ ∵ at $\frac{y}{x}\neq 0$ $y=0$ & $x=0$ inverse fn not exists so it becomes $z = x^2-y^2$ }

Q. If $z = (x^3+y^3)e^{-x/y} \Rightarrow x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial x} = 3z$

{ ∵ when $x=y=0$ the $e^{-x/y}$ does not exist. ∴ $z = x^3+y^3$ homogeneous of order 3 }

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Application of Euler's Theorem :-

If z is an homogeneous fn. of degree n in x and y then —

$$x^2\frac{\partial^2 z}{\partial x^2} + 2xy\frac{\partial^2 z}{\partial x \partial y} + \frac{y^2\partial^2 z}{\partial y^2} = n(n-1)z$$

Note :-

This formula is applicable directly if z is an algebraic fn.

Ex:

Q. If $z = (x^2+y^2)^{1/3}$ then $x^2\frac{\partial^2 z}{\partial x^2} + 2xy\frac{\partial^2 z}{\partial x\partial y} + y^2\frac{\partial^2 z}{\partial y^2} = \frac{2}{3}\left(\frac{2}{3}-1\right)z$

$= -\frac{2}{9}z$

Q If $z = \frac{x^3y^3}{x^2+y^2}$ then $x^2\frac{\partial^2 z}{\partial x^2} + 2xy\frac{\partial^2 z}{\partial x\partial y} + y^2\frac{\partial^2 z}{\partial y^2} = 4(4-1)z$

$= 12z$

↳ $6-2=4$

Q. If $z = \tan^{-1}\left(\frac{x^3+y^3}{x-y}\right)$ then $x^2\frac{\partial^2 z}{\partial x^2} + 2xy\frac{\partial^2 z}{\partial x\partial y} + y^2\frac{\partial^2 z}{\partial y^2} =$ ——

Shortcut

Note :-

If z is not algebric then

$$x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = \frac{n\phi(z)}{\phi'(z)} = F(z)$$

and $$x^2\frac{\partial^2 z}{\partial x^2} + 2xy\frac{\partial^2 z}{\partial x\partial y} + y^2\frac{\partial^2 z}{\partial y^2} = F(z)\left[F'(z)-1\right].$$

$\therefore F(z) = 2\cdot\frac{\tan z}{\sec^2 z} = \sin 2z$

$\therefore F(z)\left[F'(z)-1\right] = \sin 2z\left[2\cos 2z - 1\right]$

$= \sin 4z - \sin 2z$ Ans

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Q. If $\log\left(\frac{x^4+y^4}{x+y}\right)$ then $x^2\frac{\partial^2 z}{\partial x^2} + 2xy\frac{\partial^2 z}{\partial x \partial y} + y^2\frac{\partial^2 z}{\partial y^2} = ?$

$\rightarrow$ $F(z) = 3 \cdot \frac{\theta^2}{\theta^2} = 3$

$F(z)[F'(z)-1] = 3[0-1] = \boxed{-3}$ Ans

Q.

Explicit Function :-

Any function which we can express in the form of either $y = f(x)$ or $x = g(y)$ etc is called an explicit fn.

means y = purely fn of x
x = purely fn of y
$\Rightarrow$ Explicit fn.

ex:- $y = ax^2 + bx + c$

Ordinary Derivatives upto n^{th} order can be easily evaluated.

Implicit function :-

Any function which is not explicit is said to be an implicit function.

ex:- $y = ax^2 + bxy + c$

Ordinary Derivatives upto n^{th} order evaluation becomes complex.

$\therefore$ Partial Derivative is used.

Shortcut

$$\boxed{\frac{dy}{dx} = -\frac{\partial f/\partial x}{\partial f/\partial y}}$$

$$\boxed{\frac{d^2y}{dx^2} = -\frac{[q^2r - 2pqs + p^2t]}{q^3}}$$

$p = \frac{\partial z}{\partial x}$
$q = \frac{\partial z}{\partial y}$
$r = \frac{\partial^2 z}{\partial x^2}$
$s = \frac{\partial^2 z}{\partial x \partial y}$ & $t = \frac{\partial^2 z}{\partial y^2}$

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* If $ax^2 + 2hxy + by^2 = 1$.

Shortcut

$$\frac{d^2y}{dx^2} = \frac{h^2 - ab}{(hx + by)^3}$$

Ex:-

Q. $2x^2 + 4xy + 3y^2 = 1$

$a = 2$
$b = 3$
$h = 2$

$$\Rightarrow \frac{d^2y}{dx^2} = \frac{2^2 - 2\times 3}{(2x + 3y)^3} = \frac{-2}{(2x+3y)^3}$$ Ans

Q. $x^2 + xy + 2y^2 = 1$

$$\frac{d^2y}{dx^2} = \frac{\left(\frac{1}{2}\right)^2 - 2}{\left(\frac{1}{2}x + 2y\right)^3} = \frac{-14}{(x+4y)^3}$$ Ans

* If $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

Shortcut

$$\frac{d^2y}{dx^2} = \frac{abc + 2fgh - af^2 - bg^2 - ch^2}{(hx + by + f)^3}$$

ex:-

Q. $2x^2 + 4xy + 3y^2 + 2x + 4y + 1 = 0$

$a = 2$
$b = 3$
$c = 1$
$h = 2$
$g = 1$
$f = 2$

$$\therefore \frac{d^2y}{dx^2} = \frac{6 + 8 - 8 - 3 - 4}{(2x + 3y + 2)^3} = \frac{-1}{(2x+3y+2)^3}$$

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Q. $y^2 - 5x + 4x^2 = 8$

$a = 1$
$b = 1$
$c = -8$
$h = 0$
$g = -\frac{5}{2}$
$f = 0$.

$$\frac{d^2y}{dx^2} = \frac{-32 + \frac{25}{4}}{(1y)^3} = \frac{-153}{4y^3}$$

Alternate method (By Explicit)

$$y^2 = 8 + 5x - 4x^2$$

$$y = \sqrt{8 + 5x - 4x^2}$$

$\frac{d^2y}{dx^2}$ will become complex.

∴ Implicit shortcut is very Easy to use.

COMPOSITE FUNCTION

→ $(gof)x = g[f(x)]$.

→ Any function in function is said to be a composite fn.

or

→ If z is a fn in x and y and x, y are functions in t then z is called a composite fn in 't'.

$z = f(x, y)$ → t

* Total Derivative of a composite fn :-

If $z = f(x, y)$ and $x = g(t)$, $y = h(t)$ then the total derivative of z w.r. to 't' is denoted by —

$$\boxed{\frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt}}$$

→ Total Derivative means combination of ordinary + partial derivative.

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$$\boxed{\frac{du}{dt} = \frac{\partial u}{\partial x}\frac{dx}{dt} + \frac{\partial u}{\partial y}\frac{dy}{dt} + \frac{\partial u}{\partial z}\frac{dz}{dt}}$$

Q. $z = e^x \sin y$

$x = \log t$

$y = t^2$

$\frac{dz}{dt} = ?$

$\rightarrow$ $\frac{\partial z}{\partial x} = e^x \sin y$

$\frac{\partial z}{\partial y} = e^x \cos y$

$\frac{\partial x}{\partial t} = \frac{1}{t}$

$\frac{dy}{dt} = 2t$

$\therefore \frac{dz}{dt} = e^x \sin y\left(\frac{1}{t}\right) + e^x \cos y\,(2t)$

$= \frac{e^x}{t}\left[\sin y + 2t^2 \cos y\right]$.

Q. $u = x^2 + y^2 + z^2$

$x = e^{2t}$, $y = e^{2t}\cos 3t$, $z = e^{2t}\sin 3t$

then $\frac{du}{dt} = ?$

$\frac{dy}{dt}$ & $\frac{dz}{dt}$ will take time as in uv form.

Alternate Method $u = (e^{2t})^2 + (e^{2t}\cos 3t)^2 + (e^{2t}\sin 3t)^2$

$= e^{4t}\left[1 + \cos^2 3t + \sin^2 3t\right]$

$u = 2e^{4t}$

$\frac{du}{dt} = 8e^{4t}$ Ans

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Q. $u = x^2 - y^2$

$x = e^t \cos t$, $y = e^t \sin t$

$\left.\frac{du}{dt}\right|_{t=0} = ?$

$\rightarrow u \quad e^{2t}\cos^2 t - e^{2t}\sin^2 t$

$= e^{2t}(\cos^2 t - \sin^2 t)$

$u = e^{2t} \cdot \cos 2t$

$\Rightarrow \left.\frac{du}{dt}\right|_{t=0} = 2e^{2t}(-\sin 2t) + 2\cos 2t \cdot e^{2t}$

$= 2 \cdot 1 =$

$= 2$ Ans

Q. $u = x^3 y e^z$

$x = t$, $y = t^2$, $z = \log t$

$\left.\frac{du}{dt}\right|_{t=2} = ?$

$u = t^3 t^2 e^{\log t}$

$u = t^6$

$\left.\frac{du}{dt}\right|_{t=2} = 6t^5 = 6 \times 2^5 = 192$ Ans

Q. If $x = r\cos\theta$, $y = r\sin\theta$..

then ① $\frac{\partial^2\theta}{\partial x^2} + \frac{\partial^2\theta}{\partial y^2} =$ ——

② $\frac{\partial^2\theta}{\partial x \partial y} =$ ——

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$$\frac{r\sin\theta}{r\cos\theta} = \frac{y}{x}$$

$$\Rightarrow \tan\theta = \frac{y}{x}.$$

$$\theta = \tan^{-1}\frac{y}{x}.$$

$$\frac{\partial\theta}{\partial x} = \frac{1}{1+\frac{y^2}{x^2}} \times \frac{-y}{x^2}$$

$$\frac{\partial\theta}{\partial x} = \frac{-y}{x^2+y^2}$$

$$\frac{\partial^2\theta}{\partial x^2} = \frac{-y\times -1}{(x^2+y^2)^2} \times 2x.$$

$$= \frac{2xy}{(x^2+y^2)^2}$$

Similarly $\frac{\partial^2\theta}{\partial y^2} = \frac{-2xy}{(x^2+y^2)^2}$

$$\therefore \boxed{\frac{\partial^2\theta}{\partial x^2} + \frac{\partial^2\theta}{\partial y^2} = 0.}$$

$\because \frac{\partial\theta}{\partial x} = \frac{-y}{x^2+y^2}$

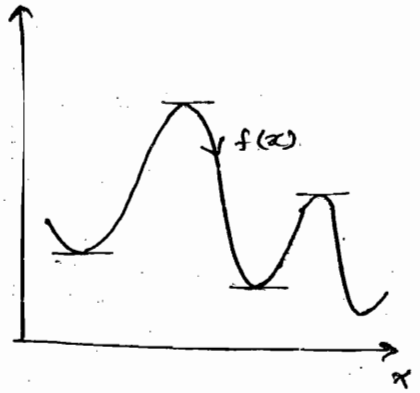
$$\frac{\partial\theta}{\partial x\partial y}$$

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Maxima and Minima.

Maximum Value :-

If a continuous fn $f(x)$ increases to a certain value and then decreases that value is called maximum value of the function.



Minimum Value :-

If a continuous fn $f(x)$ decreases to a certain value and then increases, that value is called minimum value of the fn.

Note :-

i) The maxima and minima occurs alternatively.

ii) A fn can have several maximum and several minimum values.

iii) The min^m value may be greater than the max^m value.

iv) The least min^m value is called the global minimum or universal minimum and the highest max^m value is called the global max^m or universal max^m.

→ For a maxima or minima —

$f'(x) = 0$

$x = \alpha, \beta, \lambda \ldots\ldots$

$x = \alpha$, $f''(\alpha) = 0$, f is minimum at $x = \alpha$. Min. value $= f(\alpha)$

$x = \beta$, $f''(\beta) < 0$, f is maximum at $x = \beta$. Max^m value $= f(\beta)$

$x = \lambda$, $f''(\lambda) = 0$, f is stationary.

if $f'''(\lambda) \neq 0$ then 'λ' is said to be inflation point.

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Q. $f(x) = x^x$

$f'(x) = x^x(1+\log x)$

$f''(x) = x^x\left(\frac{1}{x}\right) + (1+\log x)\,x^x(1+\log x)$

$f''(x) = x^{x-1} + x^x(1+\log x)^2$

$f'(x) = 0$

$\Rightarrow x^x(1+\log x) = 0$

↳ can't be never 0

$\therefore 1 + \log_e x = 0$

$\Rightarrow \log_e x = -1$

$\Rightarrow x = e^{-1} = \frac{1}{e}$.

$f''(x) = f''\left(\frac{1}{e}\right) = \left(\frac{1}{e}\right)^{\frac{1}{e}-1} > 0$ $\left\{\begin{array}{l} 2^2 = 4 > 0 \\ 2^{-2} = \frac{1}{4} > 0. \end{array}\right\}$

for $f(x) = x^x$, F is min^m at $x = \frac{1}{e}$

Imp Result

$\therefore$ Min^m value $= f\left(\frac{1}{e}\right) = \left(\frac{1}{e}\right)^{1/e} = e^{-\frac{1}{e}}$ Ans.

Q. $f(x) = x^{\frac{1}{x}}$ has its max^m at $x = e$.

Max^m value $= f(e) = e^{\frac{1}{e}}$.

Imp Result

****

fn	Min value	Max value.	At.
x^x	$e^{-\frac{1}{e}}$	—	$x = \frac{1}{e}$.
$x^{\frac{1}{x}}$	—	$e^{\frac{1}{e}}$	$x = e$.

f

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Q. $f(x) = \frac{\log x}{x} = \frac{1}{x}\log x$

$f'(x) = \frac{1}{x}\left(\frac{1}{x}\right) + \log x\left(-\frac{1}{x^2}\right)$

$f'(x) = \frac{1}{x^2}(1-\log x)$

$f''(x) = -\frac{2}{x^3}(1-\log x) + \frac{1}{x^2}\left(-\frac{1}{x}\right)$

$f'(x) = 0.$

$\Rightarrow \frac{1}{x^2}(1-\log x) = 0.$

$\Rightarrow \log x = 1$

$\therefore x = e.$

$f''(x) = f''(e) = 0 - \frac{1}{e^3} < 0$

$\therefore f$ is max^m at $x = e$.

Max^m value $= f(e) = \frac{\log e}{e} = \frac{1}{e}.$

Q. $f(x) = x^3 + \frac{3}{x}$

$f'(x) = 3x^2 - \frac{3}{x^2}$

$f''(x) = 6x + \frac{3 \times 2}{x^3} = 6x + \frac{6}{x^3}$

$3x^2 - \frac{3}{x^2} = 0$

$\Rightarrow x^2 = \frac{1}{x^2}$

$\Rightarrow x^4 - 1 = 0$

$\Rightarrow (x^2-1)(x^2+1) = 0$

$\Rightarrow x^2 - 1 = 0$

$\Rightarrow x = \pm 1$

$f''(1) = 6 + 6 = 12 > 0 \rightarrow$ min^m at $x = 1$ → Min. value = 4.

$f''(-1) = -6 + 6 = -12 < 0 \rightarrow$ max^m at $x = -1$

↳ Max^m value $= -4$

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Q. $f(x) = x^2 e^{-x}$

$f'(x) = e^{-x} \times (2x) + x^2(-e^{-x}) = 0$

$\Rightarrow 2xe^{-x} - x^2 e^{-x} = 0$

$\Rightarrow e^{-x}(2x - x^2) = 0$

$\Rightarrow e^{-x} \times x(2-x) = 0$

$x = 0, 2.$

$f''(x) = -2xe^{-x} + e^{-x} \times 2 - [e^{-x}(2x) + x^2(-e^{-x})]$

$= e^{-x}(2 - 4x + x^2)$

at $x = 0$ $f''(x) = 2 > 0 \rightarrow$ min^m at $x = 0$ (min value = 0)

at $x = 2$ $f''(x) = \frac{-2}{e^2} < 0 \rightarrow$ max^m at $x = 2$

Max value $= 4e^{-2}$

Q. $f(x) = x^1 e^{-x}$ has its max at $x = \underline{1}$.

Trick *** Remember in this type max^m occurs at degree.

Q.

Shortcut

$f(x) = a\cos x + b\sin x + c$

Min value $= c - \sqrt{a^2 + b^2}$

Max value $= c + \sqrt{a^2 + b^2}$

Q. $f(x) = 3\cos x + 4\sin x + 2.$

Min value $= 2 - \sqrt{3^2 + 4^2} = 2 - 5 = -3$

Max value $= 2 + \sqrt{3^2 + 4^2} = 2 + 5 = 7.$

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Q. $f(x) = 5\cos x + 3\cos\left(x+\frac{\pi}{3}\right) + 3$

$= 5\cos x + 3\cos x \cdot \cos\frac{\pi}{3} - 3\sin x \cdot \sin\frac{\pi}{3} + 3$

$= \cos x\left(5+\frac{3}{2}\right) - \frac{3\sqrt{3}}{2}\sin x + 3.$

$\therefore$ Min value

$= c - \sqrt{a^2+b^2}$

$= 3 - \sqrt{\left(\frac{13}{2}\right)^2 + \left(-\frac{3\sqrt{3}}{2}\right)^2}$

$= 3 - \sqrt{\frac{169}{4} + \frac{27}{4}}$

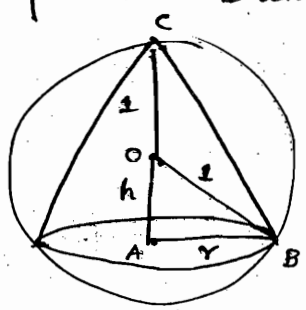
$= 3 - \sqrt{\frac{196}{4}}$

$= 3 - \frac{14}{2}$

$= -4.$

Max value $= 3+7 = 10$

Q. Find the height of the cone of maxm volume that can be inscribed in a sphere of radii 1 unit.



$r^2 = 1-h^2$

$\because$ volume of cone $= \frac{1}{3}\pi r^2 h$

$= \frac{1}{3}\pi(1-h^2)(1+h)$

$\therefore V = \frac{\pi}{3}(1+h-h^2-h^3)$

$\frac{dV}{dh} = \frac{\pi}{3}(1-2h-3h^2) = 0$

$\Rightarrow 1-2h-3h^2 = 0$

$\Rightarrow 3h^2+2h-1 = 0$

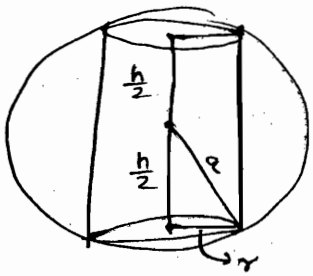
$h = \frac{-2\pm\sqrt{4+12}}{6} = \frac{-2\pm 4}{6} = \frac{-6}{6}, \frac{2}{6}$

$\therefore h = \frac{1}{3}$

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$\therefore$ height $= 1+h = 1+\frac{1}{3} = \frac{4}{3}$ ans.

Q. find the height of the cylinder of maximum volume that can be inscribed in a sphere of radii a units.



$\rightarrow$ $r^2 = a^2 - \frac{h^2}{4}$

volume of cylinder $= \pi r^2 h$

$V = \pi r^2 h$

$= \pi\left(a^2 - \frac{h^2}{4}\right) h$

$V = \pi\left(a^2 h - \frac{h^3}{4}\right)$

$\frac{dv}{dh} = \pi\left(a^2 - \frac{3h^2}{4}\right) = 0$

$\Rightarrow a^2 - \frac{3h^2}{4} = 0$

$\Rightarrow 3h^2 = 4a^2$

$\Rightarrow h^2 = \frac{4a^2}{3}$

$\boxed{h = \frac{2a}{\sqrt{3}}}$ $\rightarrow$ height for which max^m volume of cylinder occurs in sphere.

Imp Result.

If $z = f(x,y)$

$p = \frac{\partial z}{\partial x}$

$q = \frac{\partial z}{\partial y}$

$r = \frac{\partial^2 z}{\partial x^2}$

$s = \frac{\partial^2 z}{\partial x \partial y}$

$t = \frac{\partial^2 z}{\partial y^2}$

$\rightarrow$ for a maxima and minima, Let p and $q = 0$ (since 1st order derivative are only p & q)

$\rightarrow$ Since, p and q are fn. in var. x and y so, we have two eqn in x & y.

$\rightarrow$ Solve these eqns and get a relation bet^n x and y.

Ex:- $x = y$, $x = -y$, $y = 2x$, $x = 2y$ etc type

$\rightarrow$ Put this relation either in $p = 0$ or in $q = 0$; then the eqn transformed to either purely in x or in y

$\rightarrow$ Solve the Eqn and get the roots Ex $x = x_0, x_1, x_2 \ldots$ and $y = y_0, y_1, y_2 \ldots$ etc (x_0, y_0) $(x_1, y_1) \ldots$ are critical points.

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→ Take a point x_0, y_0, find the values of r, s and t

* **Case-1** → If $rt - s^2 > 0$ and $r > 0$, then the fn attains its minimum at that point.
  → Minm value = $f(x_0, y_0)$.

* **Case-2** → If $rt - s^2 > 0$ and $r < 0$, then the fn attains its maximum at that point.
  → Maxm value = $f(x_1, y_1)$

* **Case-3** → If $rt - s^2 < 0$ then the fn has no extremum.

Case-4 → If $rt - s^2 = 0$ then the case is doubtful and needs further investigation.

Q. $f(x,y) = x^2 + y^2 + 6x + 12$ has

a) minm at $(-3, 0)$
b) maxm at $(-3, 0)$
c) no extremum
d) none.

→
$$p = 2x + 6$$
$$q = \frac{\partial f}{\partial y} = 2y$$
$$r = \frac{\partial^2 f}{\partial x^2} = 2$$
$$s = \frac{\partial^2 f}{\partial x \partial y} = 0$$
$$t = \frac{\partial^2 f}{\partial y^2} = 2.$$

$rt - s^2 = 4 - 0^2 > 0$ and $r > 0$.
minimum

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Q. $f = 1 - x^2 - y^2$ has.

(a) min at (0,0)
(b) max at (0,0)
(c) no extremum
(d) none.

→ $p = -2x$
$q = -2y$
$r = -2$
$s = 0$
$t = -2.$

$rt - s^2 = 4 > 0$ and $-2 < 0.$

maximum.

Q. $f(x,y) = x^3 + y^3 - 3xy.$

(a) max at (1,-1)
(b) min at (1,1)
(c) max at (1,-1)
(d) min at (1,1)

→ $p = 3x^2 - 3y$
$q = 3y^2 - 3x$
$r = 6x$
$s = -3$
$t = 6y$

at 1,-1
$6 \times (-6) - 9 < 0$ and $r = 6 > 0.$

~~saddle~~ no extremum

or for multiple points check
P and Q must be 0 at that point
then consider max & min
at 1,1

$6 \times 6 - 9 > 0$ and $r = 6 > 0.$

minimum

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Q. $f(x,y) = x^2 + y^2 + xy + x - 4y + 5$ has _____

(a) min at $(2,-3)$

(b) min at $(-2,3)$

(c) max at $(2,-3)$

(d) max at $(-2,3)$

→ $p = 2x + y + 1$

$q = 2y + x - 4$

$r = 2$

$s = 1$

$t = 2$

$rt - s^2 = 4 - 1 > 0$ & $r = 2 > 0$.

minimum

Q. $f(x,y) = 4x^2 + 6y^2 - 8x - 4y + 8$ has _____

(a) min $\frac{8}{3}$ (b) max $\frac{8}{3}$ (c) min $\frac{10}{3}$ (d) max $\frac{10}{3}$.

→ $p = 8x - 8 = 0 \Rightarrow x = 1$

$q = 12y - 4 = 0 \Rightarrow y = \frac{1}{3}$

$r = 8$

$s = 0$

$t = 12$

$rt - s^2 = 96 - 0 > 0$ & $r = 8 > 0$

minimum

$\therefore f\left(1, \frac{1}{3}\right) = 4 + 6\left(\frac{1}{3^2}\right) - 8 - \frac{4}{3} + 8$

$= 4 + \frac{2}{3} - \cancel{8} - \frac{4}{3} + \cancel{8}$

$= 4 - \frac{2}{3}$

$= \frac{10}{3}$

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Q. $f(x,y) = \sin x + \sin y + \sin(x+y)$ has ____

(a) min $\frac{\sqrt{3}}{2}$ (b) max $\frac{\sqrt{3}}{2}$ (c) min $\frac{3\sqrt{3}}{2}$ (d) max $\frac{3\sqrt{3}}{2}$

→ $P = \cos x + \cos(x+y)$

$q = \cos y + \cos(x+y)$

$r = \frac{\partial^2 f}{\partial x^2} = -\sin x - \sin(x+y)\Big|_{\frac{\pi}{3},\frac{\pi}{3}} = -\sqrt{3}$

$s = \frac{\partial^2 f}{\partial x \partial y} = -\sin(x+y)\Big|_{\frac{\pi}{3},\frac{\pi}{3}} = -\frac{\sqrt{3}}{2}$

$t = \frac{\partial^2 f}{\partial y^2} = -\sin y - \sin(x+y)\Big|_{\frac{\pi}{3},\frac{\pi}{3}} = -\sqrt{3}$

→ Make $P = 0$ or $q = 0$.

$\therefore \cos x + \cos(x+y) = 0$

$\cos y + \cos(x+y) = 0$

$\cos x - \cos y = 0$

$\Rightarrow x = y$

$\therefore \cos x + \cos 2x = 0$

$\Rightarrow \cos 2x = -\cos x$

$\Rightarrow \cos 2x = \cos(\pi - x)$

$2x = \pi - x$

$\Rightarrow x = \frac{\pi}{3} = y$

$\therefore rt - s^2 = 3 - \frac{3}{4} > 0$ and $r = -\sqrt{3} < 0$

max. $\frac{3\sqrt{3}}{2}$

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Definite and Indefinite Integrals:-

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$$

$$\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$$

$$\int_a^b \frac{f(x)}{f(x)+f(a+b-x)}\,dx = \frac{b-a}{2}$$

$$\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$$

$$\int_{-a}^{a} f(x)\,dx = 2\int_0^a f(x)\,dx \quad , \text{ if } f(x) \text{ is even.}$$

$$= 0 \quad , \text{ if } f(x) \text{ is odd.}$$

$$\int_0^{2a} f(x)\,dx = 2\int_0^a f(x)\,dx, \quad \text{if } f(2a-x) = f(x)$$

$$= 0 \quad , \text{ if } f(2a-x) = -f(x)$$

$$\int_0^{\pi/2} \frac{f(\sin x)}{f(\sin x)+f(\cos x)}\,dx = \int_0^{\pi/2} \frac{f(\tan x)}{f(\tan x)+f(\cot x)}\,dx = \frac{\pi}{4}.$$

$$\int_0^{\pi/2} \sin^n x\,dx = \int_0^{\pi/2} \cos^n x\,dx = \frac{(n-1)(n-3)(n-5)\cdots}{n(n-2)(n-4)\cdots\cdots} \times K$$

$$\left\{ \begin{array}{l} K = 1 \text{ , if } n \text{ is odd} \\ K = \frac{\pi}{2} \text{ , if } n \text{ is even} \end{array} \right\}$$

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$$\int_0^{\pi/2} \sin^m x \cdot \cos^n x \, dx = \frac{(m-1)(m-3)\ldots\ldots (n-1)(n-3)\ldots\ldots}{(m+n)(m+n-2)(m+n-4)\ldots\ldots} \times K$$

$$\left\{ \begin{array}{l} K = 1, \text{ if either } m \text{ or } n \text{ or both are odd} \\ K = \frac{\pi}{2}, \text{ if both } m \text{ and } n \text{ are even} \end{array} \right\}$$

$$\int \cos x \, dx = \sin x$$

$$\int \sec^2 x \, dx = \tan x$$

$$\int \sec x \tan x \, dx = \sec x$$

$$\int \tan x \, dx = \log \sec x$$

$$\int \cot x \, dx = \log \sin x$$

$$\int \sinh x \, dx = \cosh x$$

$$\int \cosh x \, dx = \sinh x$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} \, dx = \sin^{-1}\left(\frac{x}{a}\right)$$

$$\int \frac{1}{\sqrt{a^2 + x^2}} \, dx = \sinh^{-1}\left(\frac{x}{a}\right)$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} \, dx = \cosh^{-1}\left(\frac{x}{a}\right)$$

$$\int \frac{1}{a^2 + x^2} \, dx = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right)$$

$$\int \frac{1}{a^2 - x^2} \, dx = \frac{1}{2a} \log\left(\frac{a+x}{a-x}\right)$$

$$\int \frac{1}{x^2 - a^2} \, dx = \frac{1}{2a} \log\left(\frac{x-a}{x+a}\right)$$

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$f(x)$ ②

$$\int \frac{f'(x)}{f(x)}\,dx = \log f(x)$$

$$\int \frac{f'(x)}{\sqrt{f(x)}}\,dx = 2\sqrt{f(x)}$$

$$\int \frac{f'(x)}{1+(f(x))^2}\,dx = \tan^{-1}[f(x)]$$

$$\int [f(x)]^n f'(x)\,dx = \frac{[f(x)]^{n+1}}{n+1}$$

$$\int e^{f(x)} \cdot f'(x)\,dx = e^{f(x)}$$

$$\int e^x [f(x) + f'(x)]\,dx = e^x \cdot f(x)$$

$$\int f(x)g(x)\,dx = f(x)\int g(x)dx - \int f'(x)\int g(x)dx\,dx$$

To choose $f(x)$, remember I L A T E

Shortcut *** → $\int$ algebraic $\times e^x\,dx$ = e^x (algebraic fn with derivates of alternate sign)

$$\int x e^x\,dx = e^x(x-1)$$

$$\int x^2 e^x\,dx = e^x(x^2-2x+2)$$

$$\int x^3 e^x\,dx = e^x(x^3-3x^2+6-6)$$

$$\int (x^2+x)e^x\,dx = e^x[(x^2+x)-(2x+1)+2]$$

Shortcut when e^{ax} is present.

$$\int x e^{ax}\,dx = e^{ax}\left[\frac{x}{a} - \frac{1}{a^2}\right]$$

$$\int x^2 e^{ax}\,dx = e^{ax}\left[\frac{x^2}{a} - \frac{2x}{a^2} + \frac{2}{a^3}\right]$$

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$$\int \log x \, dx = x \log x - x$$

$$\int e^{ax} \sin bx \, dx = \frac{e^{ax}}{a^2+b^2} [a \sin bx - b \cos bx]$$

$$\int e^{ax} \cos bx \, dx = \frac{e^{ax}}{a^2+b^2} [a \cos bx + b \sin bx]$$

Q. ① $\int_0^{1/2} \frac{1}{\sqrt{1-x^2}} dx = (\sin^{-1} x)_0^{\frac{1}{2}} = \sin^{-1}\frac{1}{2} - \sin^{-1} 0 = \frac{\pi}{6}$.

($\sin^{-1}\frac{1}{2} = \frac{\pi}{6}$, $\sin^{-1} 0 = 0$)

② $\int_{\log 2}^{\log 3} \frac{e^x}{1+e^x} dx = \log(1+e^x)\Big|_{\log 2}^{\log 3} = \log(1+e^{\log 3}) - \log(1+e^{\log 2})$

$= \log 4 - \log 3$

$= \log \frac{4}{3}$ Ans

③ $\int_0^1 \frac{1}{e^x + e^{-x}} dx$

$= \int_0^1 \frac{e^x}{e^{2x}+1} dx$

$= \int_0^1 \frac{e^x}{1+(e^x)^2} dx = \tan^{-1}(e^x)\Big|_0^1 = \tan^{-1} e - \tan^{-1} 1$

$= \tan^{-1} e - \frac{\pi}{4}$ Ans.

④ $\int_0^1 x e^x = e^x(x-1)\Big|_0^1 = e^1(1-1) - e^0(0-1)$

$= 1$. Ans

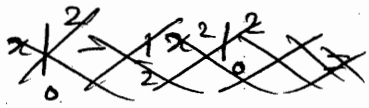
Q. $\int_a^b \frac{\sqrt{x}}{\sqrt{x}+\sqrt{a+b-x}} dx = \frac{b-a}{2} = \frac{7-2}{2} = \frac{5}{2}$

($f(x)$ → $\frac{\sqrt{x}}{\sqrt{x}+\sqrt{a+b-x}}$, $f(a+b-x)$)

Q. $\int_0^{20} \frac{f(x)}{f(x)+f(20-x)} dx = \frac{20-0}{2} = 10$

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Q. $\int_0^2 |1-x|\,dx$ → observe fn within limits when modulus present.

$= \int_0^2 |1-x|\,dx$ = 

$= \int_0^1 |1-x\,dx + \int_1^2 x-1\,dx$

$= x - \frac{x^2}{2}\Big|_0^1 + \frac{x^2}{2} - x\Big|_1^2$

$\Rightarrow 1 - \frac{1}{2} + (2-2) - (\frac{1}{2} - 1)$

$= \frac{1}{2} + \frac{1}{2}$

$= 1$. Ans

* When Greatest Integer fn present observe limit bet^n limits if integer lies → split
if no integer lie bet^n limits → no need to split.

Q. $\int_{-1.5}^{1} [x+1]\,dx$

$\int_{-1.5}^{-1} [x+1]\,dx + \int_{-1}^{0} [x+1]\,dx + \int_{0}^{1} [x+1]\,dx$

$$[x+1] = \begin{cases} -1, & -1.5 < x < -1 \\ 0, & -1 < x < 0 \\ 1, & 0 < x < 1 \end{cases}$$

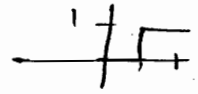
$\int_{-1.5}^{-1} -1\,dx + \int_{-1}^{0} 0\,dx + \int_{0}^{1} 1\,dx$

$= -x\Big|_{-1.5}^{-1} + x\Big|_0^1$

$= -(-1+1.5) + (1-0) = -.5+1 = 0.5$ Ans

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Q. $\int_1^2 x[x]\,dx$.



$x[x] = x$

$$\int_1^2 x\,dx = \frac{x^2}{2}\Big|_1^2 = \left(\frac{2^2}{2} - \frac{1^2}{2}\right) = \frac{3}{2}$$ Ans

Q. $\int_{-1}^{1} \frac{1}{1+x^2}\,dx$ ⟵ *** When limits −a to a check even or odd:

$$= 2\int_0^1 \frac{1}{1+x^2}\,dx$$

$$= 2\tan^{-1}x\Big|_0^1$$

$$= 2\left(\tan^{-1}1 - \tan^{-1}0\right)$$

$$= 2\left(\frac{\pi}{4} - 0\right)$$

$$= \frac{\pi}{2}$$ Ans

Q. $\int_{-a}^{a} \sqrt{\frac{a+x}{a-x}}$ ⟵ no even no odd. Always try to eliminate sq. root from numerator not from denominator WHEN $\sqrt{f(x)}$ IS PRESENT.

$$= \int_{-a}^{a} \frac{a+x}{\sqrt{a^2-x^2}}$$

$$= \int_{-a}^{a} \frac{a}{\sqrt{a^2-x^2}} + \frac{x}{\sqrt{a^2-x^2}}\,dx$$ (odd ⟶ 0)

even, odd.

$$= 2\int_0^a \frac{a}{\sqrt{a^2-x^2}}\,dx = 2a\sin^{-1}\left(\frac{x}{a}\right)\Big|_0^a = 2a\left(\sin^{-1}1 - \sin^{-1}0\right)$$

$$= 2a\left(\frac{\pi}{2} - 0\right)$$

$$= a\pi$$ Ans

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Q. $\int_0^{\pi} \frac{x \sin x}{1+\cos^2 x}\, dx$

Q. $\int_{-a}^{a} \sqrt{\frac{a-x}{a+x}} = a\pi$ (also)

→ When sign changes of x even though answer is same
→ So forget the sign but remember the answer always.

(a) $a\pi$ (b) $\frac{a}{\pi}$ (c) $\frac{\pi}{a}$ (d) $\frac{1}{\pi a}$

***

Q. $I = \int_0^{\pi} \frac{x \sin x}{1+\cos^2 x}\, dx = \frac{\pi^2}{4}$

Imp Result: For Removal of algebric a function always use —

$$\int_0^a f(x)\, dx = \int_0^a f(a-x)\, dx$$

$$I = \int_0^{\pi} \frac{(\pi - x) \sin(\pi - x)}{1+\cos^2(\pi - x)}$$

$$I = \int_0^{\pi} \frac{(\pi - x) \sin x}{1+\cos^2 x}\, dx$$

$$2I = \int_0^{\pi} \frac{(\pi + \pi - x) \sin x}{1+\cos^2 x}\, dx$$

$$2I = \pi \int_0^{\pi} \frac{-\sin x}{1+(\cos x)^2}\, dx$$

$$\int \frac{f'(x)}{1+[f(x)]^2}\, dx = \tan^{-1}[f(x)]$$

$$I = -\frac{\pi}{2} \left[\tan^{-1}(\cos x)\right]_0^{\pi} = -\frac{\pi}{2}\left[\tan^{-1}(-1) - \tan^{-1}(1)\right]$$

$$= -\frac{\pi}{2}\left(-\frac{\pi}{4} - \frac{\pi}{4}\right) = -\frac{\pi}{2} \times -\frac{\pi}{2} = \frac{\pi^2}{4}$$ Ans

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$$*** I = \int_0^{\pi/2} \log(\tan x)\,dx = 0$$

Imp Results

$$I = \int_0^{\pi/2} \log(\cot x)\,dx = 0$$

Q. $$I = \int_0^{\pi/2} \log(\tan x)\,dx$$

$$= \int_0^{\pi/2} \log\left(\tan\left(\frac{\pi}{2}-x\right)\right)dx = \int_0^{\pi/2} \log \cot x\,dx.$$

$$2I = \int_0^{\pi/2} [\log(\tan x) + \log(\cot x)]\,dx$$

$$= \int_0^{\pi/2} \log(\tan x \cdot \cot x)\,dx$$

$$= \int_0^{\pi/2} \log 1\,dx$$

$$= 0 \quad \text{Ans}$$

Whenever limit -a to a given.

Check even or odd

Q. $$\int_{-\pi}^{\pi} x^4 \sin^5 x\,dx$$

x^4 → even, $\sin^5 x$ → odd

= odd

$$= 0.$$

If formula forget

Q. $$\int_0^{\pi/2} \sin^5 x\,dx$$

for Numerator:
- if power even → write odd no
- if power odd → write even no

for Denominator:
- if power even → write even
- if power odd → write odd

$$= \frac{4 \times 2}{5 \times 3 \times 1} = \frac{8}{15}$$

- If power = odd → keep result as it is
- If power = even → multiply result by $\frac{\pi}{2}$

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Q. $\int_0^{\pi/2} \cos^7 x\, dx = \frac{6 \times 4 \times 2}{7 \times 5 \times 3} = \frac{16}{35}$

Q. $\int_0^{\pi/2} \sin^8 x\, dx = \frac{7 \times 5 \times 3 \times 1}{8 \times 6 \times 4 \times 2} \times \frac{\pi}{2} = \frac{35}{256}\pi$

Q. $\int_0^1 \frac{x^6\, dx}{\sqrt{1-x^2}}$

Let $x = \sin\theta$

$dx = \cos\theta\, d\theta$.

if $x \to 0 \quad \theta \to 0$

if $x \to 1 \quad \theta \to \frac{\pi}{2}$.

$$\int_0^{\pi/2} \frac{\sin^6\theta}{\cos\theta} \cos\theta\, d\theta.$$

$$= \frac{5 \times 3 \times 1}{6 \times 4 \times 2} \times \frac{\pi}{2} = \frac{5\pi}{32}$$

Q. $\int_0^{\pi/2} \sin^4 x \cos^5 x\, dx$ → both are even then only multiply by $\frac{\pi}{2}$.

$$= \frac{3 \times 1 \times 4 \times 2}{9 \times 7 \times 5 \times 3 \times 1} =$$ → do individually as before in numerator

→ In Denominator — start from sum of powers and continue.

$$= \frac{8}{315}$$

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Q. $\int_0^{\pi/2} \sin^6 x \cos^4 x \, dx$.

$= \frac{5 \times 3 \times 1 \times 3 \times 1}{10 \times 8 \times 6 \times 4 \times 2} \times \frac{\pi}{2} = \frac{3\pi}{512}$.

Q. $\int_0^{\pi/2} \sin^3 x \cos^5 x \, dx$

$= \frac{2 \times 4 \times 2}{8 \times 6 \times 4 \times 2} = \frac{1}{24}$.

Q. $\int_0^{\pi} \sin^3 x \, dx$

$\because \int_0^{2a} f(x)\,dx = 2\int_0^{a} f(x)\,dx$ if $f(2a-x) = f(x)$.

$\therefore f(\pi - x) = \sin^3(\pi - x) = \sin^3 \pi = f(x)$

Converted in $\pi/2$

To use formula

$\therefore 2 \int_0^{\pi/2} \sin^3 x \, dx$

$= 2 \times \frac{2}{3 \times 1} = \frac{4}{3}$

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Definite and Improper Integrals :-

An integral $\int_a^b f(x)\,dx$ is said to be an improper integral if ① f becomes infinite in the interval of integration.

② one or both of the limits are infinite.

Ex:- $\int_0^1 \frac{1}{x}\,dx$, $\int_0^1 \sqrt{\frac{1+x}{1-x}}\,dx$, $\int_0^3 \frac{1}{(x-1)^{2/3}}\,dx$.

$\int_1^\infty \frac{1}{x^3}\,dx$, $\int_{-\infty}^0 \sin hx\,dx$, $\int_{-\infty}^\infty \frac{1}{1+x^2}\,dx$.

Q. $\int_0^1 \frac{1}{x}\,dx = \log x\Big|_0^1 = \log 1 - \log 0 = \log\frac{1}{0} = \log\infty = \infty$

Divergent

Q. $\int_0^1 \sqrt{\frac{1+x}{1-x}}\,dx$.

A/c to prev. discussion when limit was -1 to 1 ans was π. Here limit is 0 to 1 ans will be $\frac{\pi}{2}$. But some extra term would be present also.

$= \int_0^1 \frac{1+x}{\sqrt{1-x^2}}\,dx = \int_0^1 \frac{1}{\sqrt{1-x^2}} + \frac{x}{\sqrt{1-x^2}}\,dx$

$= \int_0^1 \frac{1}{\sqrt{1-x^2}}\,dx + \int_0^1 \frac{-2x}{2\sqrt{1-x^2}}\,dx$

$= \left(\sin^{-1}x - \frac{1}{2}\times 2\sqrt{1-x^2}\right)_0^1$

$= (\sin^{-1}1 - \sqrt{1-1}) - (\sin^{-1}0 - \sqrt{1-0})$.

$= \frac{\pi}{2} + 1$

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Q. $\int_0^3 \frac{1}{(x-1)^{2/3}}\,dx$

$= \int_0^3 (x-1)^{-2/3}\,dx$

$= \left[\frac{(x-1)^{-2/3+1}}{-\frac{2}{3}+1}\right]_0^3$

$= \frac{(x-1)^{1/3}}{\frac{1}{3}}\Big|_0^3$

$= 3\left[2^{\frac{1}{3}} - (0)^{\frac{1}{3}}\right]$

$= 3\left[\sqrt[3]{2}+1\right]$ Ans

Q. $\int_1^\infty \frac{1}{x^3}\,dx$

$= \left(\frac{-1}{2x^2}\right)_1^\infty$

$= -\frac{1}{2}\left[\frac{1}{\infty} - \frac{1}{1}\right]$

$= -\frac{1}{2}[0-1]$

$= \frac{1}{2}$ Ans

Q. $\int_{-\infty}^0 \sinh x\,dx$

$= \cosh x\,dx\Big|_{-\infty}^0$

$= \frac{e^x + e^{-x}}{2}\Big|_{-\infty}^0$

$= \frac{e^0 + e^{-0}}{2} - \frac{e^{-\infty} + e^{-\infty}}{2} = -\infty$ Ans

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Q. $\int_{-\infty}^{\infty} \frac{1}{1+x^2}\,dx = 2\int_{0}^{\infty} \frac{1}{1+x^2}\,dx$

$= 2 \tan^{-1}x \Big|_0^{\infty}$

$= 2\left(\tan^{-1}(\infty) - \tan^{-1}0\right)$ ($\infty \to \frac{\pi}{2}$, $0 \to 0$)

$= 2 \times \frac{\pi}{2}$

$= \pi$ Ans

Q. $\int_0^{\infty} x e^{-x^2}\,dx$

$= \frac{-1}{2}\int_0^{\infty} e^{-x^2}(-2x)\,dx$ $\left\{ \because \int e^{f(x)} \cdot f'(x) = e^{f(x)} \right\}$

$= \frac{-1}{2}\left(e^{-x^2}\right)_0^{\infty}$

$= -\frac{1}{2}\left[e^{-\infty} - e^{-0}\right]$ ($e^{-\infty} \to 0$, $e^{-0} \to 1$)

$= -\frac{1}{2}\left[0 - 1\right) = \frac{1}{2}$

*** Q. $\int_0^{\infty} e^{-x^2}\,dx$

Let $t = x^2$

$dt = 2x\,dx$

$dx = \frac{dt}{2\sqrt{t}}$

$x \to 0 \to t \to 0$

$x \to \infty \to t \to \infty$.

donot take $t = -x^2$

$\Rightarrow x^2 = -t$

x^2 never becomes -ve

Wrong Idea

Imp Result

*** $\Gamma \to$ means Γ fn

$= \frac{1}{2}\int_0^{\infty} \frac{e^t}{\sqrt{t}}\,dt$

$= \frac{1}{2}\int_0^{\infty} e^t \cdot t^{-\frac{1}{2}}\,dt$

$= \frac{1}{2}\sqrt{\frac{1}{2}} = \frac{\sqrt{\pi}}{2}$

Definite Improper integral

When x becomes complex to evaluate

Take Help of Γ function.

$\Gamma(n) = \int_0^{\infty} e^{-x} x^{n-1}\,dx$

$\Gamma(n+1) = n\Gamma(n)$; if $n > 0$

$\Gamma(n+1) = n!$; if $n \in N$

$\Gamma(1) = 1$

$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$

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Do clearly Prev. problem here.

—

Q. $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)}\,dx\,dy.$

$= \int_0^\infty \int_0^\infty e^{-x^2} \cdot e^{-y^2}\,dx\,dy.$

$= \int_0^\infty e^{-x^2}\,dx \int_0^\infty e^{-y^2}\,dy$

$= \frac{\sqrt{\pi}}{2} \times \frac{\sqrt{\pi}}{2} = \frac{\pi}{4}.$

Q. $\int_0^\infty \int_0^\infty \int_0^\infty e^{-(x^2+y^2+z^2)}\,dx\,dy\,dz$

$= \frac{\sqrt{\pi}}{2} \times \frac{\sqrt{\pi}}{2} \times \frac{\sqrt{\pi}}{2} = \frac{\sqrt{\pi}}{8}$ Ans

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Q. $\int_0^\infty x^3 e^{-x}\,dx = 3!$ → Power will become factorial

Shortcut

$= \cancel{x^3}\, e^{-x}\left[\frac{x^3}{-1} - \frac{3x^2}{-1^2} + \frac{6x}{-1^3} - \frac{6}{-1^4}\right]_0^\infty = 0 \times \infty$ (indetminate form)

Q. $\int_0^\infty \frac{x}{(x^2+9)^2}\,dx$

Let $t = x^2+9$

$\Rightarrow dt = 2x\,dx \Rightarrow dx = \frac{dt}{2x}$

$x \to 0 \;,\; t \to 9$

$x \to \infty \;,\; t \to \infty$

$= \int_9^\infty \frac{\cancel{x}}{t^2} \times \frac{dt}{2\cancel{x}}$

$= \frac{1}{2}\left[-\frac{1}{t}\right]_9^\infty$

$= -\frac{1}{2}\left[\frac{1}{\infty} - \frac{1}{9}\right]$

$= -\frac{1}{2}\left[0 - \frac{1}{9}\right]$

$= \frac{1}{18}$ Ans

Q. $\int_{-\infty}^{\infty} \frac{x}{(x^2+9)^2}\,dx$ → fn is odd $\Rightarrow$ result = 0.

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Vector Calculus

Gradient of a scalar fn :-

Let $\phi(x,y,z) = c$ be any scalar fn, then the grad of ϕ is denoted by "grad ϕ" or "$\Delta\phi$" and is defined as –

$$\Delta\phi = i\frac{\partial\phi}{\partial x} + j\frac{\partial\phi}{\partial y} + k\frac{\partial\phi}{\partial z}$$

we know that –

$$\bar{r} = xi + yj + zk$$

$$|\bar{r}| = \sqrt{x^2+y^2+z^2}$$

$$r^2 = x^2 + y^2 + z^2$$

$$\Rightarrow \cancel{2}r\frac{\partial r}{\partial x} = \cancel{2}x \quad , \quad \cancel{2}r\frac{\partial r}{\partial y} = \cancel{2}y \quad , \quad \cancel{2}r\frac{\partial r}{\partial z} = \cancel{2}z$$

$$\Rightarrow \frac{\partial r}{\partial x} = \frac{x}{r}$$

$$\frac{\partial r}{\partial y} = \frac{y}{r}$$

$$\frac{\partial r}{\partial z} = \frac{z}{r}$$

$$\therefore \nabla r = i\frac{\partial r}{\partial x} + j\frac{\partial r}{\partial y} + k\frac{\partial r}{\partial z}$$

$$= i\left(\frac{x}{r}\right) + j\left(\frac{y}{r}\right) + k\left(\frac{z}{r}\right)$$

$$\nabla r = \frac{xi + yj + zk}{r}$$

$$\therefore \boxed{\text{grad } r = \nabla r = \frac{\bar{r}}{r}}$$

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$$\nabla(\log r) = \frac{1}{r}\nabla r = \frac{1}{r}\cdot\frac{\bar{r}}{r} = \frac{\bar{r}}{r^2}$$

$$\nabla\left(\frac{1}{r}\right) = \frac{-1}{r^2}\nabla r = \frac{-1}{r^2}\cdot\frac{\bar{r}}{r} = \frac{-\bar{r}}{r^3}$$

$$\nabla(r^n) = n r^{n-1}\nabla r = n r^{n-1}\frac{\bar{r}}{r} = n r^{n-2}\bar{r}$$

Tangent vector to a curve :-

Let $\bar{r}(t)$ be the given vector curve, then $\frac{d\bar{r}}{dt}$ is called the tangent vector to the given curve

$$\bar{r} = xi + yj + zK$$

(fn of x)

$$\boxed{\frac{d\bar{r}}{dt} = \frac{dx}{dt}i + \frac{dy}{dt}j + \frac{dz}{dt}K}$$

Normal to a surface :-

Let $\phi(x,y,z) = c$ be any surface then $\nabla\phi$ is called the normal to the surface ϕ.

and $\frac{\nabla\phi}{|\nabla\phi|}$ is called the unit normal vector to the Surface 'ϕ' and is denoted by 'N'

$$\boxed{\text{i.e., } \phi(x,y,z) = c \quad \text{then } N = \frac{\nabla\phi}{|\nabla\phi|}}$$

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Directional Derivative :-

Let $\phi(x, y, z) = c$ be any surface, then

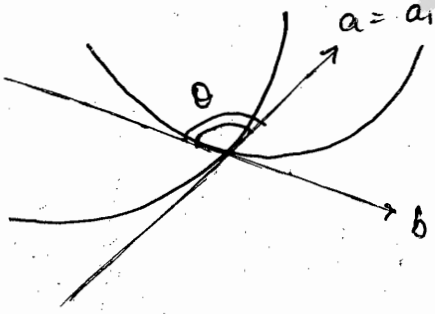
$\boxed{\nabla\phi \cdot e}$ is called the directional derivative to the surface ϕ. where e is the unit ~~normal~~ vector in the dir^n of given vector.

i.e $e = \dfrac{\bar{a}}{|a|}$

The $\boxed{\text{maxm value of the directional derivative}}$ or the greatest value of directional derivative or the magnitude of a gradient of a scalar fn ϕ is $\boxed{\text{defined by } |\Delta\phi|}$

The angle between the curves :-

The angle bet^n the two curves is the angle bet^n tangents at their point of intersection :.



$a = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ → For tangent ①

$b = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$. → for tangent ②

Note :-

If the two curves cuts each orthogonally then

$a \cdot b = 0$ $\{\because \theta$ will become 90° & $\cos\theta = \cos 90^\circ = 0\}$

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The angle betⁿ two surfaces :-

The angle betⁿ two surfaces is the angle betⁿ normals drawn at their surfaces.

If f and g are any two surfaces and 'θ' is the angle betⁿ them then

$$\cos\theta = \frac{\nabla f \cdot \nabla g}{|\nabla f|\,|\nabla g|}$$

Note :-

If the two surfaces cuts each orthogonally then $\nabla f \cdot \nabla g = 0$.

Q. Find the unit normal vector to surface at the point $(1, 2, -1)$ where $\phi = x^3 + y^3 + 3xyz = 3$ (Surface)

→ $\phi = x^3 + y^3 + 3xyz = 3$.

$$\nabla\phi = (3x^2 + 3yz)\,\hat{i} + (3y^2 + 3xz)\,\hat{j} + (3xy)\,\hat{k}$$

$$\nabla\phi\Big|_{1,2,-1} = -3\hat{i} + 9\hat{j} + 6\hat{k}.$$

$\because N = \frac{\nabla\phi}{|\nabla\phi|}$ is unit normal to surface

$$\therefore N = \frac{3(-\hat{i} + 3\hat{j} + 2\hat{k})}{3\sqrt{1+9+4}} = \frac{-\hat{i} + 3\hat{j} + 2\hat{k}}{\sqrt{14}}$$

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Q. Unit normal to $x^2y + 2xz = 4$ at $(2, -2, 3)$

(a) $\frac{i + 2j - 2k}{3}$

(b) $\frac{i - 2j + 2k}{3}$

(c) $\frac{-i + 2j + 2k}{3}$

(d) $\frac{i + j + 2k}{3}$.

$\rightarrow$ $\nabla\phi\Big|_{2,-2,3} = (2xy + 2z)i + (x^2)j + (2x)k$

$= -2i + 4j + 4k$

$\therefore N = \frac{\nabla\phi}{|\nabla\phi|} = \frac{2(-i + j + 2k)}{2\sqrt{1+4+4}} = \frac{-i + j + 2k}{3}$.

Q. U.N to $x^2 + y^2 + z^2 = 1$ at $\left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right)$

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(a) $\frac{i + j}{\sqrt{2}}$

(b) $\frac{j + k}{\sqrt{2}}$

(c) $\frac{i + k}{\sqrt{2}}$

(d) $\frac{i + j + k}{\sqrt{3}}$

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DIRECTIONAL DERIVATIVE (4 types total)

Type-1

Q. find the directional derivative of $f = xy + yz + zx$ at $(1,2,0)$ in the dir^n of $i + 2j + 2k$.

$\rightarrow$ $\nabla\phi = i\frac{\partial\phi}{\partial x} + j\frac{\partial\phi}{\partial y} + k\frac{\partial\phi}{\partial z}$

$\nabla f = i(y+z) + j(x+z) + k(y+x)$

$\nabla f\big|_{(1,2,0)} = 2i + j + 3k$

$a = i + 2j + 2k$

$e = \frac{a}{|a|} = \frac{i+2j+2k}{\sqrt{1+4+4}}$

$e = \frac{i+2j+2k}{3}$

$D \cdot D = \nabla f \cdot e$

$= 2\left(\frac{1}{3}\right) + 1\left(\frac{2}{3}\right) + 3\left(\frac{2}{3}\right)$

$= \frac{10}{3}$ ←

Q. find $D \cdot D$ of $f = 2xy + z^2$ at $(1,-1,3)$ in the dir^n of $i + 2j + 3k$.

$\rightarrow$ $\nabla f\big|_{1,-1,3} = (2y)i + (2x)j + (2z)k$

$= -2i + 2j + 6k$

$a = i + 2j + 3k$

$\hat{e} = \frac{i+2j+3k}{\sqrt{1+4+9}}$

$e = \frac{i+2j+3k}{\sqrt{14}}$

$D \cdot D = \nabla f \cdot e$

$= -2\left(\frac{1}{\sqrt{14}}\right) + 2\left(\frac{2}{\sqrt{14}}\right) + 6\left(\frac{3}{\sqrt{14}}\right)$

$= \frac{20}{\sqrt{14}}$ ←

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Type-II:

Q. find the directional derivative of $f = x^2 - y^2 + 2z^2$ at $P(1,2,3)$ in the dirⁿ of $\overline{PQ}$ where $Q(5,0,4)$

→ $a = \overline{PQ} = Q - P = 4i - 2j + k$

$$e = \frac{a}{|a|} = \frac{4i - 2j + k}{\sqrt{16+4+1}} = \frac{4i - 2j + k}{\sqrt{21}}$$

dot product

$$\nabla\phi\Big|_{1,2,3} = (2x)i - (2y)j + (4z)k$$

$$= 2i - 4j + 12k$$

$$\therefore D.D = \frac{8 + 8 + 12}{\sqrt{21}} = \frac{28}{\sqrt{21}}$$ Ans.

Q. find the D.D of $f = xy + yz + zx$ at $P(1,2,-1)$ in the dirⁿ of $\overline{PQ}$ where $Q = 1,2,3$.

→ $$\nabla f\Big|_{1,2,-1} = (y+z)i + (x+z)j + (x+y)k$$

$$= i + 3k$$

$a = \overline{PQ} = Q - P = 4k$

$\therefore e = \frac{a}{|a|} = \frac{4k}{4}$

$e = k$

$$\therefore \boxed{D.D = 3.}$$

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Type-III

Q. find the directional derivative of $f = xy^2 + yz^2 + zx^2$ at $(1,1,1)$ along the tangent to the curve $x = t,\ y = t^2,\ z = t^3$

$$\nabla f\Big|_{1,1,1} = i(y^2 + 2xz) + j(2xy + z^2) + k(2yz + x^2)$$

$$\nabla f\Big|_{1,1,1} = (3i + 3j + 3k)$$

$$\therefore D \cdot D = \nabla f \cdot e$$

$$= \frac{3 + 6 + 9}{\sqrt{14}}$$

$$= \frac{18}{\sqrt{14}}$$

$$a = \frac{d\bar{r}}{dt} = \frac{dx}{dt} i + \frac{dy}{dt} j + \frac{dz}{dt} k$$

$x = t \Big| y = t^2 \Big| z = t^3$

$\frac{dx}{dt} = 1 \Big| \frac{dy}{dt} = 2t \Big| \frac{dz}{dt} = 3t^2$

→ To take t value take easy relation.

→ If you choose complicated relation to find 't' then after varifying we know that for other 't' our point varies (Ultimately time loss)

$$\therefore a = i + 2tj + 3t^2 k$$

$$= i + 2j + 3k \quad \{\because t = 1\}$$

$$\therefore e = \frac{a}{|a|} = \frac{i + 2j + 3k}{\sqrt{14}}$$

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Type-IV

Q. find the directional derivative of surface $f = xyz^2 + zz$ at $(1,1,1)$ in the dirⁿ of normal to surface $3xy^2 + y = z$ at $(0,1,1)$

$$\nabla f = i(yz^2 + z) + j(xz^2) + k(2xyz + z)$$

$$\nabla f\Big|_{1,1,1} = 2i + j + 3k$$

$$a = \nabla g\Big|_{0,1,1} = i(3y^2) + j(6xy + 1) + k(-1)$$

$$= 3i + j - k$$

$$e = \frac{a}{|a|} = \frac{3i + j - k}{\sqrt{11}}$$

$$\therefore D \circ D = \nabla f \cdot e$$

$$= \frac{6 + 1 - 3}{\sqrt{11}}$$

$$= \frac{4}{\sqrt{11}}$$

Q. find the maximum value of the directional derivative to the surface $\phi = x^2yz^3$ at $(2,1,-1)$

→ $$\nabla\phi = i(2xyz^3) + j(x^2z^3) + k(3x^2yz^2)$$

$$\nabla\phi\Big|_{2,1,-1} = -4i - 4j + 12k$$

$$|\nabla\phi| = \sqrt{16 + 16 + 144}$$

$$= \sqrt{176}$$

$$= \sqrt{16 \times 11}$$

$$= 4\sqrt{11}$$

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Q. find the greatest value of dir^n derivative of $\phi = x^2yz$ at $(1,4,1)$

$$\rightarrow \quad \nabla\phi\Big|_{1,4,1} = (2xyz)\,i + (x^2z)\,j + (x^2y)\,k$$

$$= 8i + j + 4k$$

$$|\nabla\phi| = \sqrt{64+16+1}$$

$$= \sqrt{81}$$

$$= 9$$

Q. find the magnitude of gradient $u = \frac{x^2}{2} + \frac{y^2}{3}$ at $(1,3)$

(a) $\frac{1}{\sqrt{10}}$ (b) $\frac{1}{\sqrt{5}}$ (c) $\sqrt{5}$ (d) $\sqrt{10}$

$$\rightarrow \quad \nabla u\Big|_{1,3} = \frac{2x}{2}i + \frac{2y}{3}j$$

$$= 1^\circ + 2j$$

$$|\nabla u| = \sqrt{1+4} = \sqrt{5}$$

Q. find the angle betn surfaces $x^2+y^2+z^2 = 9$ and $x^2+y^2-z = 3$ at the point $(2,-1,2)$

$$\rightarrow \quad \nabla f\Big|_{2,-1,2} = \frac{2x}{2}i + 2yj + \frac{2z}{2}K = 4i - 2j + 4K$$

$$\nabla g = \frac{2x}{2}i + 2yj - K = 4i - 2j - K$$

$$\cos\theta = \frac{\nabla f \cdot \nabla g}{|\nabla f|\,|\nabla g|} = \frac{16+4-1}{\sqrt{36}\sqrt{21}} = \frac{+18}{36\sqrt{21}}$$

$$\therefore \theta = \cos^{-1}\left(\frac{8}{3\sqrt{21}}\right)$$

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Q. find acute angle betⁿ the surfaces $xy^2z = 3x + z^2$ and $3x^2 - y^2 + 2z = 1$ at the point $(1, -2, 1)$.

→ $\nabla f\Big|_{1,-2,1} = i(y^2z - 3) + j(2xyz) + k(xy^2 - 2z)$

$= i(-1) + j(-4) + k(2)$

$\nabla g = 6xi - 2yj + 2k$

$(\nabla g)_{1,-2,1} = 6i + 4j + 2k$

→ Due to acute angle we will take only +ve θ.

$\therefore \cos\theta = \dfrac{|6 - 16 + 4|}{\sqrt{1+16+4}\ \sqrt{36+16+4}}$

$= \dfrac{6}{\sqrt{7\times3}\ \sqrt{7\times4\times2}}$

$= \dfrac{6}{2\times7\times\sqrt{6}} = \dfrac{3}{7\sqrt{6}}$

$\theta = \cos^{-1}\left(\dfrac{3}{7\sqrt{6}}\right)$

Q. find the const. a, b so that the surfaces $ax^2 - byz = (a+2)x$ and $4x^2y + z^3 = 4$ may intersect orthogonally at the point $(1, -1, 2)$.

→ $\nabla f\Big|_{1,-1,2} = (2ax - a - 2)i - bzj - byk$

$= (a-2)i - 2bj + bk$

$\nabla g\Big|_{1,-1,2} = (8xy)i + (4x^2)j + (3z^2)k$

$= -8i + 4j + 12k$

A/c to Q –

$\nabla f \cdot \nabla g = 0 \Rightarrow -8(a-2) - 8b + 12b = 0$

$\Rightarrow \boxed{b = 2a - 4}$ — (1)

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A/c to Q— 1, −1, 2 lies also on surface ① —

$\therefore a(1)^2 - b(-1)(2) = (a+2)$ —— ①

$\Rightarrow a + 2b = a + 2$

$\Rightarrow 2b = 2$

$\Rightarrow b = 1$ Ans

$\therefore$ from ① —

$a = \frac{5}{2}$ Ans

Divergence of a vector function :—

Let $\bar{F} = F_1 i + F_2 j + F_3 k$ be any differentiable vector fn then the divergence of f is denoted by div F or $\nabla \cdot F$ and is defined as —

$$\text{div } F = \nabla \cdot F = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial xy} + \frac{\partial F_3}{\partial z}$$

Ex:- $\bar{r} = xi + yj + zk$

$\text{div } \bar{r} = 1 + 1 + 1$

$\text{div} \cdot \bar{r} = \nabla \cdot \bar{r} = 3$

Solenoidal Vector :-

The neccessary and sufficient condⁿ for a vector fn f to be solenoidal is —

$\text{div } F = 0$

$\nabla \cdot F = 0$

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Note:-

Let ϕ is a scalar and a is vector then—

$$\text{div}(\phi A) = \nabla \cdot (\phi A) = (\text{grad}\,\phi) \cdot A + \phi (\text{div}\, A)$$

$$\boxed{\nabla \cdot (\phi A) = (\nabla \phi) \cdot A + \phi (\nabla \cdot A)}$$

Q. $\nabla^2\left(\frac{1}{r}\right) = \underline{0}$ where $r = |\bar{r}|$ and $\bar{r} = xi + yj + zk$

Imp Result →

$$\nabla \cdot \nabla\left(\frac{1}{r}\right) = \nabla \cdot \left(-\frac{1}{r^2}\nabla r\right) = \nabla \cdot \left(-\frac{1}{r^2}\frac{\bar{r}}{r}\right) = \nabla \cdot \left(\underset{\phi}{-\frac{1}{r^3}} \cdot \underset{A}{\bar{r}}\right)$$

$$= \nabla\left(-\frac{1}{r^3}\right) \cdot \bar{r} + \left(-\frac{1}{r^3}\right)(\nabla \cdot \bar{r})$$

$$= \frac{3}{r^4}\nabla r \cdot \bar{r} - \frac{3}{r^3}$$

$$= \frac{3}{r^4}\frac{\bar{r}}{r} \cdot \bar{r} - \frac{3}{r^3}$$

$$= \frac{3}{r^4} \times r^2 - \frac{3}{r^3}$$

$$= \frac{3}{r^3} - \frac{3}{r^3}$$

$$= 0$$

****

SHORTCUT — $\nabla^2[f(r)] = f''(r) + \frac{2}{r} f'(r)$

① $\nabla^2(\log r) = -\frac{1}{r^2} + \frac{2}{r} \cdot \frac{1}{r} = \frac{1}{r^2}$

② $\nabla^2(r^n) = n(n-1)r^{n-2} + \frac{2}{r} n r^{n-1}$

$$= r^{n-2}[n(n-1) + 2n]$$

$$= r^{n-2}[n(n+1)]$$

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Curl of a vector function :-

Let $F = F_1 i + F_2 j + F_3 K$ be any differentiable vector function then the curl of F is denoted by Curl F or $\nabla \times F$ and is defined as —

$$\nabla \times F = \begin{vmatrix} i & j & K \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix}$$

Ex:- $\bar{r} = xi + yj + zK$

$$\text{Curl } \bar{r} = \nabla \times r = \begin{vmatrix} i & j & K \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & z \end{vmatrix}$$

$$= i(0-0) - j(0-0) + K(0-0)$$

$$= 0$$

$$\therefore \boxed{\text{Curl } \bar{r} = 0}$$

Irrotational Vector :-

The necessary and sufficient condⁿ for a vector fn F to be irrotational is —

$$\boxed{\nabla \times F = 0}$$

Note :-

Let F be any non zero vector function then —

$$\text{div}(\text{Curl } F) = 0$$

$$\nabla \cdot (\nabla \times F) = 0.$$

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Q. If $F = (x+2y)\,i + (y+2z)\,j + (x+\lambda z)\,k$ is solenoidal then value of λ = ?

→ $\nabla \cdot F = 0$ (for Solenoidal)

$\Rightarrow \nabla \cdot [(x+2y)\,i + (y+2z)\,j + (x+\lambda z)\,k] = 0$

$\Rightarrow 1 + 1 + \lambda = 0$

$\Rightarrow \lambda = -2$

Q. Find the value of λ so that —

$F = (\lambda x^2 y + yz)\,i + (4xy^2 + xz)\,j + 2xyz\,k$ is zero divergence.

→ $\nabla \cdot F = 0$

$\Rightarrow 2\lambda xy + 8xy + 2xy = 0$

$\Rightarrow xy(2\lambda + 8 + 2) = 0$

$\Rightarrow \lambda + 4 + 1 = 0$

$\Rightarrow \lambda = -5$ Ans

Q. If $f = x^3 + y^3 + z^3 - 3xyz$ then div(grad F) = ___

→ $\nabla \cdot (\nabla f) = \nabla^2 f$

$= 6x + 6y + 6z$

$= 6(x+y+z)$

Q. If $F = 2xy\,i - x^2 z\,j$ then $(\text{curl } F)_{(1,1,1)}$ = ___

$$\nabla \times F = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xy & -x^2 z & 0 \end{vmatrix} = i(0+x^2) - j(0-0) + k(-2xz-2x)$$

$= i - 4k$ @ point (1,1,1)

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Q. If $F = 3x^2 i + 5xy^2 j + xyz^3 k$
then $(\text{div } F)_{1,2,3} =$ ____

→ $\nabla \cdot F = 6x + 10xy + 3xyz^2$

$= 6 + 20 + 54$

$= 80$ Ans

Q. If $\nabla \cdot (r^n \bar{r}) = 0$ then $n =$ ____

→ ∵ $\nabla \cdot (\phi A) = (\nabla \phi) \cdot A + \phi (\nabla \cdot A)$

$\Rightarrow (\nabla r^n) \cdot \bar{r} + r^n (\nabla \cdot \bar{r}) = 0$ (3)

$\Rightarrow n r^{n-1} \nabla r \cdot \bar{r} + 3 r^n = 0$

$\Rightarrow n r^{n-1} \frac{\bar{r}}{r} \cdot \bar{r} + 3r^n = 0$

$\Rightarrow n r^{n-2} r^2 + 3r^n = 0$

$\Rightarrow n r^n + 3r^n = 0$

$\Rightarrow n + 3 = 0$

$\Rightarrow n = -3$

Q. Find the constants a, b, c so that the vector for $F = (x + 2y + az) i + (bx - 3y - z) j + (4x + cy + 2z) k$ is irrotational.

→ $\nabla \times F = 0 \Rightarrow$

$$\begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x + 2y + az & bx - 3y - z & 4x + cy + 2z \end{vmatrix}$$

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$$\Rightarrow i(c+1) - j(4-a) + k(b-2) = 0$$

$$\Rightarrow \quad c+1 = 0 \Rightarrow c = -1$$
$$\text{and } 4-a = 0 \Rightarrow a = 4$$
$$\text{and } b-2 = 0 \Rightarrow b = 2$$

} Ans

Vector Integration :-

① Line integral $\int$

② Surface integral $\iint$

③ Volume integral $\iiint$

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(i). Line Integral :-

Any integral which is evaluated along the curve is called the line integral

Let F be any differentiable vector function defined along 'curve 'c' , then the line integral of F is defined by — $\int_c F \cdot d\bar{r}$

If c is a closed curve , it is denoted by — $\oint_c F \cdot d\bar{r}$

Cartesian form :-

Let $F = F_1 i + F_2 j + F_3 k$

$\bar{r} = xi + yj + zk$

$d\bar{r} = dx\,i + dy\,j + dz\,k$

$$\int_c F \cdot dr = \int_c (F_1\,dx + F_2\,dy + F_3\,dz)$$ is called Cartesian form of line integral

Note :- If $F(x, y, z)$ are fn. of t, then

$$\int_c F \cdot d\bar{r} = \int \left[F_1 \frac{dx}{dt} + F_2 \frac{dy}{dt} + F_3 \frac{dz}{dt}\right] dt$$

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Q. Evaluate $\int_C F \cdot d\bar{r}$ where $F = (2y+3)\hat{i} + xz\hat{j} + (yz-x)\hat{k}$

① where c is the curve $x = 2t^2$ $y = t$, $z = t^3$ joining $(0,0,0)$ to $(2,1,1)$

② where c is the curve of line joining $(0,0,0) \to (0,0,1) \to (0,1,1) \to (2,1,1)$

→ (1). $$\int_C F \cdot d\bar{r} = \int_C \left[F_1 \frac{dx}{dt} + F_2 \frac{dy}{dt} + F_3 \frac{dz}{dt}\right] dt$$

$$= \begin{array}{lll} x = 2t^2 & y = t & z = t^3 \\ \frac{dx}{dt} = 4t & \frac{dy}{dt} = 1 & \frac{dz}{dt} = 3t^2 \end{array}$$

$\because y = t$

$\therefore y \to 0, t \to 0$

$y \to 1, t \to 1$

$$= \int_{t=0}^{1} \left[(2y+3)\frac{dx}{dt} + xz\frac{dy}{dt} + (yz-x)\frac{dz}{dt}\right] dt$$

$$= \int_{t=0}^{1} \left[8t^2 + 12t + 2t^5 + 3t^6 - 6t^4\right] dt$$

$$= \left[\frac{8t^3}{3} + \frac{12t^2}{2} + \frac{2t^6}{6} + \frac{3t^7}{7} - \frac{6t^5}{5}\right]_0^1$$

$$= \frac{8}{3} + 6 + \frac{1}{3} + \frac{3}{7} - \frac{6}{5}$$

$$= 9 + \frac{3}{7} - \frac{6}{5}$$

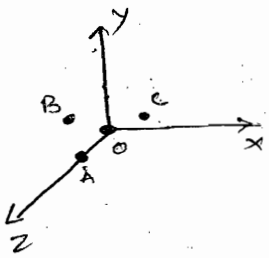
$$= \frac{288}{35}$$

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② $\int_C F \cdot dr = \int_C F_1 \cdot dx + F_2\, dy + F_3\, dz$.

Choose dx, dy, dz such that only that derivative variable varies other remains const. or 1 remains const. & other two varies

① along $\overline{OA}$ $\quad x=0,\ y=0$
$dx=0 \quad dy=0.$
as $z=0$ to $z=1$



$= \int_{OA} (yz - x)\, dz = 0 \quad \{ \because x=0, y=0 \}$

② along $\overline{AB}$ $\quad x=0,\ z=1$
$dx=0 \quad dz=0.$
as $y=0$ to 1.

$= \int_{AB} xz\, dy = 0 \quad \{ \because x=0 \}$

③ along ~~AB~~ $\overline{BC}$ — $\quad y=1,\ z=1$
$dy=0,\ dz=0$
as $x=0$ to 2

$= \int_{x=0}^{2} (2y+3)\, dx = \int_{x=0}^{2} [2(1)+3]\, dx = 5[x]_0^2 = 10$

$\therefore \int_C F \cdot dr = 0+0+10 = \underline{10}$

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Q. $\int F \cdot dr$ where $F = 3x^2 i + (2xz - y) j + zk$ where c is the st. line joining $(0,0,0)$ to $(2,1,3)$

→ eqn of OA $= \frac{x-0}{2-0} = \frac{y-0}{1-0} = \frac{z-0}{3-0} = t$

$\Rightarrow \frac{x}{2} = \frac{y}{1} = \frac{z}{3} = t$

$\Rightarrow x = 2t \;,\; y = t \;,\; z = 3t$

The eqn of a st. line joining (x_1, y_1, z_1) and (x_2, y_2, z_2) in the symmetric form is $\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1} = t$

where t is some scalar.

$$\therefore \int_C F \cdot dr = \int_C \left[F_1 \frac{dx}{dt} + F_2 \frac{dy}{dt} + F_3 \frac{dz}{dt} \right] dt$$

$$= \int_{t=0}^{1} \left[3(4t^2)(2) + (2(2t)(3t) - t)(1) + 3t(3) \right] dt$$

$$= \int_{t=0}^{1} \left(24t^2 + 12t^2 - t + 9t \right] dt$$

$$= \left| 36\frac{t^3}{3} + \frac{8t^2}{2} \right|_0^1 = 12 + 4 = 16$$

Q. $\int_C F \cdot dr = ?$ where $F = 2xyz\, i + x^2 y\, j + x^2 z\, k$ where C is the st. line joining $(0,0,0)$ to $(1,1,1)$

→ eqn of st line —

$$\frac{x}{1} = \frac{y}{1} = \frac{z}{1} = t$$

$$x = t \;,\; y = t \;,\; z = t$$

$$\frac{dx}{dt} = \frac{dy}{dt} = \frac{dz}{dt} = 1$$

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$$\therefore \int F \cdot dr = \int_C F_1 \frac{dx}{dt} + F_2 \frac{dy}{dt} + F_3 \frac{dz}{dt}$$

$$= \int_{t=0}^{1} \left[2t^3(1) + t^3(1) + t^3(1) \right] dt$$

$$= \int_{t=0}^{1} 4t^3 dt = 4 \frac{t^4}{4} \Big|_0^1 = 1 - 0 = 1$$

Q. $\int_C F \cdot dr$, where $F = 3xy\,i - y^2 j$ and C is the curve $y = 2x^2$ in the xy plane joining $(0,0)$ to $(1,2)$

$$\rightarrow \int_C F \cdot dr = \int \left[3xy\,dx - y^2 dy \right]$$

जो variable के term में change करेंगे limit उसी की लेंगे

$y = 2x^2$

$dy = 4x\,dx$

$x \rightarrow 0$ to 1

$$= \int_{x=0}^{1} 3x(2x^2)\,dx - (2x^2)^2 4x\,dx$$

$$= \left| \frac{6x^4}{4} - \frac{16x^6}{6} \right|_0^1 = \frac{3}{2} - \frac{8}{3} = \frac{-7}{6}$$

Q. $\int_C F \cdot dr = ?$ where $F = (5xy - 6x^2)\,i + (2y - 4x)\,j$ and C is the curve $y = x^3$ in the xy plane joining $(1,1)$ to $(2,8)$

$$\rightarrow \int_C F \cdot dr = \int (5xy - 6x^2)\,dx + (2y - 4x)\,dy$$

$y = x^3$

$dy = 3x^2 dx$

$x \Rightarrow 1$ to 2

$$= \int_{x=1}^{2} (5x^4 - 6x^2)\,dx + (2x^3 - 4x)\,3x^2 dx$$

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$$= \int_{x=0}^{2} [5x^4 - 6x^2 + 6x^5 - 12x^3]\, dx$$

$$= \left[\frac{5x^5}{5} - \frac{2x^3}{3} + \frac{6x^6}{6} - \frac{3x^4}{4} \right]_0^2$$

$$= [\text{illegible}]$$

$$= \text{illegible}$$

$$= (32 - 16 + 64 - 48) - (1 - 2 + 1 - 3)$$

$$= 32 + 3 = 35 \quad \text{Ans}$$

Q. $\oint_C F \cdot d\bar{r} = ?$ where $F = (2x - y + z)\,i + (x + y - z^2)\,j + (3x + 2y - 4z)\,k$

Where C is the circle in xy plane having the centre at origin and radius 3 units.

$$\rightarrow \int F \cdot dr = \int (2x - y + z)\,dx + (x + y - z^2)\,dy$$

{ Circle is in xy plane ∴ do not consider dz }

Our Relation is —

$$x^2 + y^2 = 9$$

$$x = r\cos\theta = 3\cos\theta$$

$$y = r\sin\theta = 3\sin\theta$$

$$\Rightarrow dx = -3\sin\theta\, d\theta$$

$$dy = 3\cos\theta\, d\theta$$

as $\theta \rightarrow 0$ to 2π to complete a circle

$$= \int_{\theta=0}^{2\pi} 2(3\cos\theta - 3\sin\theta)(-3\sin\theta\, d\theta) + (3\cos\theta + 3\sin\theta) \times (3\cos\theta)\, d\theta$$

$$= \int_{\theta=0}^{2\pi} (-9(\sin\theta)\cos\theta + 9)\, d\theta$$

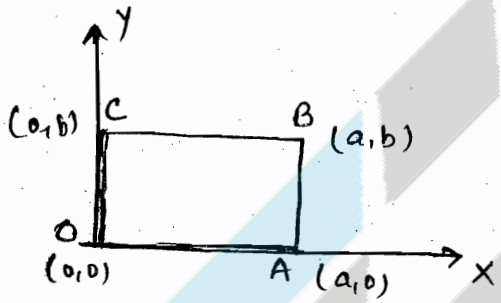
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$$= \left[-9 \frac{\sin 2\theta}{2} + 9\theta \right]_0^{2\pi}$$

$$= 18\pi$$

Q. $\oint_C F \cdot d\bar{r}$ Where $F = (x^2+y^2)\,i - 2xy\,j$ where C is the rectangle bounded by $x=0$, $x=a$, $y=0$, $y=b$.

→



$$\int F \cdot dr = \oint (x^2+y^2)\,dx - 2xy\,dy$$

① along OA — $y=0 \Rightarrow dy=0$ (not changing)

$x \to 0$ to a.

$$\int F \cdot dr = \int_{x=0}^{a} (x^2+0)\,dx = \left(\frac{x^3}{3}\right)_0^a = \frac{a^3}{3}$$

② along AB — $x=a \Rightarrow dx=0$ (not changing)

$y \to 0$ to b

$$\int F \cdot dr = \int_{y=0}^{b} -2ay\,dy = -2a\left(\frac{y^2}{2}\right)_0^b = -ab^2$$

③ along BC — $y=b \Rightarrow dy=0$ (not changing)

$x \to a$ to 0

$$\int F \cdot dr = \int_{x=a}^{0} (x^2+b^2)\,dx = 0 - \frac{a^3}{3} - ab^2$$

④ along CO — $x=0 \Rightarrow dx=0$

$$\int F \cdot dr = \int 0 - 0 = 0$$

$$\therefore \int F \cdot dr = \frac{a^3}{3} - ab^2 + \frac{a^3}{3} - ab^2 + 0 = -2ab^2$$

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Surface Integral :—

Any integral which is evaluated over surface is called a surface integral.

Let F be any differentiable vector fn. defined over a surface S then the surface integral is defined by $\iint_S F \cdot N \, ds$ where N is the outer unit normal vector to given surface.

$$\iint_S F \cdot N \, ds = \iint_{R_1} F \cdot N \frac{dx\,dy}{|N \cdot k|}$$ where R_1 is projection in xy plane.

$$= \iint_{R_2} F \cdot N \frac{dy\,dz}{|N \cdot i|} \quad \text{---} \quad R_2 \text{ --- } yz \text{ plane}$$

$$= \iint_{R_3} F \cdot N \frac{dz\,dx}{|N \cdot j|} \quad \text{---} \quad R_3 \text{ --- } zx \text{ plane}$$

Q. Evaluate $\iint_S F \cdot N \, ds$ where $F = zi + xj - 3y^2z\,k$ and S is the surface of cylinder $x^2+y^2=16$ included in the Ist octant bounded by $z=0$ & $z=5$;

∵ z is given ∴ either xz plane or yz plane will be choosen for Projection

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Q. Evaluate $\iint_S F \cdot N \, ds$ where $F = 4xz\,i - y^2 j + yz\,K$ and S is the cube bounding $0 \le x, y, z \le 1$.

→ for **POAR** —

$N = i$ & $x = 1$ (const)

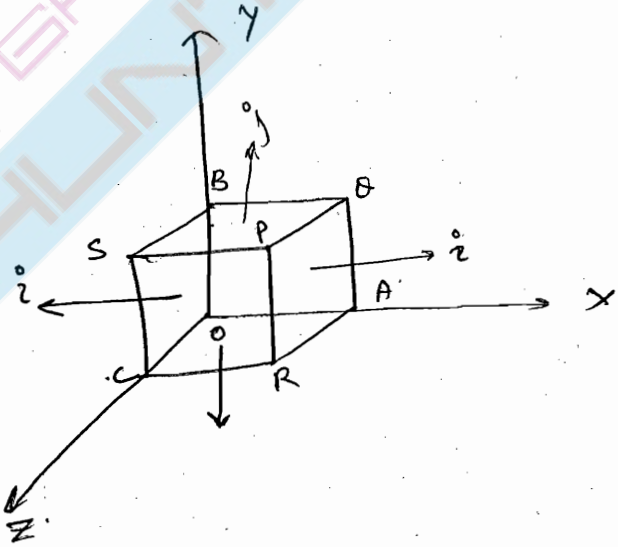
$F \cdot N = 4xz = 4z$

$$\int_S F \cdot N \, ds = \int_{y=0}^{1} \int_{z=0}^{1} 4z \, dy \, dz$$

$$= 4 \frac{z^2}{2} \Big|_0^1 \, y \Big|_0^1$$

$$= 4 \times \frac{1}{2} \times 1$$

$$= 2$$



For **SBOC** —

$N = -i$, $x = 0$.

$F \cdot N = -4xz = 0$

$\therefore \iint F \cdot N \, dS = 0$

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For PQBS —

$N = j, \quad y = 1.$

$$\int_S F \cdot N \, ds = \int_{x=0}^{1} \int_{z=0}^{1} -y \, dx \, dz = -x\Big|_0^1 z\Big|_0^1 = -1$$

For OCRA —

$N = j \quad y = 0$

$$\int_S F \cdot N = 0$$

For SPRC —

$N = K \qquad z = 1.$

$$\int_S F \cdot N \, ds = \int_{x=0}^{1} \int_{y=0}^{1} yz \, dx \, dy$$

$$= \frac{y^2}{2}\Big|_0^1 x\Big|_0^1$$

$$= \frac{1}{2}$$

For BQOA — $z = 0, \quad N = -K$

$$\int F \cdot N \, ds = 0$$

$$\therefore \iint_S F \cdot N \, ds = 2 + 0 + (-1) + 0 + \frac{1}{2}$$

$$= 1 + \frac{1}{2} = \frac{3}{2} \quad \text{Ans}$$

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Volume Integral :-

Any integral which is evaluated over a volume is called the volume integral.

Q. $\iiint_V \nabla \cdot F \, dv$ where $F = (2x^2 - 3z)\,i - 2xy\,j + 2xk$ where V is the region bounded by $x=0, y=0, z=0$ and $2x+2y+z=4$.

$$\nabla \cdot F = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$$

→ $\nabla \cdot F = 4x - 2x$

z → in terms of x & y ——→ 0 to $4-2x-2y$

y → ——→ 0 to $2-x$

x → ——— Const ——→ 0 to 2.

$$\iiint \nabla \cdot F \, dv = \int_{x=0}^{2} \int_{y=0}^{2-x} \int_{z=0}^{4-2x-2y} 2x \, dx\, dy\, dz$$

$$= 2 \int_{x=0}^{2} \int_{y=0}^{2-x} x(4-2x-2y)\, dx\, dy$$

$$= 2 \int_{x=0}^{2} \left[4x - \frac{2x^2}{2} - 2yx \right]_0^{2-x} dx$$

$$= 2 \int_{x=0}^{2} (8 - 4x - 4 + x^2 - 4x - 4y + 2yx)\, dx$$

$$= 2 \int_{x=0}^{2} [x^2 - 8x - 4y + 2yx + 4]\, dx$$

$$= 2 \int_{x=0}^{2} \int_{y=0}^{2-x} [4x - 2x^2 - 2xy]\, dx\, dy$$

$$= 2 \int_{0}^{2} 4xy - 2x^2 y - xy^2 \Big|_0^{2-x} dx$$

$$= 2 \int_{0}^{2} x(2-x)^2 \, dx$$

$$= 2 \left[4\frac{x^2}{2} + \frac{x^4}{4} - 4\frac{x^3}{3} \right]_0^2$$

$$= 2 \left[4 \times \frac{4}{2} + \frac{16}{4} - \frac{32}{3} \right] = \frac{8}{3}$$ Ans

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Vector Transformation

Gauss Divergence Theorem :-

→ The gauss divergence theorem use the connection betⁿ surface to volume integrals.

→ Let S be the closed surface enclosed by a volume 'V' and Let F be any differentiable vector fn then

$$\iint_S F \cdot N \, ds = \iiint_V (\text{div } F) \, dV = \iiint_V (\nabla \cdot F) \, dV.$$

* Cartesian form of Divergence Theorem :-

$$\iint_S (F_1 \, dy\,dz + F_2 \, dz\,dx + F_3 \, dx\,dy) = \iiint_V \left(\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}\right) dx\,dy\,dz$$

Q. The value of $\int_S (ax\,i + by\,j + cz\,k) \cdot N \, ds$, where S is the surface of sphere $x^2 + y^2 + z^2 = 1$.

→ $\nabla \cdot F = a + b + c$

$\therefore \int_S F \cdot N \, ds = \int_V (\text{div } F) \, dV$

$= (a+b+c) \frac{4}{3} \pi r^3$

$= \frac{4\pi}{3} (a+b+c)$ Ans

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Q. Find $\int_S (axi + byj + czk) \cdot N ds$ where S is sphere $x^2+y^2+z^2 = 1$

Same Problem →

or $\int_S (ax^2 + bx^2 + cz^2) ds$ where S is sphere $x^2+y^2+z^2=1$

Imp Result $= (a+b+c)\frac{4\pi}{3}$

Q. $\int_S \bar{r} \cdot N ds$, where 's' is the closed region enclosed by a volume 'V'.

→ $\nabla \cdot \bar{r} = 3$.

$\therefore \int_S \bar{r} \cdot N ds = \int_V (\nabla \cdot \bar{r}) dv = 3V$ Ans

Q. $\int_S \bar{r} \cdot N ds$ where S is sphere $x^2+y^2+z^2 = a^2$

→ $\nabla \cdot \bar{r} = 3$

$\therefore \int_S \bar{r} \cdot N ds = \int_V (\nabla \cdot \bar{r}) dv = \frac{4}{3}\pi a^3 \times 3 = 4\pi a^3$ Ans

Q. $\int_S (x\,dy\,dz + y\,dz\,dx + z\,dx\,dy)$ where S is the cube of unit length.

→ $\nabla \cdot F = 1 + 1 + 1 = 3$

$\int_S F \cdot N ds = \int_V (\text{div } F) dv = 3V = 3(1)^3 = 3$ Ans

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Q. $\int_S (x+z)\,dy\,dz + (y+z)\,dz\,dx + (x+y)\,dx\,dy$ where S is sphere $x^2+y^2+z^2=4$.

$\Rightarrow \quad \nabla \cdot F = 1+1+0 = 2$

$\therefore \int_S F \cdot N\,ds = \int_V (\nabla \cdot v)\,dV = 2V = 2\times\frac{4}{3}\times 2^3$

$= \frac{64\pi}{3}$ Ans

Q. $\int_S (4xz\hat{i} - y^2\hat{j} + yz\hat{k}) \cdot N\,ds$ where S is cube bounded by $0 \le x, y, z \le 1$.

$\rightarrow \quad \nabla \cdot F = 4z - 2y + y = 4z - y$

$\int_S F \cdot N\,ds = \int_V (\text{div } F)\,dV = \int_{x=0}^{1}\int_{y=0}^{1}\int_{z=0}^{1} [4z-y]\,dx\,dy\,dz$

$= (x)_0^1 \left[4\left(\frac{z^2}{2}\right)_0^1 (y)_0^1 - \left(\frac{y^2}{2}\right)_0^1 (z)_0^1 \right]$

$= (1-0)\left[2 - \frac{1}{2}\right]$

$= \frac{3}{2}$ Ans

Q. $\int_S [(x^2-yz)\hat{i} + (y^2-zx)\hat{j} + (z^2-xy)\hat{k}] \cdot N\,ds$ where S is the cuboid bounded by $0 \le x \le a$, $0 \le y \le b$, $0 \le z \le c$

$\rightarrow \quad \nabla \cdot F = 2x + 2y + 2z$

$\therefore \int_S F \cdot N\,ds = \int_V (\nabla \cdot F)\,dV = \int_{x=0}^{a}\int_{y=0}^{b}\int_{z=0}^{c} (2x + 2y + 2z)\,dx\,dy\,dz$

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$$= \left[x^2yz + y^2zx + z^2xy\right]_{\substack{x=0\\y=0\\z=0}}^{\substack{y=b\\x=a}}$$

$$= a^2bc + ab^2c + abc^2$$

$$= abc(a+b+c). \quad \underline{\text{Ans}}$$

Q. A region of volume 10 cubic units is bounded by a closed surface 'S'. and N is the outer unit normal vector to S, then —

$$\int_S (4x\,i + 2y\,j - z\,k) \cdot N\,ds = ___$$

→ $\nabla \cdot F = 4 + 2 - 1 = 5$

$$\int_V 5\,dv = 5V \xrightarrow{10} = 50.$$

*** (ONLY APPLICABLE WHEN R is also in only x-y plane, no z plane)

GREEN'S THEOREM in a plane :- $(\int \to \iint)$

Let 'R' be the closed region in the xy plane bounded by a closed and non intersecting curve 'C' and Let P and Q are continuous fn of x and y possessing the first order partial derivatives, then

$$\oint_C P\,dx + Q\,dy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy.$$

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Q. The value of $\oint_C (2x-y)dx + (x+y)dy$ where C is the circle $x^2+y^2=9$

→ $\frac{\partial P}{\partial x} = -1$, $\frac{\partial Q}{\partial x} = 1$.

→ Const., so no need to consider limit.

{ Only for ▭ and □ we will take const. limit of x and y.

for other curves except rectangle & square { else where, we will take one limit const and other as a fn of former one.

$\therefore \iint_R (1-(-1))\,dx\,dy$

$= \int_R 2\,dR = 2R = 2 \times \pi r^2 = 2 \times \pi \times 3^2 = 18\pi$ Ans

Q. $\oint_C \underbrace{(x^2+y^2)}_{P}dx - \underbrace{2xy}_{Q}dy$, where C is the rectangle bounded by $x=0$, $x=a$, $y=0$, $y=b$.

→ $\frac{\partial P}{\partial y} = 2y$, $\frac{\partial Q}{\partial x} = -2y$

$\iint_R (-2y-2y)\,dx\,dy$

$= \int_{x=0}^{a}\int_{y=0}^{b} -4y\,dx\,dy$

$= (x)_0^a \left(-4\frac{y^2}{2}\right)_0^b = ax - 2b^2 = -2ab^2$

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Q. $\oint_C \overset{P}{(x^2 - xy^3)}\,dx + \overset{Q}{(y - 2xy)}\,dy$ where C is the square bounded by the points (0,0) (2,0) (2,2) (0,2).

$\rightarrow \quad \frac{\partial P}{\partial y} = -3xy^2 \quad , \quad \frac{\partial Q}{\partial x} = -2y$

$\therefore \iint_R -2y - (-3xy^2)\,dx\,dy$

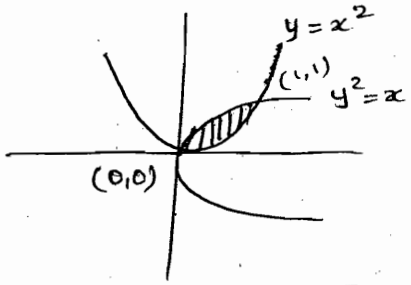
$= \int_{x=0}^{2}\int_{y=0}^{2} -xy + 3xy^2\,dx\,dy$

$= -2\left(\frac{y^2}{2}\right)_0^2 (x)_0^2 + 3\left(\frac{x^2}{2}\right)_0^2 \left(\frac{y^3}{3}\right)_0^2$

$= -8 + 16 = 8$ Ans

Q. $\oint_C (3x^2 - 8y^2)\,dx + (4y - 6xy)\,dy$ where 'C' is the region bounded by $y = \sqrt{x}$ and $y = x^2$

$\rightarrow \quad \frac{\partial P}{\partial y} = -16y \quad , \quad \frac{\partial Q}{\partial x} = -6y$



$\int_{x=0}^{1}\int_{y=x^2}^{\sqrt{x}} (-6y + 16y)\,dx\,dy$ (10y)

$= \int_0^1 10\left(\frac{y^2}{2}\right)_{x^2}^{\sqrt{x}}\,dx$ (5)

$= 5\int_0^1 (x - x^4)\,dx$

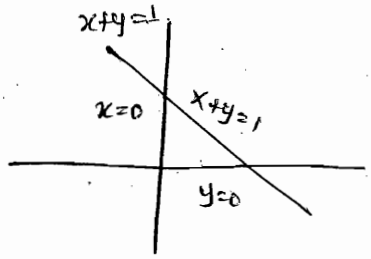
$= 5\left[\frac{x^2}{2} - \frac{x^5}{5}\right]_0^1 = 5\left(\frac{3}{10}\right) = \frac{3}{2}$ Ans

$x^2 = \sqrt{x}$
$x^4 = x$
$x(x^3 - 1) = 0$
$x = 0, 1$
$y = 0, 1$

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Q. $\oint_C (3x^2 - 8y^2)dx + (4y - 6xy)dy$ where C is the region bounded by $x=0, y=0, x+y=1$

$\rightarrow \frac{\partial P}{\partial y} = -16y \quad , \frac{\partial Q}{\partial x} = -6y$



$$\int_{x=0}^{1} \int_{y=0}^{1-x} (-6y + 16y)\, dx\, dy$$

$$= \int_0^1 \int_{y=0}^{1-x} 10y\, dx\, dy$$

$$= \int_0^1 10\left(\frac{y^2}{2}\right)_0^{1-x} dx$$

$$= 5\int_0^1 (1-x)^2 dx$$

$$= 5\left|\frac{(1-x)^3}{-3}\right|_0^1$$

$$= -\frac{5}{3}(0-1)$$

$$= \frac{5}{3} \quad \text{Ans}$$

THE GATE HUNT

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STOKE'S THEOREM :- $\int \to \iint$

Let S be an open surface bounded by a closed and a non intersecting curve 'C' and F be any differentiable vector function then —

$$\oint F \cdot d\bar{r} = \iint_S (\text{curl } F) \cdot N \, ds = \iint_S (\nabla \times F) \cdot N \, ds$$

Q. Evaluate by the Stoke's theorem $\oint_C F \cdot d\bar{r}$ where $F = -y^3 i + x^3 j$ and S is the circular disc $x^2 + y^2 \le 1$, $z = 0$.

$$\to \quad \nabla \times F = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -y^3 & x^3 & 0 \end{vmatrix}$$

$$= i(0-0) - j(0-0) + k(3x^2 + 3y^2)$$

$$= 3(x^2+y^2)k$$

$$= 3k \quad \{\because x^2+y^2 \le 1\}$$

Directly we can say $N = \vec{k}$ { $\because z = 0$ given means xy plane and normal vector will be only $\vec{k}$ }

$$\therefore \int 3k \cdot k \, ds$$

$$= 3 \int ds$$

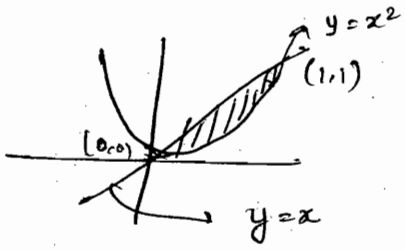
$$= 3S$$

$$= 3\pi(1)^2$$

$$= 3\pi \text{ Ans}$$

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Q. The area bounded by $y = x^2$ and $y = x$ —



$x^2 = x$

$x(x-1) = 0$

$x = 0, 1$

$y = 0, 1$

Imp Trick

from Graph

The area $= \int_0^1 (x - x^2)\,dx$ → [Area will be always upper part − lower part.]

$= \left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^1$

$= \frac{1}{2} - \frac{1}{3}$

$= \frac{1}{6}$ Ans

Q. The area bounded by $y^2 = 4ax$ & $x^2 = 4ay$ is $16a^3/3$

The area bounded by $y^2 = 8x$ & $x^2 = 8y$ is $64/3$

——— $y^2 = 4x$ & $x^2 = 4y$ is $16/3$

——— $y^2 = x$ & $x^2 = y$ is $1/3$

***Q. The arc length of the curve $y = f(x)$ between $x = a$ & $x = b$ is $\int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,dx$

Imp Result

Q. $y = \frac{2}{3}x^{3/2}$ between $x = 0$ and $x = 1$.

$\frac{dy}{dx} = \frac{2}{3} \times \frac{3}{2} x^{\frac{3}{2}-1} = \sqrt{x}$

$\therefore \int_a^b \sqrt{1 + (\sqrt{x})^2}\,dx = \int_a^b \sqrt{1+x}\,dx = \int_a^b (1+x)^{\frac{1}{2}}\,dx$

$= \frac{(1+x)^{3/2}}{3/2}\Big|_0^1 = \frac{2}{3}\left[2\sqrt{2} - 1\right]$

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Q. A parabolic arc $y = \sqrt{x}$ $1 \le x \le 2$ is revolving around x axis then the volume of the solid revolution.

→

$$\text{volume} = \int_a^b \pi y^2 dx \quad (x\text{-axis})$$

$$= \int_a^b \pi x^2 dy \quad (y\text{-axis}).$$

$\therefore$ volume $= \int_1^2 \pi (\sqrt{x})^2 dx$.

$= \pi \left(\frac{x^2}{2}\right)_1^2 = \frac{\pi}{2}(4-1) = \frac{3\pi}{2}$ Ans

WB (97) Pg-49

volume $= \int_a^b \pi x^2 dy$.

$y^2 = 8x$, $x = 2$

$y^2 = 8(2)$

$y^2 = 16$

$\Rightarrow y = \pm 4$

$\therefore$ volume $= \int_{-4}^{4} \pi x^2 dy$ $\left(x = \frac{y^2}{8}\right)$

$= \frac{\pi}{8^2} \int_{-4}^{4} (y^2)^2 dy$

$= \frac{\pi}{64} \cdot 2 \int_0^4 y^4 dy$.

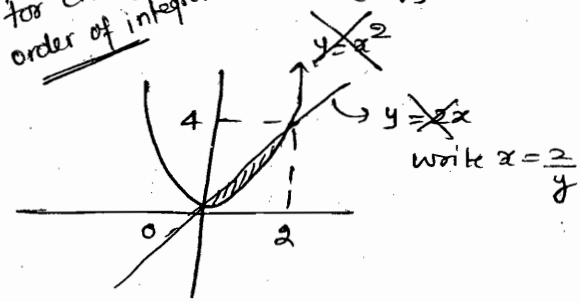
$= \frac{\pi}{32} \times \frac{y^5}{5}\Big|_0^4$

$= \frac{\pi}{32} \times \frac{4 \times 4 \times 4 \times 4 \times 4}{5} = \frac{32\pi}{5}$ Ans

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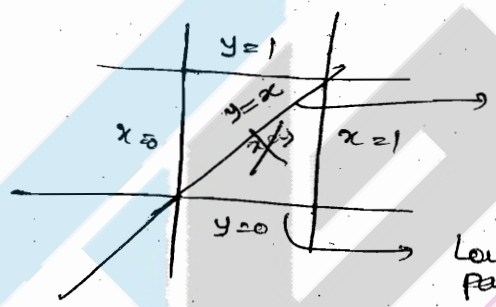
(96) $\int_0^2 \int_{y=x^2}^{y=2x} f(x,y)\, dy\, dx$ $\Rightarrow x = \frac{y}{2}$ $\Rightarrow x = \sqrt{y}$

For changing order of integration write $x = \sqrt{y}$



$$\simeq \int_0^4 \int_{y/2}^{\sqrt{y}} f(x,y)\, dx\, dy$$ Ans.

(97) $\int_0^1 \int_{x=y}^{x=1} xy \sin(xy)\, dx\, dy = \int_0^1 \int_a^b xy \sin xy\, dy\, dx$

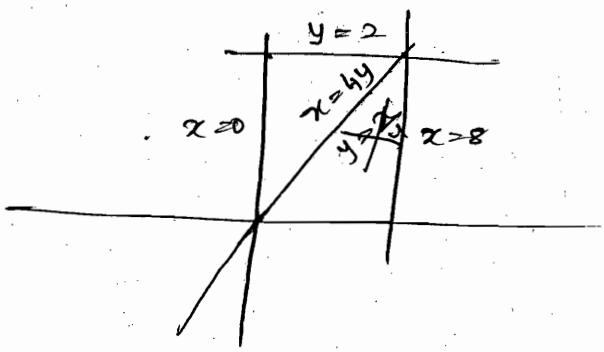


Upper part as upper limit

Lower part as lower limit

By changing order of integration $= \int_0^1 \int_0^x xy \sin(xy)\, dy\, dx$

(98) $\int_0^8 \int_{y=x/4}^{2} f(x,y)\, dy\, dx \simeq \int_\gamma^\delta \int_p^q f(x,y)\, dx\, dy$



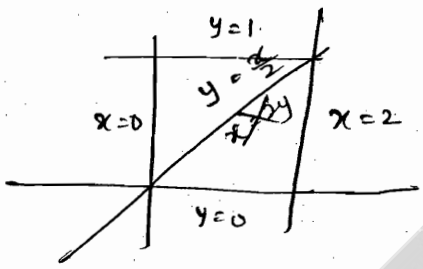
$$\simeq \int_{y=0}^{2} \int_{x=0}^{4y} f(x,y)\, dx\, dy.$$

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(85) ...

$$\int_0^1 \int_{x=2y}^{x=2} e^{x^2}\, dx\, dy$$

By changing order of integration —



$$\simeq \int_0^2 \int_0^{x/2} e^{x^2}\, dy\, dx$$

$$= \int_0^2 e^{x^2} (y)_0^{x/2}\, dx.$$

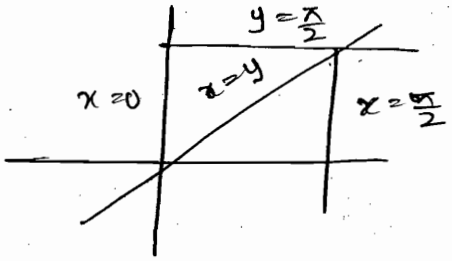
$$= \int_0^2 e^{x^2} \frac{x}{2}\, dx$$

$$= \frac{1}{2\times 2} \int_0^1 e^{x^2} 2x\, dx$$

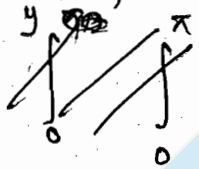
$$= \frac{1}{4} \left(e^{x^2}\right)_0^2 = \frac{e^4 - e^0}{4} = \frac{e^4 - 1}{4}$$ Ans

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(84) $\int_0^{\pi/2} \int_{y=x}^{y=\pi/2} \frac{\cos y}{y}\, dy\, dx$



By changing order of integration —



$$= \int_0^{\pi/2} \int_0^{y} \frac{\cos y}{y}\, dx\, dy$$

$$= \int_0^{\pi/2} \frac{\cos y}{y} (x)_0^y\, dy$$

$$= \int_0^{\pi} \cos y\, dy$$

$$= (\sin y)_0^{\pi/2}$$

$$= \sin\frac{\pi}{2} - \sin 0$$

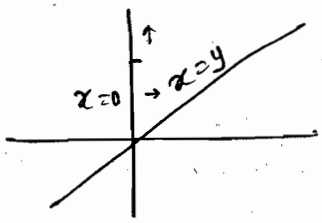
$$= 1 - 0$$

$$= 1 \quad \text{Ans}$$

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(78) $\int_0^{\infty}\int_{y=x}^{y=\infty} \frac{1}{y} e^{-y/2}\, dy\, dx$

$=$

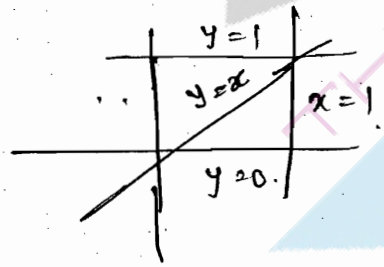


$= \int_0^{\infty}\int_0^{y} \frac{1}{y} e^{-\frac{y}{2}}\, dx\, dy$

$= \int_0^{\infty}\int_0^{y} \frac{1}{y} e^{-y/2} (x)_0^y\, dy$

$= -2\left[e^{-\frac{y}{2}}\right]_0^{\infty} = -2\left[e^{-\infty} - e^{-0}\right] = -2(0-1) = 2$

(77) $\int_{y=0}^{y=1}\int_{x=y}^{x=1} y\sqrt{1+x^3}\, dx\, dy$



$= \int_0^1\int_{y=0}^{x} y\sqrt{1+x^3}\, dy\, dx$

$= \int_0^1 \sqrt{1+x^3}\left(\frac{y^2}{2}\right)_0^x dx$

$= \int_0^1 \sqrt{1+x^3}\,\frac{x^2}{2}\, dx$

$= \frac{1}{3}\int_0^1 (1+x^3)^{\frac{1}{2}} \cdot 3x^2\, dx$

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$$= \frac{1}{6/3} \left| \frac{(1+x^3)^{3/2}}{\frac{3}{2}} \right|_0^1$$

$$= \frac{1}{9} \left[2\sqrt{2} - 1 \right].$$

(59) $f(n) = \int_0^{\pi/4} \tan^n x \, dx$ then $f(3) + f(1) = ?$

$\rightarrow$ $f(3) + f(1) = \int_0^{\pi/4} (\tan^3 x + \tan x) \, dx.$

$$= \int_0^{\pi/4} (\tan x)(\sec^2 x) \, dx$$

$$= \left(\frac{\tan^2 x}{2} \right)_0^{\pi/4} = \frac{1}{2}$$ Ans

(60) $\int_0^{\pi/2} \frac{\cos x - \sin x}{1 + \sin x \cos x} \, dx = ?$

$\because \int_0^{\frac{\pi}{2}} f(x) \, dx = \int_0^a f(a-x) \, dx$

$$I = \int_0^{\pi/2} \frac{\sin x - \cos x}{1 + \cos x \cdot \sin x} \, dx$$

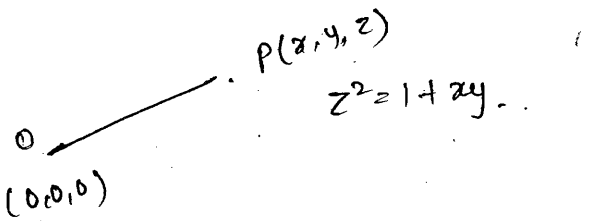
$$2I = \int_0^{\pi/2} \frac{\cos x - \sin x + \sin x - \cos x}{1 + \sin x \cos x}$$

$$= 0$$

$\Rightarrow I = 0$ Ans

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(56)



$$OP = \sqrt{x^2+y^2+z^2}$$

$$= \sqrt{x^2+y^2+1+xy}$$

$$f = x^2+y^2+xy+1$$

$$p = \frac{\partial f}{\partial x} = 2x+y = 0$$

$$q = \frac{\partial f}{\partial y} = 2y+x = 0$$

$$x - y = 0$$

$$x = y$$

Put $x = y$.

$$3x = 0$$

$$x = 0$$

$$\Rightarrow x = 0, \quad y = 0.$$

$$\therefore OP = \sqrt{0+0+0+1}$$

$$= 1$$

(53) $x = 2$ after making $f''(x) = 0$ { Since maximum of slope needed not maximum of curve demanding }

(45) $x = uv, \quad y = \frac{v}{u}$.

$$f(x,y)\,dx\,dy = \iint f(uv, \frac{v}{u}) \cdot \phi(u,v)\,du\,dv$$

$$\phi(u,v) = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} v & u \\ -\frac{v}{u^2} & \frac{1}{u} \end{vmatrix}$$

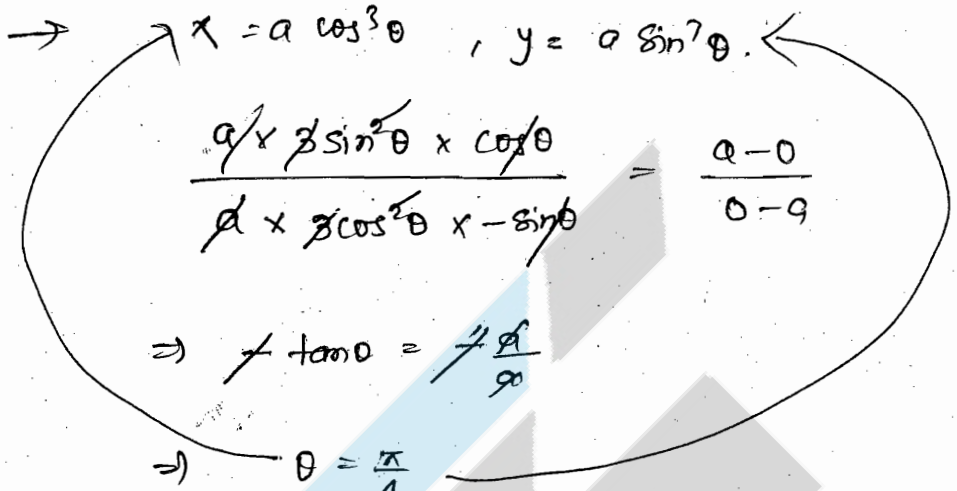
$$= \frac{2v}{u}$$

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Q. 'A curve C is defined as $x = a\cos^3\theta$

$y = a\sin^3\theta$ in $\left[0, \frac{\pi}{2}\right]$

What will be point P on curve C where tangent to curve is parallel to chord joining points (a,0) & (0,a).



$x = a\cos^3\theta$, $y = a\sin^3\theta$.

$$\frac{a \times 3\sin^2\theta \times \cos\theta}{a \times 3\cos^2\theta \times -\sin\theta} = \frac{a-0}{0-a}$$

$$\Rightarrow -\tan\theta = -\frac{a}{a}$$

$$\Rightarrow \theta = \frac{\pi}{4}.$$

∴ putting values — we get —

$$\left(\frac{a}{2\sqrt{2}}, \frac{a}{2\sqrt{2}}\right)$$

Q. 19 — Correct $y = a(1-\cos\theta)$.

Q. (8). $\lim_{n\to\infty} \left(\frac{n!}{n^n}\right)^{\frac{1}{n}}$

$$l = \left(\frac{n!}{n^n}\right)^{\frac{1}{n}}$$

$$\log l = \frac{1}{n}\left[\log n! - n\log n\right]$$

~~$\log l = \frac{1}{n}\log n! - \log n$~~

$$= \frac{1}{n}\left[\log \frac{1\times2\times3\times\cdots n}{n\times n\times n\cdots n}\right]$$

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$$= \frac{1}{n}\left[\log\left(\frac{1}{n}\right) + \log\left(\frac{2}{n}\right) + \cdots\right]$$

Imp formula

$$\lim_{n\to\infty}\left(\frac{1}{n}\sum_{r=1}^{n}\log\left(\frac{r}{n}\right)\right) = \int_0^1 \log x\, dx$$

$$= \left[x\log x - x\right]_0^1$$

$$\log l = (1\log 1 - 1]$$

($\log 1 \to 0$)

$$l = e^{-1} = \frac{1}{e} \quad \text{Ans}$$

Q⑥ $$\lim_{n\to\infty}\left[\frac{1}{1+n} + \frac{1}{2+n} + \cdots + \frac{1}{n+n}\right]$$

$$= \lim_{n\to\infty}\frac{1}{n}\left[\frac{1}{1+\frac{1}{n}} + \frac{1}{1+\frac{2}{n}} + \cdots\right]$$

$$= \lim_{n\to\infty}\left(\frac{1}{n}\sum_{r=1}^{n}\frac{1}{1+\left(\frac{r}{n}\right)}\right)$$

$$= \int_0^1 \log\cdot\frac{1}{1+x}\,dx$$

$$= \log(1+x)\Big|_0^1$$

$$= \log 2 - \log 1$$

$$= \log 2 \quad \text{Ans}$$

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(5) $\lim_{n\to\infty}\left[\frac{n}{n^2}+\frac{n}{n^2+1^2}+\cdots+\frac{n}{n^2+(n-1)^2}\right]$

$$\lim_{n\to\infty}\frac{n}{n^2}\left[\sum_{r=0}^{n-1}\frac{1}{1+\left(\frac{r}{n}\right)^2}\right]$$

$$=\int_0^1\frac{1}{1+x^2}\,dx$$

$$=\left(\tan^{-1}x\right)_0^1$$

$$=\tan^{-1}1-\tan^{-1}0$$

$$=\frac{\pi}{4}-0=\frac{\pi}{4}$$ Ans

(6) $\lim_{n\to\infty}\sum_{k=1}^{n}\frac{1}{3n+k}$

$$=\lim_{n\to\infty}\frac{1}{n}\sum_{k=1}^{n}\frac{1}{3+\left(\frac{k}{n}\right)}$$

$$=\int_0^1\frac{1}{3+x}\,dx=\left[\log(3+x)\right]_0^1=\log 4-\log 3=\log\frac{4}{3}$$ Ans

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DIFFERENTIAL EQ^{N}

An eq^{n} consisting of Differential coeff. $\frac{dy}{dx}$, $\frac{d^2y}{dx^2}$, $\frac{d^3y}{dx^3}$, ... $\frac{d^ny}{dy^n}$ is called a differential eq^{n}:-

* order - $\frac{d^ny}{dy^n}$ → is order

* Degree - The degree of diff eq^{n} is the degree of the highest order

$\frac{d^2y}{dx^2} + 5\left(\frac{dy}{dx}\right)^3 + 6y = 0$ $\quad 0 \to 2$, $D \to 1$

$\left(\frac{d^2y}{dx^2}\right)^3 + 5\frac{dy}{dx} + 2y = k\left(\frac{d^3y}{dx^3}\right)^2$ $\quad 0 - 3$, $D - 2$

$\left(\frac{d^2y}{dx^2}\right)^2 + \sqrt{1+\left(\frac{dy}{dx}\right)^2} = 5y$ $\quad D \to 4$, $0 \to 2$

$\left(\frac{d^2y}{dx^2}\right)^3 + 5\cos\left(\frac{dy}{dx}\right) + 6y = 0$ $\quad D \to$ not ~~defined~~ determined, $0 \to 2$.

Formation of a Differential Eq^{n} :-

A diff eq^{n} can be formed by eliminating arbitary const. in the solution.

Ex:- $y^2 = 4ax$

$2y\frac{dy}{dx} = 4a$..

$y\frac{dy}{dx} = x \cdot 2\frac{dy}{dx}$

$2x\frac{dy}{dx} - y = 0$

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Q. $x^2+y^2=a^2$

$2x + 2y\frac{dy}{dx} = 0.$

$$\boxed{x + y\frac{dy}{dx} = 0.}$$

Note :- Differentiate only that much time upto three no. of arbitary constants.

Q. $y = ae^x + be^{-x}$

$\frac{dy}{dx} = ae^x - be^{-x}.$

$\frac{d^2y}{dx^2} = ae^x + be^{-x}$

$$\boxed{\frac{d^2y}{dx} = y}$$

Q. $y = e^x(a\cos x + b\sin x)$

$y_1 = e^x(a\cos x + b\sin x) + e^x(-a\sin x + b\cos x)$

$y_1 - y = e^x(-a\sin x + b\cos x)$

$y_2 - y_1 = \underbrace{e^x[-a\sin x + b\cos x]}_{y_1 - y} + \underbrace{e^x[-a\cos x + b\sin x]}_{-y}$

$y_2 - y_1 = y_1 - y - y$

$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 2y = 0$

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Solution of a Differential Eqⁿ :-

* Variable Seperable Method :-

① $\frac{dy}{dx} + \sqrt{\frac{1+y^2}{1+x^2}} = 0$

$$\frac{dy}{dx} = -\frac{\sqrt{1+y^2}}{\sqrt{1+x^2}}$$

$$\int \frac{dy}{\sqrt{1+y^2}} = \int -\frac{dx}{\sqrt{1+x^2}}$$

$$\Rightarrow \boxed{\sinh^{-1} y = -\sinh^{-1} x + c}$$

② $\frac{dy}{dx} = e^{x+y}$

$$\frac{dy}{dx} = e^x \cdot e^y$$

$$\int dy\, e^{-y} = \int e^x dx$$

$$-e^{-y} = e^x + c$$

$$\Rightarrow e^x + e^{-y} + c = 0$$

③ $\frac{dy}{dx} = x^2 e^{x-y}$

$$\int dy\, e^y = \int x^2 e^x dx$$

$$e^y = e^x [x^2 - 2x + 2] + c$$

e//

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④ $\frac{dy}{dx} + y^2 = 0$

$$\int dy\, y^{-2} = -\int dx$$

$$\frac{y^{-2+1}}{-1} = -x + c$$

$$\Rightarrow -y^{-1} = -x + c$$

$$\Rightarrow y^{-1} + x + c = 0 \quad \text{or} \quad \boxed{y = \frac{1}{x+c}}$$

⑤ $e^x \tan y\, dx + (1+e^x)\sec^2 y\, dy = 0$

$$e^x \tan y\, dx = -(1+e^x)\sec^2 y\, dy$$

$$\frac{e^x dx}{1+e^x} = -\frac{\sec^2 y\, dy}{\tan y}$$

$$\int \frac{e^x}{1+e^x} dx + \int \frac{\sec^2 y}{\tan y} dy = \int 0$$

$$\log(1+e^x) + \log \tan x = \log c$$

$$\Rightarrow \boxed{(1+e^x)\tan y = c}$$

⑥

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IInd Method

* Eqn Reducable to variable - seperable form :-

If the given diff. eqn is not directly variable seperable form, by making substitution. again we can transform and under variable seperable form.

Q. $\frac{dy}{dx} = (4x+y+1)^2$

↳ Let $z = 4x+y+1$

$\frac{dz}{dx} = 4 + \frac{dy}{dx}$

$4 + \frac{dy}{dx} = 4 + (4x+y+1)^2$

$\Rightarrow \frac{dz}{dx} = 4 + z^2$

$\Rightarrow \int \frac{dz}{z^2+4} = \int dx$

$\Rightarrow \frac{1}{2}\tan^{-1}\frac{z}{2} = x + c$

$\Rightarrow \boxed{\frac{1}{2}\tan^{-1}\left(\frac{4x+y+1}{2}\right) = x + c}$

Q. $1 + \frac{dy}{dx} = 1 + \sin(x+y) + \cos(x+y)$

$x+y = z$

$1 + \frac{dy}{dx} = \frac{dz}{dx}$

$\frac{dz}{dx} = 1 + \sin z + \cos z$

$\Rightarrow \int \frac{dz}{1+\sin z+\cos z} = \int dx$

Remember

WHENEVER INTEGRAL IS in form of $\int \frac{1}{a+b\sin x}dx$; $\int \frac{1}{a+b\cos x}$, $\int \frac{1}{a\cos x + b\sin x + c}dx$

Put $t = \tan\frac{x}{2} \Rightarrow dx = \frac{2dt}{1+t^2}$

$\sin x = \frac{2t}{1+t^2}$

$\cos x = \frac{1-t^2}{1+t^2}$

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$$\int \frac{1}{1+\frac{2t}{1+t^2}+\frac{1-t^2}{1+t^2}} \times \frac{2dt}{1+t^2} = x+c$$

$$\int \frac{2\,dt}{(1+t^2+2t+1-t^2)} = x+c$$

$$\int \frac{\cancel{2}}{\cancel{2}(1+t)}\,dt = x+c$$

$$\log(1+t) = x+c.$$

$$\log\left[1+\tan\frac{x+y}{2}\right] = x+c$$

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HOMOGENEOUS DIFFERENTIABLE EQUATION :-

A diff. eqⁿ of the form —

$\frac{dy}{dx} = \frac{f(x,y)}{g(x,y)}$ is said to be homogeneous differential eqⁿ, if both $f(x,y)$ and $g(x,y)$ are homogeneous function of the same degree.

$y = vx \Rightarrow v = \frac{y}{x}$

$$\boxed{\frac{dy}{dx} = v + x\frac{dv}{dx}}$$

When $\frac{dx}{dy} = \frac{f(x,y)}{g(x,y)}$

$x = vy \Rightarrow v = \frac{x}{y}$

$$\boxed{\frac{dx}{dy} = v + y\frac{dv}{dy}}$$

ILATE

Ex:- $\frac{dy}{dx} = \frac{x^2+y^2}{2xy}$

$\frac{dx}{dy} = \frac{2xy}{x^2+y}$

→ which form should be selected for Substitution?

↳ ~~जिसके~~ numerator में ज्यादा no. of terms होगा

$\therefore \frac{dxy}{dx} = \frac{x^2+y^2}{2xy}$

$v + x\frac{dv}{dx} = \frac{x^2 + v^2x^2}{2x \cdot vx}$

$x\frac{dv}{dx} = \frac{1+v^2}{2v} - v$

$x\frac{dv}{dx} = \frac{1-v^2}{2v}$

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$$\int \frac{-2v}{1-v^2}\,dv = \int -\frac{dx}{x}$$

$\frac{1}{x} \to \log cx$

$-\frac{1}{x} \to \log \frac{c}{x}$

$\frac{2}{x} \to \log cx^2$

$-\frac{2}{x} \to \log \frac{c}{x^2}$

$\frac{3}{x} \to \log cx^3$

$-\frac{3}{x} \to \log \frac{c}{x^3}$

$$\Rightarrow \log(1-v^2) = \log \frac{c}{x}$$

$$\Rightarrow 1 - \frac{y^2}{x^2} = \frac{c}{x}$$

$$\Rightarrow \frac{x^2-y^2}{x^2} = \frac{c}{x}$$

$$\Rightarrow \boxed{x^2-y^2 = cx}$$

Q. $x^2y\,dx - (x^3+y^3)dy = 0.$

$$\frac{dy}{dx} = \frac{x^2y}{x^3+y^3}$$

$$\Rightarrow \frac{dx}{dy} = \frac{x^3+y^3}{x^2y}$$

$$v + y\frac{dv}{dy} = \frac{v^3y^3 + y^3}{v^2y^3}$$

$$v + y\frac{dv}{dy} = \frac{v^3+1}{v^2}$$

$$y\frac{dv}{dy} = \frac{1}{v^2}$$

$$\Rightarrow \int \frac{dy}{y} = \int v^2\,dv$$

$$\Rightarrow \log cy = \frac{v^3}{3}.$$

$$\Rightarrow \log cy = \frac{x^3}{3y^3}$$ Ans.

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Q. $x\frac{dy}{dx} = y + x\tan\left(\frac{y}{x}\right)$

$$\frac{dy}{dx} = \frac{y}{x} + \tan\frac{y}{x}$$

$$v + x\frac{dv}{dx} = v + \tan v$$

$$\int \frac{dv}{\tan v} = \int \frac{dx}{x}$$

$$\log \sin v \cos v\, dv = \log cx$$

$$\boxed{\sin\left(\frac{y}{x}\right) = cx}$$ Ans

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Q. $(1+e^{x/y})dx + e^{x/y}\left(1-\frac{x}{y}\right)dy = 0$

Trig → Angle ज्यादा
Algebra → ज्यादा power numerator में
exponential $\frac{x}{y} - \frac{dx}{dy}$
$\frac{y}{x} - \frac{dx}{dy}$

$$\frac{dx}{dy} = \frac{e^{x/y}\left(\frac{x}{y}-1\right)}{(1+e^{x/y})}$$

$$v + y\frac{dv}{dy} = \frac{e^v(v-1)}{1+e^v}$$

$$y\frac{dv}{dy} = \frac{e^v(v-1)}{(1+e^v)} - v$$

$$y\frac{dv}{dy} = \frac{ve^v - e^v - v - ve^v}{(1+e^v)}$$

$$y\frac{dv}{dy} = -\frac{(v+e^v)}{(1+e^v)}$$

$$\int \frac{(1+e^v)}{v+e^v}\,dv = \int -\frac{dy}{y}$$

$$\log(v+e^v) = \log\frac{c}{y}$$

$$\Rightarrow \frac{x}{y} + e^{x/y} = \frac{c}{y} \Rightarrow \boxed{x + ye^{\frac{x}{y}} = c}$$ Ans

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NON HOMOGENEOUS DIFFERENTIAL FUNCTION :-

A diff. eqn of the form —

$$\frac{dy}{dx} = \frac{a_1x + b_1y + c_1}{a_2x + b_2y + c_2}$$ is said to be a non homogeneous differential eqn

Case-1 :- If $a_1b_2 - a_2b_1 \neq 0$, then the substitution is —

$$x = X + h$$
$$y = Y + K$$

where h, K are constants to be determined.

Case-2 :- If $a_1b_2 - a_2b_1 = 0$, then some part of numerator and denominator are equal and that part is considered as substitution.

Q. Find the substitution that transforms a non homogeneous diff eqn $\frac{dy}{dx} = \frac{2x + 2y - 2}{3x + y - 5}$ to homogeneous form.

$\hookrightarrow \neq 0 \rightarrow$ 1st form

→

$$\frac{dy}{dx} = \frac{2(X+h) + 2(Y+K) - 2}{3(X+h) + (Y+K) - 5}$$

h	K	1	h	K	
2	2	-2	2	2	→ write like this only
3	1	-5	3	1	→ numerator → Denominator

$$\frac{h}{-10+2} = \frac{k}{-6+10} = \frac{1}{2-6}$$

$$\frac{h}{-8} = \frac{k}{4} = \frac{1}{-4}$$

$h = 2$, $K = -1$ ✓ $\Rightarrow$ $x = X + 2$, $y = Y - 1$ ✓

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Q. Find Substitution ——

$$\frac{dy}{dx} = \frac{x+2y-3}{2x+y-3}$$

$\hookrightarrow \neq 0 \rightarrow 1^{st}$ form.

	h	k	1	h	k
1	2	−3	1	2	
2	1	−3	2	1	

$$\frac{h}{-6+3} = \frac{k}{-6+3} = \frac{1}{1-4}$$

$$\frac{h}{-3} = \frac{k}{-3} = \frac{1}{-3}$$

$h = 1, \quad k = 1$ ←

$$\boxed{\begin{aligned} x &= X+1 \\ y &= Y+1 \end{aligned}}$$ Ans.

Q. Find the Substitution ——

$$\frac{dy}{dx} = \frac{2x+3y+1}{4x+6y+1}$$

$\hookrightarrow = 0 \rightarrow 2^{nd}$ form

$$\frac{dy}{dx} = \frac{(2x+3y)+1}{2(2x+3y)+1}$$

$$\boxed{z = 2x+3y}$$

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Q. $\frac{dy}{dx} = \frac{x+2y+3}{y+2x+5}$

$\frac{dy}{dx} = \frac{x+2y+3}{2x+y+5}$

$\hookrightarrow \neq 0 \rightarrow$ & 1st form.

h	k	1	h	k.
1	2	3	1	2
2	1	5	2	|

$$\frac{h}{10-3} = \frac{k}{6-5} = \frac{1}{1-4}$$

$$\frac{h}{7} = \frac{k}{1} = \frac{1}{-3}$$

$h = -\frac{7}{3}$, $k = -\frac{1}{3}$

$$x = X + \frac{7}{3}$$
$$y = Y - \frac{1}{3}$$

Ans

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EXACT DIFFERENTIAL EQN :-

A diff. eqn of the form $M\,dx + N\,dy = 0$ is said to be exact if $\boxed{\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}}$

$$\text{Gen. Soln} \rightarrow \int^{x} M\,dx + \int (\text{terms of } N \text{ not containing } x)\,dy = c$$

Ex:- $(x^2 - ay)\,dx + (y^2 - ax)\,dy = 0$

$\frac{\partial M}{\partial y} = -a$, $\frac{\partial N}{\partial x} = -a$

exact.

$$\int (x^2 - ay)\,dx + \int y^2\,dy = c$$

$$\frac{x^3}{3} - axy + \frac{y^3}{3} = c$$

$$\Rightarrow x^3 + y^3 - 3axy = 3c \quad \text{Ans}$$

MORE Efficient Shortcut

GENERALIZATION :-

Ex:- $(ax + hy + g)\,dx + (hx + by + f)\,dy = 0$

$\frac{\partial M}{\partial y} = h$, $\frac{\partial N}{\partial x} = h$.

$$\boxed{\frac{ax^2}{2} + hxy + gx + \frac{by^2}{2} + fy = c} \quad \text{Ans}$$

Q. $e^x \tan y\,dx + (1 + e^x)\sec^2 y\,dy = 0$

$\frac{\partial M}{\partial y} = e^x \sec^2 y$, $\frac{\partial N}{\partial x} = e^x \sec^2 y$

exact.

Soln

$\therefore \quad e^x \tan y + \cdot \tan y = c$

$$\Rightarrow \boxed{(e^x + 1)\tan y = c}$$

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Q. $(x^2+y^2)dx + 2xy\,dy = 0$

$\frac{\partial M}{\partial y} = 2y \quad , \quad \frac{\partial N}{\partial x} = 2y$

exact

Soln

$$\frac{x^3}{3} + y^2 x = C$$

Q. $(1+e^{\frac{x}{y}})dx + e^{x/y}(1-\frac{x}{y})dy = 0$

exact

$\frac{\partial M}{\partial y} = e^{x/y} \cdot x \cdot (-\frac{1}{y^2})$

$\frac{\partial N}{\partial x} = e^{x/y}(\frac{-1}{y}) + (1-\frac{x}{y})e^{x/y}(\frac{1}{y})$

$= e^{x/y}(-\frac{1}{y} + \frac{1}{y} - \frac{x}{y^2})$

Soln.

$$x + y e^{\frac{x}{y}} = C$$

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Non - Exact Differential Eqⁿ :-

A diff eqⁿ $Mdx + Ndy = 0$ is said to be non exact, if $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$

METHOD-1 (Method to form a suitable Integrating factor)

IMPORTANT INTEGRATING FACTORS :-

***

① $x dy + y dx = d(xy)$

② $\frac{x dy - y dx}{x^2} = d\left(\frac{y}{x}\right)$

③ $\frac{x dy - y dx}{xy} = d\left(\log \frac{y}{x}\right)$

④ $\frac{x dy - y dx}{x^2+y^2} = d\left(\tan^{-1} \frac{y}{x}\right)$

⑤ $\frac{y dx - x dy}{y^2} = d\left(\frac{x}{y}\right)$

⑥ $\frac{y dx - x dy}{xy} = d\left(\log \frac{x}{y}\right)$

⑦ $\frac{y dx - x dy}{x^2+y^2} = d\left(\tan^{-1} \frac{x}{y}\right)$

⑧ $\frac{y e^x dx - e^x dy}{y^2} = d\left(\frac{e^x}{y}\right)$

⑨ $\frac{2xy dx - x^2 dy}{y^2} = d\left(\frac{x^2}{y}\right)$

⑩ $\frac{x dx + y dy}{x^2+y^2} = d\left[\frac{1}{2} \log (x^2+y^2)\right]$

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Q. $y(1+xy)dx + x(1-xy)dy = 0$.

$(y+xy^2)dx + (x-x^2y)dy = 0$.

$\frac{\partial M}{\partial y} = 1+2xy$, $\frac{\partial N}{\partial x} = 1-2xy$

Non exact

→ we can make it perfect I.F but considering 1st term carefully.

$y\,dx + xy^2dx + xdy - x^2y\,dy = 0$

Perfect I.F

$\frac{d(xy)}{(xy)^2} + \frac{xy[y\,dx - x\,dy]}{(xy)^2} = \frac{0}{(xy)^2}$

$\Rightarrow \int \frac{1}{(xy)^2}\,d(xy) + \int d\left(\log\frac{x}{y}\right) = 0$

$\Rightarrow \boxed{\frac{-1}{xy} + \log\frac{x}{y} = C}$ $\left\{\because \frac{dx}{x^2} = -\frac{1}{x}\right\}$

Q ① $y\,dx - x\,dy + (1+x^2)dx + x^2\sin y\,dy = 0$.

by taking $-$ common it is also integrable

$\frac{y\,dx - x\,dy}{x^2} + \frac{(1+x^2)\,dx}{x^2} + \frac{x^2\sin y\,dy}{x^2}$

$\Rightarrow -\int \frac{(x\,dy - y\,dx)}{x^2} + \int\left(\frac{1}{x^2}+1\right)dx + \int \sin y\,dy = \int 0$.

$\Rightarrow -\int d\left(\frac{y}{x}\right) + \int\left(\frac{1}{x^2}+1\right)dx + \int \sin y\,dy = \int 0$

$\Rightarrow \boxed{-\frac{y}{x} - \frac{1}{x} + x - \cos y = C}$ ⇐

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Q. $\int \frac{axy^2dx}{y^2} + \int \frac{ye^x dx - e^x dy}{y^2} = \int 0$

$\Rightarrow \int axdx + \int d\left(\frac{e^x}{y}\right) = \int 0$

$\Rightarrow \boxed{a\frac{x^2}{2} + \frac{e^x}{y} = c}$ ✓

Q. $(y^2e^x + 2xy)dx - x^2dy$.

$\Rightarrow \frac{y^2e^x dx}{y^2} + \frac{2xy\,dx - x^2dy}{y^2} = 0$

$\Rightarrow \int e^x dx + \int d\left(\frac{x^2}{y}\right) = \int 0$

$\Rightarrow \boxed{-e^x + \frac{x^2}{y} = c}$ ✓

Q. $\frac{2x^2y^2dx}{y^2} + \frac{ye^x dx - e^x dy}{y^2} - \frac{y^3dy}{y^2} = 0$

$\Rightarrow \int 2x^2dx + \int d\left(\frac{e^x}{y}\right) - \int y\,dy = \int 0$

$\Rightarrow \boxed{2\frac{x^3}{3} + \frac{e^x}{y} - \frac{y^2}{2} = c}$ ✓

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NON EXACT DIFFERENTIAL EQN :-
METHOD-2

If $Mdx + Ndy = 0$ is non exact, but homogeneous and if $Mx + Ny \neq 0$, then $\frac{1}{Mx+Ny}$ is an I.F.

After multiplying in diff. eqn → non exact becomes exact.
Then for soln will follow procedure of Exact diff. eqn.

→ Q. (1) $(x^2+y^2)dx - 2xydy = 0$ (Homogeneous, $d=2$, $d \neq 2$)

$$\frac{\partial M}{\partial y} = +2y \quad , \quad \frac{\partial N}{\partial x} = 2y$$

$$\therefore \text{I.F.} = \frac{1}{Mx+Ny} = \frac{1}{x(x^2+y^2) + y(-2xy)} = \frac{1}{x^3+xy^2-2xy^2}$$

$$= \frac{1}{x^3 - xy^2}$$

$$= \frac{1}{x(x^2-y^2)}$$

Now multiply by I.F —

$$\int \frac{x^2+y^2}{x(x^2-y^2)}dx - \frac{2xy}{x(x^2-y^2)}dy = 0.$$

$$\Rightarrow \int \left[\frac{2x}{x^2-y^2} - \frac{1}{x}\right]dx = \log c$$

$$\Rightarrow \log \frac{x^2-y^2}{x} = \log c$$

$$\therefore \boxed{x^2-y^2 = cx}$$

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Calculate I.F —

(A)

(1) $x^2y\,dx - (x^3+y^3)\,dy = 0.$

$\frac{\partial M}{\partial y} = 0 \quad Mx + Ny \neq 0$

$\therefore x^3y - x^3y + y^4 \neq 0$

$\therefore \frac{1}{Mx+Ny} = \frac{-1}{y^4}$

Note:-

In Exam option will be provided as $\frac{1}{y^4}$

Coz, (−) and const will not effect integration and in process of solving '−' or const can be taken out common and can be send to right where 0 is given.

(2) $(x^2y - xy^2)\,dx - (x^3 - 2x^2y)\,dy$

$Mx + Ny = x^3y - x^2y^2 - x^3y + 2x^2y^2$

$= x^2y^2$

$I.F = \frac{1}{Mx+Ny} = \frac{1}{x^2y^2}$

(3) $xy\,dx - (x^2+2y^2)\,dy = 0.$

$Mx + Ny = x^2y - x^2y - 2y^3$

$\therefore I.F = \frac{1}{Mx+Ny} = \frac{1}{-2y^3} \sim \frac{1}{y^3}$ Ans

no importance

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NON EXACT DIFFERENTIAL EQN :-

METHOD - 3 :-

If $Mdx + Ndy = 0$ is non exact, but is in the form of $y f(xy)dx + x g(xy)dy = 0$ and if $Mx - Ny \neq 0$

then $\frac{1}{Mx-Ny}$ is an I.F.

$\Rightarrow$ $y(1+xy)dx + x(1-xy) = 0$

$Mx - Ny = xy + x^2y^2$

$\therefore \frac{1}{Mx-Ny} = \frac{1}{xy(1+xy)}$

Q ① $y(x^2y^2 + xy + 1)dx + x(x^2y^2 - xy + 1)dy = 0$

for fast calculation of $\frac{1}{Mx-Ny}$ — cancel x & y terms having same order and some const. and increase power or multiply by $\frac{1}{0} \times \frac{1}{xy}$

$\therefore \frac{1}{Mx-Ny} = \frac{1}{x^2y^2}$

② $y(x^3y^3 + x^2y^2 + xy + 1)dx + x(x^3y^3 - x^2y^2 - xy + 1)dy = 0$

$\frac{1}{Mx-Ny} = \frac{1}{x^3y^3 + x^2y^2}$

③ $y(x^2y^2 + 2)dx + x(2 - 2x^2y^2)dy = 0$

$\frac{1}{Mx-Ny} = \frac{1}{x^3y^3}$.

④ $y[xy\sin xy + \cos xy]dx + x[x\sin xy - \cos xy]dy = 0$

$\frac{1}{xy\cos xy}$

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NON EXACT DIFFERENTIAL EQ^N :-

METHOD-4 :-

If $Mdx + Ndy = 0$ is non exact and if

$$\frac{1}{N}\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right) = f(x)$$

Tip: N से divisible तो ∂N पहले नहीं लिखना है

$$I.F = e^{\int f(x)\,dx}$$

Tip: M से divisible तो ∂M पहले नहीं लिखना है

$$\frac{1}{M}\left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) = g(y)$$

$$I.F = e^{\int g(y)\,dy}.$$

fn	I.F
If $f(x) = \frac{1}{x}$	x
$\frac{1}{x}$	$\frac{1}{x}$
$\frac{2}{x}$	x^2
$-\frac{2}{x}$	$\frac{1}{x^2}$
$\frac{3}{x}$	x^3
$-\frac{3}{x}$	$\frac{1}{x^3}$

Q. $(x^2+y^2+1)\,dx - 2xy\,dy = 0$.

$\frac{\partial M}{\partial y} = 2y$, $\frac{\partial N}{\partial x} = -2y$. → 1 term ∴ divisible by N

$$\frac{1}{N}\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right)$$

$$= \frac{1}{-2xy}\left(2y + 2y\right)$$

$$= \frac{4y}{-2xy} = \frac{-2}{x} \rightarrow I.F = \frac{1}{x^2}$$

Q. (a) $(y + xy^3)\,dx + 2(x^2y^2 + x + y^4)\,dy = 0$.

$\frac{\partial M}{\partial y} = 1 + 3xy^2$ $\quad$ $\frac{\partial N}{\partial x} = 2y^2 + 1 + 0$

2 terms ∴ divisible by M

$$\frac{1}{M}\left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right)$$

$$= \frac{1}{(y+xy^3)}\left[4xy^2 + 2 + 3xy^2\right] = \frac{1}{y} \rightarrow I.F = y$$

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(2) $(y^4+xy)\,dx + (xy^3+2y^4-4x)\,dx = 0$

$\frac{\partial M}{\partial y} = 4y^3+2 \qquad \frac{\partial N}{\partial x} = y^3-4$

2 terms → ∴ divisible by M as it M has also 2 terms

$\frac{1}{M}\left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right)$

$= \frac{1}{y^4+xy}\left(y^3-4-4y^3-2\right)$

$= \frac{1}{y^4+xy}\left(-3y^3-6\right)$

$= \frac{-3}{y} \rightarrow I.F = \frac{1}{y^3}$

(3) $\left(y+\frac{y^3}{3}+\frac{x^2}{2}\right)dx + \left(\frac{x+xy^2}{4}\right)dy = 0$

$\frac{\partial M}{\partial y} = 1+\frac{xy^2}{3} \qquad \frac{\partial N}{\partial x} = \frac{1}{4}+\frac{y^2}{4}$

$\frac{1}{N}\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right)$

$= \frac{4}{x+xy^2}\left(1+y^2-\frac{1}{4}-\frac{y^2}{4}\right)$

$= \frac{4}{x(1+y^2)}\,3\left(\frac{1}{3}+\frac{1}{3}y^2\right)$

$= \frac{3}{x} \rightarrow I.F = x^3$

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Q. $(3x^2y^4 + 2xy)\,dx + (2x^3y^3 - x^2)\,dy = 0.$

$$\frac{\partial M}{\partial y} = 12x^2y^3 + 2x$$

$$\frac{\partial N}{\partial x} = 6x^2y^3 - 2x$$

बीच में + sign आ रहा तो divisible by M not 'N'

$$\frac{1}{M}\left(\frac{\partial M}{\partial x} - \frac{\partial M}{\partial y}\right)$$

$$= \frac{1}{3x^2y^4 + 2xy}\left(-12x^2y^3 + 4x\right)$$

$$= \frac{-2}{y} \longrightarrow I.F \rightarrow \frac{1}{y^2}$$

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Linear Eqⁿ:-

$\frac{dy}{dx} + Py = Q$, where P and Q are said to be fn of x, called a Linear eqⁿ.

$I.F = e^{\int P dx}$

General Solⁿ—

$$y e^{\int P dx} = \int (Q \cdot e^{\int P dx}) dx + c$$

(I.F) (I.F)

Ex:-

(1)

$\frac{dx}{dy} + Px = Q$, where P and Q are said to fn. of y

$I.F = e^{\int P dy}$

General Solⁿ—

$$x e^{\int P dy} = \int (Q\ e^{\int P dy}) dy + c$$

(1) $\frac{dy}{dx} + \frac{y}{x} = \frac{\log x}{x}$

$P = \frac{1}{x}$

$Q = \frac{\log x}{x}$

$I.F = e^{\int P dx} = e^{\int \frac{1}{x} dx} = x$

Solⁿ—

$y \cdot x = \int \log x\, dx + c$

$\Rightarrow yx = x \log x - x + c$ Ans

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(2) $(2-x^2)\frac{dy}{dx} - 2xy = x^2$

$\frac{dy}{dx} = \frac{2xy}{2-x^2} = \frac{x^2}{2-x^2}$ (P, Q)

Now

$I.F = e^{\int P dx} = e^{\int \frac{2x}{2-x^2} dx} = e^{\log(2-x^2)}$

$= 2-x^2$

G. soln –

$y(2-x^2) = \int \frac{x^2}{2-x^2}(2-x^2)\cdot dx + C$

$= \frac{x^3}{3} + C.$

(3) $\frac{dy}{dx} + 2xy = e^{-x^2}$ (P, Q), $y(0) = 1$

$I.F = e^{\int P dx} = e^{\int 2x\,dx} = e^{\frac{2x^2}{2}} = e^{x^2}$

G. soln –

$y e^{x^2} = \int e^{-x^2} \cdot e^{x^2} dx + C$

$\boxed{y e^{x^2} = x + C}$

When $y(0) = 1$

$\therefore 1 \cdot e^0 = C \Rightarrow C = 1$

$\therefore \boxed{y e^{x^2} = x + 1}$ ←

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④ $\frac{dy}{dx} + \underbrace{\frac{1}{x\log x}}_{P} y = \underbrace{\frac{2\log x}{x\log x}}_{Q}$

$I.F = e^{\int \frac{1}{x\log x} dx} = e^{\int \frac{1}{\log x} \cdot dx} = e^{\log(\log x)} = \log x.$

G. Soln

$y \log x = \int \frac{2\log x}{x \log x} \times \log x \, dx + C$

$y \log x = \frac{(\log x)^2}{2} + C$

$y \log x = 2\int.$

$\boxed{y = \log x + \frac{C}{\log x}}$

⑤ $(x + 2y^3)\frac{dy}{dx} = y.$

$\frac{dx}{dy} = \frac{x}{y} + \frac{2y^3}{y}$

$\frac{dx}{dy} - \underbrace{\frac{1}{y}}_{P} x = \underbrace{2y^2}_{Q}$

$I.F = e^{\int -\frac{1}{y} dy} = \frac{1}{y}$

G. Soln

$x \cdot \frac{1}{y} = \int 2y^2 \times \frac{1}{y} dy + C.$

$\frac{x}{y} = y^2 + C$

$\Rightarrow \boxed{x = y^3 + Cy}$

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Q ⑤ $(1+y^2)dx + (x - e^{\tan^{-1}y})dy = 0$

$$(1+y^2)dx = (e^{\tan^{-1}y} - x)dy$$

$$\Rightarrow \frac{dx}{dy} = \frac{e^{\tan^{-1}y}}{1+y^2} - \frac{x}{1+y^2}$$

$$\Rightarrow \frac{dx}{dy} + \frac{x}{1+y^2} = \frac{e^{\tan^{-1}y}}{1+y^2}$$

$$\therefore I.F = e^{\int P\,dy} = e^{\int \frac{1}{1+y^2}dy} = e^{\tan^{-1}y}$$

General Soln —

$$x \cdot e^{\tan^{-1}y} = \int \frac{e^{\tan^{-1}y}}{1+y^2} \times e^{\tan^{-1}y}\,dy + c.$$

$$= \frac{1}{2}\int \frac{e^{2\tan^{-1}y}}{1+y^2} \times 2\,dy + c$$

$$\boxed{x\,e^{\tan^{-1}y} = \frac{1}{2}e^{2\tan^{-1}y} + c.}$$

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BERNOULLI'S EQN :-

A diff. eqn of the form $\frac{dy}{dx} + Py = Qy^n$, where P and Q are fn. of x is said to be a bernaulli's eqn

$$\therefore \quad y^{-n}\frac{dy}{dx} + P y^{1-n} = Q$$

$$\Rightarrow \quad z = y^{1-n}$$

$$\frac{dz}{dx} = (1-n)y^{-n}\frac{dy}{dx}$$

$$y^{-n}\frac{dy}{dx} = \frac{1}{(1-n)}\frac{dz}{dx}$$

$$\frac{1}{(1-n)}\frac{dz}{dx} + Pz = Q.$$

$$\Rightarrow \frac{dz}{dx} + P(1-n)z = Q(1-n)$$

* $I.F = e^{\int P(1-n)dx}$

Gen. soln.

* $y^{1-n} \cdot e^{\int P(1-n)dx} = \int \left(Q(1-n)\, e^{\int P(1-n)dx} \right) dx + C$

($e^{\int P(1-n)dx}$: I.F)

(a) $\frac{dx}{dy} + Px = Q^n$

$I.F = e^{\int P(1-n)dy}$

Gen soln

$x^{1-n} \times I.F = \int (Q(1-n) \times I.F)\, dy + C$

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Find I.F.

① $\frac{dy}{dx} + \frac{y}{x} = x^2y^3$ (P, Q)

$I.F = e^{\int \frac{1}{x}(1-n)\,dx} = e^{(1-n)\log x} = x^{1-n} = x^{-2} = \frac{1}{x^2}$

② $\frac{dy}{dx}(xy + x^2y^3) = 1$

$\Rightarrow \frac{dx}{dy} - xy = y^3$ (P, Q)

$I.F = e^{\int \frac{x}{y}(1-n)\,dy} = e^{\frac{x^2}{2}\times -1} = e^{-x^2}$

$\Rightarrow \frac{dx}{dy} - xy = y^3x^{2\to n}$ (P, Q)

$I.F = e^{\int -y(1-n)\,dy} = e^{\int -y(-2)\,dy} = e^{\int +y\,dy} = e^{\frac{y^2}{2}}$

Differential Eqn with constant coefficient :—

A diff eqn of the form —

$$a_0\frac{d^ny}{dx^n} + a_1\frac{d^{n-1}y}{dx^{n-1}} + a_2\frac{d^{n-2}y}{dx^{n-2}} + \dots + a_ny = Q.$$

Where $a_0, a_1, a_2 \dots a_n$ are constants and Q be the function of x is said to be differential equation with constant coefficient.

Let $\frac{d}{dx} = D$

$$(a_0D^n + a_1D^{n-1} + a_2D^{n-2} + \dots + a_n)y = Q.$$

$$\boxed{f(D)\,y = Q}$$

This eqn may contain two types of sol^n

→ one is called complementary fn

→ other is called particular integral.

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Note:-

If $Q = 0$, then complementary fn is called the general soln.

If $Q \neq 0$, then only particular integral must be existed.

To Find Complimentary fn (C.F) :-

Replace 'D' by m, $f(m) = 0$ is called auxilary eqn

$$a_0 m^n + a_1 m^{n-1} + a_2 m^{n-2} + \dots + a_n = 0.$$

Case-1:

Roots are real and Unequal.

Let $m = \alpha, \beta, \gamma \dots$

$$\therefore y_c = c_1 e^{\alpha x} + c_2 e^{\beta x} + c_3 e^{\gamma x} + \dots$$

Case-2

Roots are real and equal.

Let $m = \alpha, \alpha, \beta$

$$y_c = (c_1 + c_2 x) e^{\alpha x} + c_3 e^{\beta x}$$

Let $m = \alpha, \alpha, \alpha, \beta$.

$$y_c = (c_1 + c_2 x + c_3 x^2) e^{\alpha x} + c_4 e^{\beta x}$$

Let $m = \alpha, \alpha, \beta, \beta$

$$y_c = (c_1 + c_2 x) e^{\alpha x} + (c_3 + c_4 x) e^{\beta x}$$

Ex:-

$$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 0.$$

$$m^2 - 4m + 4 = 0$$

$$(m-2)^2 = 0$$

$$m = 2, 2$$

$$y = (c_1 + c_2 x) e^{2x}$$

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Case-3

Roots are complex —

Let $m = \alpha \pm i\beta$

$$y_c = [c_1 \cos\beta x + c_2 \sin\beta x] e^{\alpha x}$$

Let $m = \alpha \pm i\beta, \alpha \pm i\beta$.

$$y_c = [(c_1 + c_2 x)\cos\beta x + (c_3 + c_4 x)\sin\beta x] e^{\alpha x}$$

Ex:-

$$\frac{d^2y}{dx^2} - \frac{2dy}{dx} + 4y = 0$$

$$m^2 - 2m + 4 = 0$$

$$m = \frac{+2 \pm \sqrt{4-16}}{2}$$

$$= \frac{+2 \pm 2\sqrt{-3}}{2}$$

$$= 1 \pm i\sqrt{3}$$

$$= 1 \pm \sqrt{3}\, i$$

$$\therefore y = [c_1 \cos\sqrt{3}x + c_2 \sin\sqrt{3}x] e^{x}$$

($\alpha = 1$, $\beta = \sqrt{3}$)

To Find the roots of Eqn:-

For 3rd degree

i). Every 3rd degree eqn must contain at least 1 real root.

ii). If sum of all the coeff = 0, then $(m=1)$ satisfies eqn
↳ one root
or $(m-1)$ is a factor.

iii). If sum of coeff. of even powers = sum of coeff of odd powers then $(m=-1)$ satisfies the eqn. or $(m+1)$ will be factor.
↳ one root

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① $(D^3 - 3D + 2)y = 0$

$m^3 - 3m + 2 = 0$

↳ Sum = 0 → $m = 1$ is root ∴

Coeff.

root		1	0	−3	2
$m=1$			1	1	−2
	Sum→	1	1	−2	0
0 always	$m=1$		1	1	2
		1	2	0	

→ m^3

→ m^2

→ m

$m + 2 = 0$

$m = -2$

∴ $m = 1, 1$ & -2 are roots.

② $[D^3 - 4D^2 + 5D - 2]y = 0.$

$m^3 - 4m^2 + 5m - 2 = 0$

$1 - 4 + 5 - 2 = 0.$

		1	−4	5	−2
$m=1$			1	−3	2
		1	−3	2	
$m=1$			1	−2	
		1	−2	0	

$m - 2 = 0$

$m = 2$

∴ $m = 1, 1, 2$ ∆

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② $[D^4 - 2D^3 + 2D - 1]\, y = 0$.

$m^4 - 2m^3 + 2m - 1 = 0$

$1 - 2 + 2 - 1 = 0$.

$\therefore m = 1$ ✓

$m=1$	1	−2	0	2	−1
	0	1	−1	−1	1
$m=1$	1	−1	−1	1	0
	0	1	0	−1	
$m=-1$	1	0	−1	0	
	0	1	1		
	1	1	0		

$m + 1 = 0$

$m = -1$.

③ $[D^3 + 6D^2 + 11D + 6]\, y = 0$

$m^3 + 6m^2 + 11m + 6 = 0$.

12 = 12 → $m = -1$

$m=-1$	1	6	11	6
	0	−1	−5	−6
	1	5	6	0

$m^2 + 5m + 6 = 0$.

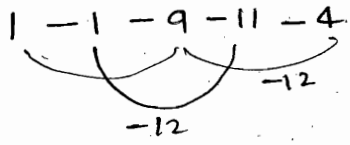
$(m+3)(m+2) = -3, -2$

$m = -1, -2, -3$

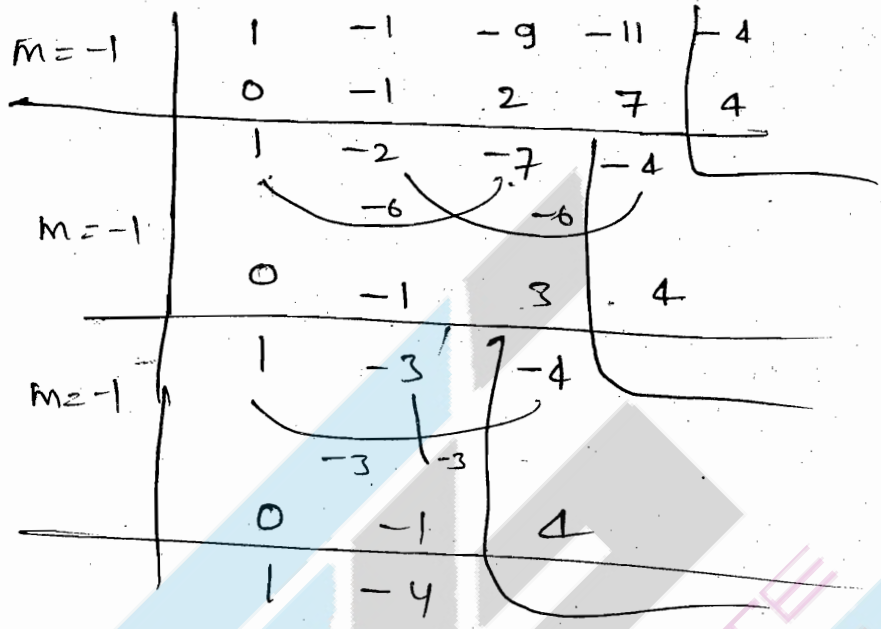
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④ $(D^4 - D^3 - 9D^2 - 11D - 4)\, y = 0$

$m^4 - m^3 - 9m^2 - 11m - 4 = 0.$



$\Rightarrow m = -1$



$m - 4 = 0$

$m = 4$

$\therefore m = -1, -1, -1, 4$

$y = (C_1 + C_2 x + C_3 x^2) e^{-x} + C_4 e^{4x}$

⑤ $(D^2 - 3D + 4)\, y = 0$

$m^2 - 3m + 4 = 0.$

$m = \dfrac{3 \pm \sqrt{9 - 16}}{2}$

$= \dfrac{3 \pm \sqrt{7}\, i}{2}$

$y = \left(C_1 \cos\frac{\sqrt{7}}{2}x + C_2 \sin\frac{\sqrt{7}}{2}x\right) e^{\frac{3}{2}x}$

⑥ $[D^2 - (a+b)D + ab]\, y$

$m^2 - (a+b) + ab = 0.$

$(m-a)(m-b) = 0$

$m = a, b.$

$\therefore \boxed{y = C_1 e^{ax} + C_2 e^{bx}}$

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To find the particular Integral:-

$$F(D)y = \theta$$

then $y_p = PI = \frac{1}{F(D)} Q$

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Case-III. To find P.I of the form $\frac{1}{f(D)}\sin ax$, $\frac{1}{f(D)}\cos ax$, replace D^2 by $-a^2$ if $\phi(-a^2) \neq 0$.

$$\frac{1}{D^2-2D+5}\sin 2x = \frac{1}{-4-2D+5}\sin 2x = \frac{1}{1-2D} \times \frac{1+2D}{1+2D}\sin 2x$$

$$= \frac{1+2D}{1-4D^2}\sin 2x = \frac{(1+2D)}{1-4(-4)}\sin 2x$$

$$= \frac{\sin 2x + 4\cos 2x}{17}.$$

Case-IV. To find P.I of the form $\frac{1}{D^2+a^2}\sin ax$, $\frac{1}{D^2+a^2}\cos ax$,

$$\frac{1}{D^2+a^2}\sin ax = \frac{x}{2D}\sin ax = \frac{x}{2}\left(\frac{-\cos ax}{a}\right)$$

$$\therefore \boxed{\begin{aligned} \frac{1}{D^2+a^2}\sin ax &= \frac{-x}{2a}\cos ax \\ \frac{1}{D^2+a^2}\cos ax &= \frac{x}{2a}\sin ax \end{aligned}}$$

$$\frac{1}{D^2+4}\sin 2x = \frac{-x}{4}\cos 2x$$

$$\frac{1}{D^2+9}\cos 3x = \frac{x}{6}\sin 3x$$

$$\frac{1}{D^2+16}\sin 4x = -\frac{x}{8}\cos 4x$$

$$\frac{1}{D^2+1}\cos 3x = \frac{1}{-9+6}\cos 3x = \frac{1}{8}\cos 3x$$

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① $(D^2 - D - 2)y = \cos 2x$

② $(D^2 - 4)y = \sin^2 x$

③ $(D^2 + 1)y = \sin 2x \cdot \cos x$

→ ① $\frac{1}{D^2 - D - 2} \cos 2x = \frac{1}{-4 - D - 2} \cos 2x = \frac{1}{-6 + D} \times \frac{-6 + D}{-6 + D} \cos 2x$

$= \frac{-6 + D}{36 - D^2} \cos 2x$

$= \left[\frac{(-6 + D)}{36 - (-4)} \cos 2x\right]$

$= \frac{-6 \cos 2x - 2 \sin 2x}{40}$

$= -\frac{2}{40} (3 \cos 2x + \sin 2x)$

$= -\frac{1}{20} (3 \cos 2x + \sin 2x).$

② $\frac{1}{D^2 - 4} \sin^2 x = \frac{1}{D^2 - 4}\left[\frac{1 - \cos 2x}{2}\right] = \frac{1}{2}\left[\frac{1}{D^2 - 4}(1) - \frac{1}{D^2 - 4}(\cos 2x)\right]$

$= \frac{1}{D^2 - 4} e^{0x}$

$= \frac{1}{D^2 - D}(5) = \frac{x}{2D - 1}(5)$

$= \frac{1}{2}\left[\frac{1}{0 - 4} \cdot 1 - \frac{1}{-4 - 4} \cos 2x\right]$

$= -\frac{1}{8} + \frac{1}{16} \cos 2x$

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③ $(D^2+1)y = \sin 2x \cdot \cos x$

$$= \frac{1}{2(D^2+1)} \; 2\sin 2x \cdot \cos x$$

$$= \frac{1}{2}\left[\frac{1}{D^2+1}\sin 3x + \frac{1}{D^2+1}\sin x\right]$$

$$= \frac{1}{2}\left[\frac{1}{-9+1}\sin 3x - \frac{x}{2(1)}\cos x\right]$$

$$= -\frac{1}{16}\sin 3x - \frac{x}{4}\cos x$$

SHORTCUT

→ If $(D^2+a^2)y = \tan ax$

$$y_p = \frac{-1}{a^2}\cos ax \cdot \log\left(\tan\left(\frac{\pi}{4} + \frac{ax}{2}\right)\right)$$

→ If $(D^2+a^2)y = \sec ax$

$$y_p = \frac{x}{a}\sin ax + \frac{1}{a^2}\cos ax \log(\cos ax)$$

ex:- $(D^2+4)y = \tan 2x$

$$y_p = \frac{-1}{4}\cos 2x \cdot \log\left[\tan\left(\frac{\pi}{4} + x\right)\right]$$

ex:- $(D^2+9)y = \sec 3x$

$$y_p = \frac{x}{3}\sin 3x + \frac{1}{9}\cos 3x \cdot \log(\cos 3x)$$

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Case-V

To find the P.I of the form $\frac{1}{f(D)}(x^m)$.

$(1+D)^{-1} = 1-D+D^2-D^3+\cdots$

$(1-D)^{-1} = 1+D+D^2+D^3+\cdots$

$(1+D)^{-2} = 1-2D+3D^2-4D^3+\cdots$

$(1-D)^{-2} = 1+2D+3D^2+4D^3+\cdots$

Remember

$D^3(x^2) = 0$

$D^n(x^n) = n!$

$D^{n+1}(x^n) = 0$

① $(D^2-4)y = x^2$

$\frac{1}{D^2-4}x^2$

$= \frac{1}{-4\left[1-\frac{D^2}{4}\right]}x^2$

$= -\frac{1}{4}\left[1-\frac{D^2}{4}\right]^{-1}x^2$

$= -\frac{1}{4}\left[1+\frac{D^2}{4}+\left(\frac{D^2}{4}\right)^2+\cdots\right]x^2$ (the $\left(\frac{D^2}{4}\right)^2$ term $\to 0$)

$= -\frac{1}{4}\left[x^2+\frac{2!}{4}\right]$

$= -\frac{1}{4}\left[x^2+\frac{1}{2}\right]$

② $[D^3+8]y = x^4+2x+1$

$= \frac{1}{D^3+8}(x^4+2x+1)$

$= \frac{1}{8\left[1+\frac{D^3}{8}\right]}(x^4+2x+1)$

$= \frac{1}{8}\left[1+\frac{D^3}{8}\right]^{-1}(x^4+2x+1)$

$= \frac{1}{8}\left[1-\frac{D^3}{8}+\frac{D^6}{8}+\cdots\right](x^4+2x+1)$ ($\frac{D^6}{8} \to 0$)

$= \frac{1}{8}\left[(x^4+2x+1)-\frac{1}{8}(24x+0+0)\right]$

$= \frac{1}{8}(x^4-x+1)$

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③ $(D^2+2D+1)y = x^2+x$

$\Rightarrow \frac{1}{(D+1)^2}(x^2+x)$

$= [1+D]^{-2}(x^2+x)$

$= [1-2D+3D^2](x^2+x)$

$= (x^2+x) - 2(2x+1) + 3(2+0)$

$= x^2+x-4x-2+6$

$= x^2-3x+4$

Case-VI. To find PI of the form $\frac{1}{f(D)} e^{ax} \cdot v$, where v is a fn. of x.

$$\frac{1}{f(D)} e^{ax} \cdot v = e^{ax} \frac{1}{f(D+a)} \cdot v$$

① $(D^2-2D+1)y = x^2e^x$

② $(D^2-2D+5)y = e^x \sin x$

③ $(D^2-4D+4)y = e^{2x}\cos 3x$

④ $(D^3+3D^2+3D+1)y = x^2e^{-x}$

→ ① $\frac{1}{(D-1)^2} e^x x^2 = e^x \cdot \frac{1}{(D+1-1)^2} x^2 = e^x \left[\frac{1}{D^2} x^2\right]$

(D represents integration)

$= e^x \frac{x^4}{12}$

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(2) $\frac{1}{D^2-2D+5}(e^x \sin x)$

$= e^x \left[\frac{1}{(D+1)^2-2(D+1)+5} \sin x \right]$

$= e^x \left[\frac{1}{D^2+1+2D-2D-2+5} \sin x \right]$

$= e^x \left[\frac{1}{D^2+4} \sin x \right]$

$= e^x \left[\frac{1}{-1+4} \sin x \right]$

$= \frac{e^x \sin x}{3}$.

(3) $\frac{1}{(D^2-4D+4)} e^{2x} \cos 3x$

$= e^{2x} \left[\frac{1}{(D+2-2)^2} \cos 3x \right]$

$= e^{2x} \left[\frac{1}{D^2} \cos 3x \right]$

$= e^{2x} \frac{1}{-9} \cos 3x$

$= \frac{-e^{2x}}{9} \cos 3x$

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(f) $\frac{1}{(D^3+3D^2+3D+1)} e^{-x} x^2$

$= \frac{1}{(D+1)^3} e^{-x} x^2$

$= e^{-x} \frac{1}{(D-1+1)^3} x^2$

$= e^{-x} \left[\frac{1}{D^3} x^2\right]$

$= e^{-x} \left(\frac{x^5}{60}\right)$ Ans

$\int\int\int x^2 \cdot \frac{x^3}{3} \cdot \frac{x^4}{12} \cdot \frac{x^5}{60}$

Case-VII To find the P.I of the form $\frac{1}{f(D)} x \cdot v$, where v is a fn of x (Trigonometric fn)

$$\frac{1}{f(D)} x \cdot v = x \cdot \frac{1}{f(D)} \cdot v - \frac{f'(D)}{[f(D)]^2} \cdot v$$

(1) $(D^2+4)y = x\cos x$

(2) $(D^2-2D+1)y = xe^x \sin x$

(1) $\frac{1}{D^2+4} (x\cos x)$

$= x \cdot \frac{1}{(D^2+4)} \cos x - \frac{2D}{(D^2+4)^2} \cdot \cos x$

$= x \cdot \frac{1}{-1+4} \cos x - \frac{2(-\sin x)}{(-1+4)^2}$

$= \frac{x}{3}\cos x + \frac{2}{9}\sin x$

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② $\frac{1}{(D-1)^2} e^x \cdot x \cdot \sin x$

$= e^x \left[\frac{1}{(D+1-1)^2} x \sin x \right]$

$= e^x \left[\frac{1}{D^2} x \sin x \right]$

$= e^x \left[x \frac{1}{-a^2} \sin x - \frac{2D}{(D^2)^2} \cdot \sin x \right]$

$= e^x \left[x \frac{1}{-1} \sin x - \frac{2\cos x}{(-1)^2} \right]$

$= e^x (-x \sin x - 2\cos x)$ —

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Differential Eqⁿ with variable co-efficients :-

or

Cauchy Euler form :-

→ A diff eqⁿ is of the form —

$$a_0 x^n \frac{d^n y}{dx^n} + a_1 x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + a_2 x^{n-2} \frac{d^{n-2} y}{dx^{n-2}} + \cdots + a_n y = Q$$

where $a_0, a_1, a_2 \cdots a_n$ are constants and Q be the function of x.

is said to be Diffⁿ Eqⁿ with Variable Coefficients.

$x = e^z \Rightarrow z = \log x$.

$$\frac{dz}{dx} = \frac{1}{x} \quad \& \quad \frac{dx}{dz} = x$$

$$\frac{d}{dz} = \frac{d}{dx} \cdot \frac{dx}{dz} = x \frac{d}{dx} = xD = \frac{d}{dz} = \theta$$

Now —

$$\frac{d}{dz}\left(x \frac{d}{dx}\right) = \frac{d^2}{dz^2}$$

$$\frac{d}{dx}\left(x \frac{d}{dx}\right) \cdot \frac{dx}{dz} = \theta^2$$

$$x\left(x \frac{d^2}{dx^2} + \frac{d}{dx} \cdot 1\right) = \theta^2$$

$$\therefore x^2 \frac{d^2}{dx^2} + x \frac{dx}{dx} = \theta^2$$

$$x^2 \frac{d^2}{dx^2} + \theta = \theta^2$$

$$x^2 \frac{d^2}{dx^2} = \theta(\theta - 1)$$

$$\therefore x^2 \frac{d^2}{dx^2} = x^2 D^2 = \theta(\theta - 1)$$

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Imp Substitution to Remember

Let $x\frac{d}{dx} = xD = \theta$.

$x^2\frac{d^2}{dx^2} = x^2D^2 = \theta(\theta-1) = \theta^2-\theta$

$x^3\frac{d^3}{dx^3} = x^3D^3 = \theta(\theta-1)(\theta-2) = \theta^3-3\theta^2+2\theta$

Variable Coeff.

① $(x^2D^2 - xD - 3]y = 0$

$= [\theta^2-\theta-\theta-3]y = 0$

$= m^2-2m-3 = 0$

$= (m-3)(m+1) = 0$.

$m = 3, -1$

$y = c_1e^{3z} + c_2e^{-z}$

$= c_1(e^z)^3 + c_2(e^z)^{-1}$

$\boxed{y = c_1x^3 + c_2x^{-1}}$

observe the diff^n bet^n const coeff approach

$D^2-D+2 = 0$

$m^2-m+2 = 0$ ——

② $[x^3D^3 + 3x^2D^2 - 2xD + 2]y = 0$.

$[\theta^3 - 3\theta^2 + 2\theta + 3\theta^2 - 3\theta - 2\theta + 2]y = 0$

$[\theta^3 - 3\theta + 2]y = 0$

$m^3 - 3m + 2 = 0$

~~$m = x, y = x^2, z = x^3$~~ $y - 3 + 2 = 0$

$\therefore m = 1$ satisfies.

	1	0	−3	2
m=1		0	1	1	−2
	1	1	−2	0
m=1		0	1	1
	1	2	0	

$m+2=0 \Rightarrow m = -2$.

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$\therefore y = (c_1 + c_2 z)e^{z} + c_3 e^{-2z}$

$\Rightarrow \boxed{y = (c_1 + c_2 \log x)x + c_3 x^{-2}}$

(3) $[x^2D^2 + xD - 4]y = 0$, $x(0) = 0$, $y(1) = 1$

$[\theta^2 - \theta + \theta - 4]y = 0$

$m^2 - 4 = 0$

$m = \pm\sqrt{4} = \pm 2 = 2, -2$

$\therefore y = c_1 e^{2z} + c_2 e^{-2z}$

$\Rightarrow y = c_1 x^2 + c_2 x^{-2}$

$\Rightarrow y = c_1 x^2 + \frac{c_2}{x^2}$

$\Rightarrow x^2 y = c_1 x^4 + c_2$

$\because x(0) = 0$

$\Rightarrow 0.0 = c_1(0) + c_2$

$\Rightarrow \boxed{c_2 = 0}$

$y(1) = 1$

$\Rightarrow 1^2 \cdot 1 = c_1(1)^4 + c_2$

$1 = c_1 + 0$

$\boxed{c_1 = 1}$

$\therefore y = 1(x^2) + \frac{0}{x^2}$

$\Rightarrow \boxed{y = x^2}$

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Q. $\left[x^4D^3 + 2x^3D^2 - x^2D + x\right] y = 1$

$\Rightarrow (x^3D^3 + 2x^2D^2 - xD + 1)\, y = \frac{1}{x}$

$\Rightarrow (\theta^3 - 3\theta^2 + 2\theta + 2\theta^2 - 2\theta - \theta + 1)\, y = e^{-z}$

$\Rightarrow (\theta^3 - \theta^2 - \theta + 1) y = e^{-z}$

$m^3 - m^2 - m + 1 = 0$

$m^2(m-1) - 1(m-1) = 0$

$(m-1)(m^2-1) = 0$

$(m-1)(m-1)(m+1) = 0$

$m = 1, 1, -1$

$\therefore y_c = (c_1 + c_2 z)e^z + c_3 e^{-z}$

$\Rightarrow y_c = (c_1 + c_2 \log x)\, x + c_3 x^{-1}$

$\Rightarrow y_p = \frac{1}{\theta^3 - \theta^2 - \theta + 1} e^{-z}$

$= \frac{z}{3\theta^2 - 2\theta - 1} e^{-z}$

$= \frac{z}{3(-1)^2 - 2(-1) - 1} e^{-z} = \frac{z}{4} e^{-z}$

$= \frac{\log x}{4} \times \frac{1}{x} = \frac{1}{4x} \log x$

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⑤ $(x^2D^2 - xD + 1)y = 0$

$\theta^2 - \theta - \theta + 1 = 0$

$m^2 - 2m + 1 = 0$

$m = +1, +1$

$= (c_1 + c_2 z)e^{z}$ (log m, x)

$= (c_1 + c_2 \log x)x$

⑥ $x^2\frac{d^3y}{dx^3} + 2x\frac{d^2y}{dx^2} - 2\frac{dy}{dx} = 0$

$x^3\frac{d^3y}{dx^3} + 2x^2\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} = 0$

$\Rightarrow (x^3D^3 + 2x^2D^2 - 2xD)y = 0$

$\Rightarrow [\theta^3 - 3\theta^2 + 2\theta + 2\theta^2 - 2\theta - 2\theta]y = 0$

$\Rightarrow m^3 - 3m^2 - 2m = 0$

$\Rightarrow m(m^2 - m - 2) = 0$

$\Rightarrow m(m-2)(m+1) = 0$

$m = 0, 2, -1$

$\therefore y = c_1e^{0z} + c_2e^{2z} + c_3e^{-z}$

$\boxed{y = c_1 + c_2x^2 + c_3x^{-1}}$

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(7) $4x^2y'' + 12xy' + 3y = 0$

$(4x^2D^2 + 12xD + 3)y = 0$

$\Rightarrow (4\theta^2 - 4\theta + 12\theta + 3)y = 0$

$\Rightarrow 4m^2 + 8m + 3 = 0.$

$m = \frac{-8 \pm \sqrt{64 - 48}}{2} = \frac{-8 \pm \sqrt{16}}{2} = \frac{-8 \pm 4}{2}$

$\frac{-8+4}{2}, \frac{-8-4}{2}$

$-2, -6$

$m = -\frac{1}{2}, -\frac{3}{2}$

$\therefore y = c_1 x^{-\frac{1}{2}} + c_2 x^{-\frac{3}{2}}$

(8) $x^2y'' + xy' - y = 0$

$[\theta^2 - \theta + \theta - 1]y = 0$

$m^2 - 1 = 0$

$m = \pm 1$

$y = c_1 e^z + c_2 e^{-z}$

$\boxed{y = c_1 x + c_2 x^{-1}}$ ✓

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Orthogonal Trajectories :-

→ The family of curves which cuts every member of the given family orthogonally is called the orthogonal Trajectory of the given family.

⟶ $f(x, y, c) = 0$ —— ①

replace

⟶ $F\left(x, y, \frac{dy}{dx}\right) = 0$ —— ②

replace $\left(\frac{dy}{dx}\right)$ by $\left(\frac{-dx}{dy}\right)$

⟶ $g\left(x, y, -\frac{dx}{dy}\right) = 0$ —— ③

The sol^n of eq^n ③ is called the orthogonal trajectory of the given family.

Note :-

If eq^n ② and eq^n ③ are identical, then the curve is said to be self orthogonal.

Q. Find the Orthogonal Trajectory of family of lines passing through the origin —

i) family of lines

ii) family of Circles having the centre at the origin.

iii) family of Parabolas

iv) family of ellipses.

***

If fn given, then

Steps —

find $\frac{dy}{dx}$

replace $\frac{dy}{dx}$ by $\frac{-dx}{dy}$

and integrate & solve.

that will represent orthogonal of that fn

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① → $y = mx$ ——— (1)

$\frac{dy}{dx} = m$

$y = x\frac{dy}{dx}$ ——— (2)

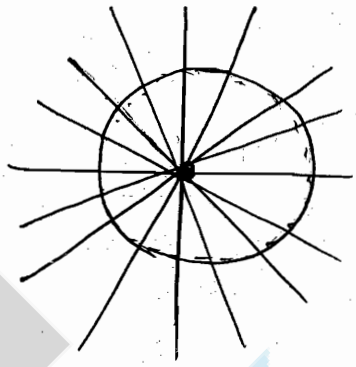
$y = -x\frac{dx}{dy}$ ——— (3)

$\int y\,dy + \int x\,dx = \int 0$

$\frac{y^2}{2} + \frac{x^2}{2} = c$

$x^2 + y^2 = 2c$

$\boxed{x^2 + y^2 = r^2}$ → Circle



② $y^2 = 4ax$ ——— (1)

$2y\frac{dy}{dx} = 4a$

$y^2 = 2y\frac{dy}{dx}\,x$

$2\int \frac{dy}{y} = \int \frac{dx}{x}$

$2x\frac{dy}{dx} = y$ ——— (2)

$y = -2x\frac{dx}{dy}$ ——— (3)

$\int y\,dy = \int -2x\,dx$

$\frac{y^2}{2} = -x\frac{x^2}{x} + c$

$\Rightarrow \boxed{x^2 + \frac{y^2}{2} = c}$ → Ellipse

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$\rightarrow \quad x^2 - y^2 = a^2 \quad \text{——①}$

$$2x - 2y\frac{dy}{dx} = 0$$

$$x + y\frac{dx}{dy} = 0$$

$$x\,dy + y\,dx = 0$$

$$\int d(xy) = \int 0$$

$$\boxed{xy = c}$$

Conclusion – Orthogonal Trajectories of family of —

Circle $\xrightarrow{\text{is}}$ st line.

st line $\xrightarrow{\text{is}}$ circle

Parabola $\xrightarrow{\text{is}}$ ellipse

Hyperbola $\xrightarrow{\text{is}}$ Rectangular Hyperbola.

(2) $\quad y^2 = 4a(x+a) \quad \text{——①}$

$$2y y_1 = 4a \cdot 1 \Rightarrow a = \frac{y y_1}{2}$$

$$y^2 = 2y y_1 \left(x + \frac{y y_1}{2}\right)$$

$$y = 2x y_1 + y y_1^2 \quad \text{——②}$$

replace y_1 by $-\frac{1}{y_1}$

$$y = -\frac{2x}{y_1} + \frac{y}{y_1^2}$$

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$$y y_1^2 = -2xy_1 + y$$

$$y y_1^2 + 2xy_1 = y \quad \text{——— (3)}.$$

$\because$ eqn (2) = eqn (3).

$\therefore$ Curve is said to be self orthogonal.

$\boxed{\frac{x^2}{a^2+\lambda^2} + \frac{y^2}{b^2+\lambda^2} = 1}$ where λ is a parameter

Imp Result → is also self-Orthogonal.

$x^2+y^2+2gx+c=0$.

Imp Concept $x^2+y^2+2fy+c=0$.

O.T of $x^2+y^2+2gx-5=0$ is —

$x^2+y^2-2fy+5=0$ → always take opposite coefficient sign.

#

$$f(r,\theta,c)=0 \quad \text{——— (1)}$$

$$F\left(r,\theta,\frac{dr}{d\theta}\right)=0 \quad \text{——— (2)}$$

replace $\frac{dr}{d\theta}$ by $-r^2\frac{d\theta}{dr}$

$$g\left(r,\theta,-r^2\frac{d\theta}{dr}\right)=0 \quad \text{——— (3)}$$

The soln of eqn (3) is called the Orthogonal of the given family.

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① $r = a\theta$

$\log r = \log a + \log\theta$

$\frac{1}{r}\frac{dr}{d\theta} = 0 + \frac{1}{\theta}$

$\frac{1}{r} \times -r^2\frac{d\theta}{dr} = \frac{1}{\theta}$

$\int \frac{dr}{r} = -\int \theta\, d\theta$

$\log r = -\frac{\theta^2}{2} + C$

$r = e^{-\frac{\theta^2}{2}+C}$

$r = e^{-\frac{\theta^2}{2}} \cdot e^C$

$\boxed{r = k\, e^{-\frac{\theta^2}{2}}}$

② $r = \frac{2a}{1-\cos\theta}$

$\log r = \log 2a - \log(1-\cos\theta)$

$\frac{1}{r}\frac{dr}{d\theta} = 0 - \frac{1}{1-\cos\theta}\cdot\sin\theta$

$\frac{1}{r} \times -r^2\frac{d\theta}{dr} = -\frac{\sin\theta}{1-\cos\theta}$

$r\frac{d\theta}{dr} = \frac{\sin\theta}{1-\cos\theta}$ $\quad 1-\cos\theta = 2\sin^2\frac{\theta}{2}$

$\Rightarrow \frac{dr}{r} = \frac{1-\cos\theta}{\sin\theta}\, d\theta$ $\quad \sin\theta = 2\sin\frac{\theta}{2}\cdot\cos\frac{\theta}{2}$

$\Rightarrow \int \frac{dr}{r} = -2\int \frac{-\frac{1}{2}\sin\theta/2}{\cos\theta/2}\, d\theta$

$\Rightarrow \log r = -2\log\cos\frac{\theta}{2} + \log C.$

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$$\Rightarrow \log r = \log \frac{c}{\cos^2\frac{\theta}{2}}$$

$$\Rightarrow r = \frac{c}{\cos^2\frac{\theta}{2}}$$

$$\Rightarrow r = \frac{c}{\frac{1+\cos\theta}{2}}$$

$$\Rightarrow \boxed{r = \frac{2c}{1+\cos\theta}}$$

i.e, $\boxed{r = \frac{2a}{1-\cos\theta}}$ $\xleftrightarrow{\text{Orthogonal}}$ $\boxed{r = \frac{2c}{1+\cos\theta}}$

③ $r^n = a^n \sin n\theta$

$$n \log r = n \log a + \log \sin n\theta$$

$$\frac{n}{r}\frac{dr}{d\theta} = n\frac{\cos n\theta}{\sin\theta}$$

$$\frac{n}{r}(-r^2)\frac{d\theta}{dr} = n\frac{\cos n\theta}{\sin n\theta}$$

$$-nr\frac{d\theta}{dr} = n\frac{\cos n\theta}{\sin n\theta}$$

$$\Rightarrow \frac{dr}{r} = \frac{1}{n}\int -n\frac{\sin n\theta}{\cos n\theta}\,d\theta$$

$$\log r = \frac{1}{n}\log\cos n\theta + \log c$$

$$n\log r = \log\cos n\theta + n\log c$$

$$\log r^n = \log\cos n\theta + \log c^n$$

$$\boxed{r^n = c^n \cos n\theta}$$

i.e, $\boxed{r^n = a^n \sin n\theta}$ $\xleftrightarrow{\text{Orthogonal}}$ $\boxed{r^n = c^n \cos n\theta}$

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Q. The diff eqⁿ of family of curves is given by $\frac{dy}{dx} = \frac{x}{y}$ then the system of orthogonal Trajectories of the given family of curves —

→ $\frac{dy}{dx} = \frac{x}{y}$

$-\frac{dx}{dy} = \frac{x}{y}$

$-\frac{dx}{x} = \frac{dy}{y}$

$-\log x = \log y + \log c$

$\Rightarrow \boxed{xy = c}$

This is the orthogonal Trajectory of family $\frac{dy}{dx} = \frac{x}{y}$ //

Differential Eqⁿ of First Order but not first Degree :-

- A differential eqⁿ of the form —

$$a_0\left(\frac{dy}{dx}\right)^n + a_1\left(\frac{dy}{dx}\right)^{n-1} + a_2\left(\frac{dy}{dx}\right)^{n-2} + \cdots + a_n = 0.$$

where $a_0, a_1, a_2 \cdots a_n$ are fn of x and y or constants is said to be differential eqⁿ of 1st order but not first degree.

Let $\frac{dy}{dx} \equiv p$

$$\boxed{a_0 p^n + a_1 p^{n-1} + a_2 p^{n-2} + \cdots + a_n = 0}$$

$$f(x, y, p) = 0$$

- This eqⁿ containing 3 variables x, y and p, therefore it may contain 3 types of solution:

→ It may be solvable for p, may be solvable for y, may be solvable for x.

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* The eqn solvable for p —

→ If it is solvable for p, then the eqn must be factorized.

→ $[p - f_1(x,y)][p - f_2(x,y)][p - f_3(x,y)] \ldots [p - f_n(x,y)] = 0$

then $p - f_1(x,y) = 0$

replace by $\frac{dy}{dx}$

$$\boxed{\frac{dy}{dx} = f_1(x,y)}$$

ex:- ① $p^2 - 5p + 6 = 0$ where $p = \frac{dy}{dx}$.

$(p-3)(p-2) = 0$.

$p - 2 = 0$

$\frac{dy}{dx} - 2 = 0$

$\int dy - \int 2\,dx = \int 0$

$y - 2x - c_1 = 0$ —— ①

$y - 3x - c_2 = 0$ —— ②

$$\boxed{(y - 2x - c_1)(y - 3x - c_2) = 0}$$

② $x^2p^2 + xyp - 6y^2 = 0$

$\Rightarrow x^2p^2 + 3xyp - 2xyp - 6y^2 = 0$

$\Rightarrow xp(xp + 3y) - 2y(xp + 3y) = 0$

$\Rightarrow (xp + 3y)(xp - 2y) = 0$

$xp + 3y = 0$

$x\frac{dy}{dx} + 3y = 0$

$\frac{dy}{dx} + \frac{3}{x}y = 0$

$y = e^{\int P dx} = \int Q e^{\int P dx} + c$

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$y(x^3) - c_1 = 0$ ——— ①

$x\frac{dy}{dx} - xy = 0.$

$\frac{dy}{dx} - \frac{2}{x}y = 0.$

$y\left(\frac{1}{x^2}\right) - c_2 = 0$ ——— ②.

Soln obtained $\therefore \boxed{(x^3y - c_1)(y - c_2x^2) = 0}$

② $xyp^2 - (x^2+y^2)p + xy = 0.$

$xyp^2 - x^2p - y^2p + xy = 0.$

$xp(yp - x) - y(yp - x) = 0$

$(xp - y)(yp - x) = 0.$

$xp - y = 0.$

$x\frac{dy}{dx} - y = 0$

$\frac{dy}{dx} - \frac{1}{x}y = 0.$

$y\left(\frac{1}{x}\right) - c_1 = 0$ ——— ①

$yp - x = 0.$

$y\frac{dy}{dx} - x = 0$

$y\,dy - x\,dx = 0$

$\frac{y^2}{2} - \frac{x^2}{2} = c_2$

$\Rightarrow y^2 - x^2 - 2c_2 = 0$ ——— ②

$\therefore \boxed{(y - c_1x)(y^2 - x^2 - 2c_2) = 0}$

Trick

When to apply linear approach variable seperable?

→ $x\frac{dy}{dx}$ or $y\frac{dx}{dy}$ → Linear

→ $x\frac{dx}{dy}$ or $y\frac{dy}{dx}$ → Variable Seperable

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* When factorization is not possible, then solvable for P is not possible two case arises —

When solvable for y ?	When solvable for x ?
① The degree of y must be 1.	① The degree of x must be 1
② Seperate y from x & P $y = f(x,P)$	② Seperate x from y & P $x = f(y,P)$
③ diff. both side w.r.to 'x'	③ Diff. both side w.r.to 'y'.
④ Replace $\frac{dy}{dx}$ by P	④ Replace $\frac{dx}{dy}$ by $\frac{1}{P}$.
⑤ neglect the terms or cancel the terms which doesnot contain $\frac{dP}{dx}$ on both sides.	⑥ Neglect the term or Cancel the term which does not contain $\frac{dP}{dy}$, on both sides.

ex:- $y + px = p^2x^4$ —— ①

∵ degree of y = 1 → solvable for y ✓

∵ all coeff are 1, 1, 1 ⇒ means factorization not possible ⇒ not solvable for P ✗

$y = p^2x^4 - px$

$\frac{dy}{dx} = p^2(4x^3) + x^4\left(2p\frac{dp}{dx}\right) - p(1) - x\left(\frac{dp}{dx}\right)$

$p - 4p^2x^3 + p = (2px^4 - x)\frac{dp}{dx}$

$2p(1 - 2px^3) = -x(1 - 2px^3)\frac{dp}{dx}$

$2p = -x\frac{dp}{dx}$

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$\Rightarrow \int \frac{dp}{p} = \int -2\frac{dx}{x}$

$\Rightarrow \log p = \log \frac{c}{x^2}$

$\boxed{p = \frac{c}{x^2}}$ 2 methods now to get soln of diff^n eq^n in Q.

either put $p = \frac{dy}{dx}$

$\frac{dy}{dx} = \frac{c}{x^2}$

and integrate and solve.

$dy - \frac{c\,dx}{x^2} = 0$

$\boxed{y + \frac{c}{x} = c}$ soln

Put in eq^n of Question.

$\because y + px = p^2x^4$.

$\frac{c}{x^2}$ $\left(\frac{c}{x^2}\right)^2$

$\Rightarrow \boxed{y + \frac{c}{x} = c}$

soln

Ans

(8) $y = 2px + y^2p^3$.

$2px = y - y^2p^3$

$x = \frac{y}{2p} - \frac{y^2p^3}{2p}$

$\frac{dx}{dy}$ ⊙

$2x = \frac{1}{p}\cdot y - y^2p^2$

$2\frac{dx}{dy} = \frac{1}{p}\cdot 1 + y\left(-\frac{1}{p^2}\frac{dp}{dy}\right) - 2yp^2 - y^2\left(2p\frac{dp}{dy}\right)$

$2\cdot\frac{1}{p} = \frac{1}{p} - \frac{y}{p^2}\frac{dp}{dy} - 2yp^2 - 2py^2\frac{dp}{dy}$

$\frac{2}{p} - \frac{1}{p} + 2yp^2 = \left(-\frac{y}{p^2} - 2py^2\right)\frac{dp}{dy}$

$\frac{(1 + 2yp^3)}{p} = -y\left(\frac{1 + 2yp^3}{p^2}\right)\frac{dp}{dy}$

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$$1 = -\frac{y}{P}\frac{dP}{dy}$$

$$\int \frac{dP}{P} = \int -\frac{dy}{y}$$

$$\log P = \log \frac{c}{y}$$

$$\boxed{P = \frac{c}{y}}$$ Put in Q.

$$\because \quad y = 2x\cdot\frac{c}{y} + y^2\frac{c^3}{y^3}$$

$$\boxed{y^2 = 2cx + c^3}$$

Shortcut

$$\boxed{y = Px + \phi(P)}$$

replace P by c

$$\boxed{y = cx + \phi(c)}$$

ex:- ① $y = Px - P^2$

gen soln = $\boxed{y = cx - c^2}$

② $\log(Px - y) = P$

$\therefore Px - y = e^P$

$y = Px - e^P$

$\boxed{y = cx - e^c}$

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Numerical Methods

I: Soln of non-linear equation :-

non linear means ?
linear → degree 1.
non-linar → degree ≠ 1

Let $f(x) = 0$ be any non linear equation to find the approximate root of $f(x) = 0$ choose any two values of $x = a, b$, particularly adjacent values in such a way that their functional values must have opposite signs.

ex:- $f(a) < 0$, $f(b) > 0$

Then, we can say that root lies betⁿ a and b.
This approximate root can be find by —

1. Method of bisection —
2. Regula Falsi method
3. Newton - Raphson

* Method of Bisection:-

$x = a, b$

$f(a) < 0$, $f(b) > 0$.



$x_1 = \frac{a+b}{2}$

$f(x_1) < 0$

$f(a), f(x_1), f(b)$

$<\quad <\quad >$

$x_2 = \frac{x_1 + b}{2}$

$f(x_2) > 0$

$x.$

$x_3 = \frac{x_1 + x_2}{2}$

upto 13th – 14th step

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The rate of convergence in bisection method is very-very slow.

→ Comparitively, the Bisection method, Regula-Falsi method has a little fast & rate of convergence.

* Regula-Falsi Method :-

$$x_1 = \frac{a f(b) - b f(a)}{f(b) - f(a)}$$

* Newton-Raphson method :-

Imp 1-marks

→ has a very fast rate of Convergence.

→ and the convergence of Newton Raphson method is 2nd order convergence or Quadratic convergence.

initial guess (99% chance given in Q.)

1% chance not given

then how to get x_0

↳ take $f(1) \rightarrow x_0$, $f(2) \rightarrow x_0$

whatever is nearer to origin that value will be taken.

when f(1) and f(2) must have opposite signs.

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$

$$\vdots$$

$$\boxed{x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}}$$

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① $f(x) = x^3 + 2x - 5 = 0$

$x_0 = 1$ find x_1 :

$f'(x) = 3x^2 + 2$

$$x_1 = 1 - \frac{x^3 + 2x - 5}{3x^2 + 2}$$

$$= 1 - \left[\frac{1 + 2 - 5}{5}\right]$$

$$= 1 + \frac{2}{5}$$

$x_1 = 1.4$ Ans

② $f(x) = x^3 + 3x - 7 = 0$

$x_0 = 1.$

$x_1 = ?$

$$x_1 = 1 - \frac{1^3 + 3 - 7}{3 \cdot 1 + 3}$$

$$= 1 - \frac{-3}{2}$$

$$= 1 + \frac{1}{2}$$

$= 1.5$ Ans

③ $f(x) = x^3 - 2x - 5 = 0.$

$f(0) = -5 < 0$

$f(1) = 1 - 2 - 5 = -6 < 0$

$f(2) = 8 - 4 - 5 = -1 < 0$ → $x_0 = 2$ is choosen

$f(3) = 27 - 6 - 5 = 16 > 0.$

$x_0 = 2$

$$x_1 = 2 - \frac{(-1)}{3(2)^2 - 2} = 2 + \frac{1}{10} = 2.1$$

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④ $f(x) = e^x - 1 = 0$ @ 0.71828 @ 0.34567 @ 0.12326 @ 0.00000.

$x_0 = -1$

$x_1 = ?$

→ $f'(x) = e^x$.

$$x_1 = -1 - \frac{e^{-1} - 1}{e^{-1}}$$

$$= -1 - 1 + e$$

$$= e - 2$$

$$= 2.71 - 2$$

$$= 0.71.$$

⑤ $f(x) = x^2 - 2x - 1$

$x_0 = 2$

$x_2 = $ ____ .

ⓐ 2.425 ⓑ 2.423 ⓒ 2.419 ⓓ 2.417 .

→ $f'(x) = 2x - 2$.

$$x_1 = 2 - \frac{-1}{2} = 2 + .5 = 2.5$$

$$x_2 = 2.5 - \frac{6.25 - 5 - 1}{5 - 2}$$

$$= 2.5 - \frac{.25}{3}$$

$$= 2.5 - .083$$

$$= 2.417$$

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Q. $f(x) = x^4 - 3x + 1 = 0$

$x_0 = 0$

$x_1 = $ ——

(a) $-\frac{1}{3}$ (b) $\frac{1}{3}$ (c) 3 (d) -3

→ $f'(x) = 4x^3 - 3$

$$x_1 = 0 - \frac{1}{-3}$$

$$= \frac{1}{3}$$ Ans

Q: The approximate root of eq^n $f(x) = (x+1)^2 + x - 3 = 0$ is to be solved by the Newton Raphson method. The method will fail in the very first iteration if the initial guess $x_0 =$ ——

(a) 0 (b) 0.5 (c) 1.0 (d) 1.5

→ $f(x) = x^2 - 2x + 1 + x - 3$

$= x^2 - x - 2$

$f'(x) = 2x - 1$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$= x_0 - \frac{x_0^2 - x_0 - 2}{2x_0 - 1}$$ → 0 method will fail.

i.e. method will fail if $2x_0 - 1 = 0$

$\Rightarrow x_0 = 0.5$

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Q The Newton Raphson iteration $x_{n+1} = \frac{1}{2}\left[x_n + \frac{R}{x_n}\right]$ can be used to compute the—

Imp Concept

a) sq. root of R

b) sq. of R

c) logrithmic of R

d) Reciprocal of R.

→ $x_{n+1} = x_n = \alpha$.

$\therefore \alpha = \frac{1}{2}\left[\alpha + \frac{2}{\alpha}\right]$.

$\Rightarrow 2\alpha - \alpha = \frac{R}{\alpha}$

$\Rightarrow \alpha^2 = R$

$\Rightarrow \alpha = \sqrt{R}$

↳ used to compute sq. root of no.

ex:- $x_{n+1} = \frac{1}{2}\left[x_n + \frac{117}{x_n}\right]$ converges to $\sqrt{117}$.

Q $f(x) = x^3 - x^2 + 4x - 4 = 0$

$x_0 = 2$

$x_1 =$ ——

a) $\frac{2}{3}$, b) $4/3$ c) 1 d) $\frac{3}{2}$.

→ $f'(x) = 3x^2 - 2x + 4$

$x_1 = 2 - \left[\frac{8^2}{12}\right]$

$= \frac{4}{3}$

$=$

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Q. The recursion relation $x = e^{-x}$ is to be solved by Newton Raphson method, then which of the following is true?

a) $x_{n+1} = e^{-x_n}$.

b) $x_{n+1} = x_n - e^{-x_n}$

c) $x_{n+1} = \dfrac{(1+x_n)e^{-x_n}}{(1+e^{-x_n})}$

d) $x_{n+1} = \dfrac{x_n^2 - e^{-x_n}(1+x_n) - 1}{x_n - e^{-x_n}}$

→ $f(x) = x - e^{-x}$.

$f'(x) = 1 + e^{-x}$

$f'(x_n) = 1 + e^{-x_n}$ → This will come into denominator.

∴ option c

Q. Consider the series —

$x_{n+1} = \dfrac{x_n}{2} + \dfrac{9}{8x_n}$

$x_0 = 0.5$ is obtained from the Newton Raphson method.

The series converges to —

→ $x_{n+1} = x_n = \alpha$ .. (Take it always when given Series Converges)

$\Rightarrow \alpha = \dfrac{\alpha}{2} + \dfrac{9}{8\alpha}$

$\Rightarrow \alpha = \dfrac{4\alpha^2 + 9}{8\alpha}$

$\Rightarrow 4\alpha^2 = 9$

$\Rightarrow \alpha^2 = \dfrac{9}{4}$

$\Rightarrow \alpha = \sqrt{9/4} = \dfrac{3}{2} = 1.5$ Ans

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Q. The sol^n of the variables x_1 and x_2 for the equations –

$$u = 10x_2 \cdot \sin x_1 - 0.8 = 0 \quad \text{——} ①$$

$$V = 10x_2^2 - 10x_2 \cos x_1 - 0.6 = 0 \quad \text{——} ②$$

is to be solved by the Newton Raphson method.

⇒ Assuming the initial value, $x_1 = 0.0$ $\quad x_2 = 1.0$.

Jacobian Matrix = ____

→ Jacobian Matrix – $\begin{bmatrix} \frac{\partial u}{\partial x_1} & \frac{\partial u}{\partial x_2} \\ \frac{\partial v}{\partial x_1} & \frac{\partial v}{\partial x_2} \end{bmatrix}$

$$\frac{\partial u}{\partial x_1} = 10x_2 \cos x_1 = 10(1)(1) = 10$$

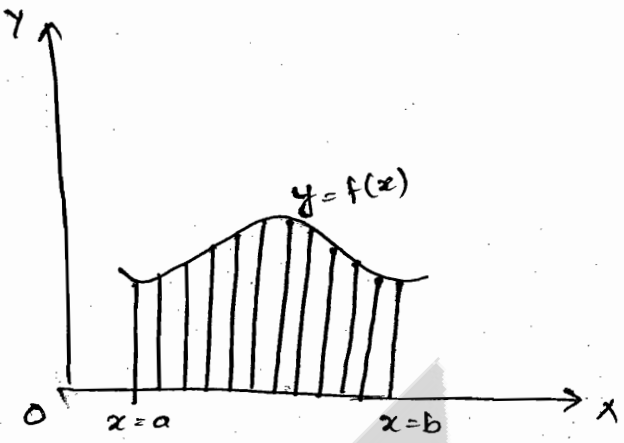
$$\frac{\partial u}{\partial x_2} = 10 \sin x_1 = 10(0) = 0.$$

$$\frac{\partial v}{\partial x_1} = 10x_2 \sin x_1 = 10(1)(0) = 0.$$

$$\frac{\partial v}{\partial x_2} = 20x_2 - 10 \cos x_1 = 20(1) - 10(1) = 10.$$

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Numerical Integration :-



$$h = \frac{b-a}{n}$$

① Trapezoidal Rule :-

$$\int_a^b f(x)\,dx = \frac{h}{2}\left[(y_0+y_n) + 2(y_1+y_2+y_3+\cdots+y_{n-1})\right]$$

② Simpson's $\frac{1}{3}$rd Rule :-

$$\int_a^b f(x)\,dx = \frac{h}{3}\left[(y_0+y_n) + 4(y_1+y_3+y_5+\cdots) + 2(y_2+y_4+y_6+\cdots)\right]$$

③ Simpson's $\frac{3}{8}$th Rule :-

$$\int_a^b f(x)\,dx = \frac{3h}{8}\left[(y_0+y_n) + 3(y_1+y_2+y_4+y_5+y_7+\cdots) + 2(y_3+y_8+y_9+\cdots)\right]$$

Note :-

The accuracy of Trapezoidal rule or order of integration of Trapezoidal rule is $O(h^2)$ → accuracy at 2nd point of decimal

The accuracy of Simpson's $\frac{1}{4}$ rule = $O(h^4)$ → accuracy upto 4th place of decimal

The accuracy of Simpson's $\frac{3}{8}$th rule is $O(h^5)$ → accuracy upto 5th place of decimal

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Q. $\int_0^1 \frac{1}{1+x^2}\,dx$.

By Trapezoidal & Simpson's rule and deviding the interval into 4 equal parts. and Derieve the value of π.

$\rightarrow \int_0^1 \frac{1}{1+x^2}\,dx$

$h = \frac{b-a}{n} = \frac{1-0}{4} = \frac{1}{4} = 0.25$

$y = \frac{1}{1+x^2}$

x	0	0.25	0.50	0.75	1.
y	1.0000	0.9412	0.8000	0.6400	0.5000.

$\frac{1}{1+\frac{1}{16}}$ $\frac{1}{1+\frac{1}{4}}$ $\frac{1}{1+\frac{9}{16}}$

$\therefore \int_0^1 \frac{1}{1+x^2}\,dx = \frac{0.25}{2}\left[(1.0000+0.5000)+2(0.9412+0.8000+0.6400)\right]$

$\tan^{-1}x\Big|_0^1 = 0.7828$

$\tan^{-1}1 - \tan^{-1}0 = 0.7828$ ($\tan^{-1}0 \to 0$)

$\Rightarrow \frac{\pi}{4} = 0.7828$

$\Rightarrow \pi = 3.1312$ we know $\pi = 3.1415$

difference occured at 2nd place.

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Simpson $\frac{1}{3}$rd Rule —

$$\int_0^1 \frac{1}{1+x^2}\,dx = \frac{0.25}{3}\left[(1.0000+0.5000)+4(0.9412+0.6400)+2(0.8000)\right]$$

$$\tan^{-1}\Big|_0^1 = 0.7854$$

$$\Rightarrow \tan^{-1}1 - \tan^{-1}0 = 0.7854$$

$$\Rightarrow \frac{\pi}{4} = 0.7854$$

$$\Rightarrow \pi = 0.7854 \times 4$$

$\pi = 3.1416$ we know $\pi = 3.1415$

diff. occur at 4th place.

By Shortcut methods.

Q. $\int_0^6 \frac{1}{1+x^2}\,dx$ by Trapezoidal rule by dividing the interval into 6 equal parts.

a) 1.4132 b) 1.3756 c) 1.2326 d) 1.0986.

→ $\int_0^6 \frac{1}{1+x^2}\,dx$

$= \tan^{-1}6 - \tan^{-1}0$

$= 1.40$

Compare with option → diff^n will be at 2nd place.

∴ option (a) is correct.

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② $\int_1^3 \log_e x \, dx$, $n=2$ = _____ By Simpson $\frac{1}{4}$th rule.

ⓐ 0.50
ⓑ 0.80
ⓒ 1.00.
ⓓ 1.29

→ Directly integrate and Compare ~~the~~, value at 4th place will differ.

③ $\int_0^{\pi/4} \frac{\sin x}{\cos^3 x} dx$, $n=4$, By Simpsons $\frac{3}{8}$ rule.

ⓐ 0.3883
ⓑ 0.4344
ⓒ 0.5026.
ⓓ 0.5589

→ $\int_0^{\pi/4} \frac{\sin x}{\cos^3 x} dx = \int_0^{\pi/4} \frac{\sin x}{\cos x} \cdot \frac{1}{\cos^2 x} dx = \int_0^{\pi/4} \tan x \cdot \sec^2 x \, dx.$

$= \left(\frac{\tan^2 x}{2}\right)_0^{\pi/4}$

$= 0.5000.$

④ $\int_1^3 \frac{1}{x} dx$, $n=2$, By Simpson. $\frac{1}{3}$rd rule.

a). 1.000
b). 1.098
c). 1.111
d) 1.120

→ Comparison ⇒ Use only Numeric method
Do not ever integrate or Shortcut method

→ $y = \frac{1}{x}$

x	1	2	3
y	1	0.50	0.333

$\Rightarrow \int_1^3 \frac{1}{x} dx = \frac{1}{3}[1.333 + 4(0.5)]$

$= \frac{3.333}{3} = 1.111.$

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Q. Torque Exerted on a flywheel over a cycle is listed in the table —

Angle (^oC)	0°	60°	120°	180°	240°	300°	360°
Torque (Nm)	0	1066	323	0	-323	-355	0
	y_0	y_1	y_2	y_3	y_4	y_5	y_6.

Estimate the value of the fn by Simpson $\frac{1}{3rd}$ rule.

(a) 542 (b) 993 (c) 1444 (d) 1986

$$\rightarrow \int_0^{2\pi} f(x)\,dx$$

$$= \frac{60^\circ}{3}\left[(0+0) + 4(1066+0-355) + 2(323-323)\right.$$

$$= \frac{\frac{\pi}{3}}{3} \times 4(711)$$

$$= \frac{\pi}{9} \times 2844$$

$$= 992.7.$$

$$\simeq 993.$$

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Numerical Soln of Ordinary Differential Eqn :-

$$\frac{dy}{dx} = f(x,y), \quad x = x_0, \quad y = y_0, \quad y(x_1) = ___$$

Methods –

1. Taylor's series method
2. Pi Card's method
3. Runge – Kutta method.

Taylor Series –

$$y = y_0 + (x-x_0)(y')_0 + \frac{(x-x_0)^2}{2!}(y'')_0 + \frac{(x-x_0)^3}{3!}(y''')_0 + \dots\infty.$$

Q. $\frac{dy}{dx} = -xy$, $y(0) = 1$, $y(0.1) = ___$

$x+y=4$
$y = 1-0.1$
$= 0.9$ ans

→

$$y' = -xy$$
$$y'' = -xy' - y.1$$
$$y''' = -xy'' - 2y'$$
$$y^{IV} = -xy''' - 3y''$$

$$y = 1 + x(0) + \frac{x^2}{2!}(-1) + \frac{x^3}{3!}(0) + \frac{x^4}{4!}(3) + \dots$$

$$= 1 - \frac{x^2}{2} + \frac{x^4}{8} - \dots \longrightarrow$$ Consider only upto 4th order rest can be neglected

$$\therefore y(0.1) = 1 - \frac{0.01}{2} + \cancel{\frac{0.0001}{8}} \text{ neglected}$$

$$= 1 - 0.005$$

$$= 0.995 \text{ Ans}$$

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② Picard's method :-

$$\frac{dy}{dx} = f(x,y) \;,\; x = x_0 \;,\; y = y_0.$$

$$\Rightarrow \int_{y_0}^{y} dy = \int_{x_0}^{x} f(x,y)\,dx.$$

$$\Rightarrow y - y_0 = \int_{x_0}^{x} f(x,y)\,dx$$

$$\Rightarrow y = y_0 + \int_{x_0}^{x} f(x,y)\,dx.$$

$$y_1 = y_0 + \int_{x_0}^{x} f(x,y_0)\,dx$$

$$y_2 = y_0 + \int_{x_0}^{x} f(x,y_1)\,dx$$

$$y_3 = y_0 + \int_{x_0}^{x} f(x,y_2)\,dx.$$

- - - -

- - - -

$$y = (\quad) + \text{- - -}$$

Q. $\frac{dy}{dx} = -xy$ ($f(x,y)$), $y(0) = 1$, $y(0.1) =$ ____

(x_0, y_0 ; $y_0 = 1$)

$$y_1 = 1 + \int_0^x -x(1)\,dx$$

$$y_1 = \left(1 - \frac{x^2}{2}\right)$$

$$y_2 = 1 + \int_0^x -x\left(1 - \frac{x^2}{2}\right)dx \qquad (y_1)$$

$$y_2 = 1 - \frac{x^2}{2} + \frac{x^4}{8}$$

$$\therefore y = 1 - \frac{x^2}{2} + \frac{x^4}{8} + \text{- - - - -}$$

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Q. $\frac{dy}{dx} = \frac{x^2}{1+y^2}$, $y(0)=0$, $y(0.25) =$ ____

When derivatives are convinient Taylor Series

When ~~Taylor~~ Ser $\int$ is convinient Picard's method

$\rightarrow$ $y_1 = 0 + \int \frac{x^2}{1+0^2} = \frac{x^3}{3}$

$y_2 = 0 + \int \frac{x^2\,dx}{1+\left(\frac{x^3}{3}\right)^2}$

$= \tan^{-1}\left(\frac{x^3}{3}\right)$

$y_3 = 0 + \int \frac{x^2}{1+\left(\tan^{-1}\frac{x^3}{3}\right)^2}$ $\rightarrow$ $\int$ becomes difficult.

$\therefore$ Take previous one as approx soln.

$\therefore$ $y = \tan^{-1}\frac{x^3}{3}$

$y(0.25) = \tan^{-1}\frac{(0.25)^3}{3}$

Q. $\frac{dy}{dx} = -xy$, $y(0)=1$, & $y(0.1) =$ ____

$\frac{dy}{dx} + xy = 0$

$I.F = e^{\int P\,dx} = e^{\int x\,dx} = e^{\frac{x^2}{2}}$.

$y \cdot e^{x^2/2} = c$.

$1 \cdot e^0 = c \Rightarrow c = 1$

$y e^{\frac{x^2}{2}} = 1$

$y = \frac{1}{e^{\frac{x^2}{2}}} = e^{-\frac{x^2}{2}} = e^{-\frac{(0.1)^2}{2}} = e^{-\frac{0.01}{2}} = e^{-0.0005}$

$= 0.995$

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Runge - Kutta Method :-

$$y = y_0 + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

where $k_1 = h \cdot f(x_0, y_0)$ ⟶ where $h = x_1 - x_0$.

$$k_2 = h\, f\left(x_0 + \frac{h}{2},\ y_0 + \frac{k_1}{2}\right)$$

$$k_3 = h f\left(x_0 + \frac{h}{2},\ y_0 + \frac{k_2}{2}\right)$$

$$k_4 = h f\left(x_0 + h,\ y_0 + k_3\right)$$

Q. (1) $\frac{dy}{dx} = \overbrace{x+y}^{f(x,y)}$ $\quad y(\underset{x_0}{0}) = \underset{y_0}{1}$, $y(\underset{x_1}{0.1}) =$ ——

$h = x_1 - x_0 = 0.1 - 0 = 0.1$

$k_1 = h f(x_0, y_0) = 0.1(0+1) = 0.1$

$k_2 = h f\left(x_0 + \frac{h}{2},\ y_0 + \frac{k_1}{2}\right) = 0.1\left[0.05 + 1.05\right] = 0.11$

($x_0 = 0$, $\frac{h}{2} = 0.05$; $y_0 = 1$, $\frac{k_1}{2} = 0.05$)

$k_3 = h f\left(x_0 + \frac{h}{2},\ y_0 + \frac{k_2}{2}\right) = 0.1\left[0.05 + 1.055\right] = 0.1105$

($x_0 = 0$, $\frac{h}{2} = 0.05$; $y_0 = 1$, $\frac{k_2}{2} = 0.055$)

$k_4 = h f(x_0 + h,\ y_0 + k_3) = 0.1\left[0.1 + 1.1105\right] = 0.12105$

Q Match the following —

E: Newton Raphson
F: Runge Kutta
G: Simpson rule
H: Gauss Elimination.

1) Soln of non linear eqn
2) Soln of linear algebraic
3) Soln of O.D. eqn.
4) ~~Soln of linear Algebraic eqn~~ Numeric Integration.

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Complex Variable

① $z = x + iy$

$\bar{z} = x - iy$

$z \cdot \bar{z} = |z|^2 = x^2 + y^2$

② $x + iy = r(\cos\theta + i\sin\theta) = re^{i\theta}$ is called the modulus-Amplitude form or Polar form

$r = |z| = \sqrt{x^2+y^2}$

$\theta = \tan^{-1}\left(\frac{y}{x}\right)$

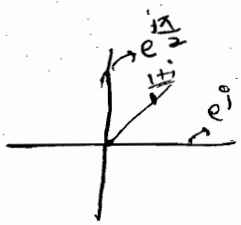
③ $\cos\theta + i\sin\theta = e^{i\theta}$

$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$.

$1 = \cos 0^\circ + i\sin 0^\circ = e^{i0}$

$i = \cos\frac{\pi}{2} + i\sin\frac{\pi}{2} = e^{i\pi/2}$

$1+i = \sqrt{2}\left[\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right] = \sqrt{2}\,e^{i\pi/4}$



Q. $(i)^i = $ ____ where $i = \sqrt{-1}$

$(x)^x = $ ____ where $x = \sqrt{-1}$

$(i)^i = \left(e^{i\pi/2}\right)^i = e^{-\frac{\pi}{2}}$.

Q. ① $(1+i)(2-5i) = $ ____

② $\left|\frac{3+4i}{1-2i}\right| = $ ____

③ $\frac{i+3}{i+1} = $ ____

④ $\frac{1+2i}{i-2} = $ ____

⑤ $\frac{-5+i10}{3+i4} = $ ____

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① $(1+i)(2-5i)$

$= 2+2i-5i-5i^2$

$= 7-3i$

② $\left|\frac{3+4i}{1-2i}\right| = \frac{\sqrt{9+16}}{\sqrt{1+4}} = \sqrt{\frac{25}{5}} = \sqrt{5}$

③ $\frac{i+3}{i+1} \times \frac{i-1}{i-1}$

$= \frac{i^2+3i-i-3}{-1-1}$

$= \frac{-4+2i}{-2}$

$= 2-i$

④ $\frac{1+2i}{i-2} = ?$

$= \frac{1+2i}{i-2} \times \frac{i+2}{i+2}$

$= \frac{i+2i^2+2+4i}{-1-4} = -i$

⑤ $\frac{-5+i10}{3+i4}$

$= \frac{-5+i10}{3+i4} \times \frac{3-i4}{3-i4}$

$= \frac{-15+30i+20i-40i^2}{9+16}$

$= \frac{-15+50i+40}{25} = 1+2i$

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Q. The value of $(1+i)^8$ = ____

(a) 2 (b) 4 (c) 8 (d) 16.

→ $(1+i)^8$

$= \left[(1+i)^2\right]^4$

$= \left[1+1+2i\right]^4$

$= 16\, i^4$

$= 16$ Ans

Q. The Polar form of $2+2i$ is ____

$r = |z| = \sqrt{4+4} = 2\sqrt{2}$.

$\theta = \tan^{-1} 1 = \frac{\pi}{4}$.

$\therefore \quad 2+2i = 2\sqrt{2}\, e^{i\frac{\pi}{4}}$

Q.

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Analytic function:-

A single valued fn which is defined and differentiable at each point of domain D is said to be an analytic fn in that domain.

Note:-

The necessary and sufficient cond^n for a function $\boxed{f(z) = u(x,y) + iv(x,y)}$ to be analytic is it should satisfy the **Cauchy-Riemann eq^n.**

Cond^n to be Analytic

$$\boxed{\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad , \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}} \quad \text{or} \quad \boxed{u_x = v_y \quad , \quad u_y = -v_x}$$

Q. $f(x) = (x^2 - y^2) + i(2xy)$

$u = x^2 - y^2$, $v = 2xy$.

$\frac{\partial u}{\partial x} = 2x$	$\frac{\partial v}{\partial y} = 2x$
$\frac{\partial u}{\partial y} = -2y$	$\frac{\partial v}{\partial x} = 2y$

Satisfying C-R eq^n

$\therefore$ Analytic in nature

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Harmonic function :-

— Any function of x and y satisfies the Laplace's eqn is said to be an harmonic function.

represented by

$$\nabla^2 u = 0$$

ex:- ① $u = x^2 - y^2$

$$\frac{\partial u}{\partial x} = 2x \qquad \frac{\partial u}{\partial y} = -2y$$

$$\frac{\partial^2 u}{\partial x^2} = 2 \qquad \frac{\partial^2 u}{\partial y^2} = -2$$

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 2 - 2 = 0 \quad \therefore \text{Harmonic} \checkmark$$

Note :-

Real and Imj part of an Analytic eqn satisfies the Laplacian Equation.

i.e, $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$

If $f(z) = u(x,y) + iv(x,y)$ ~~Given~~ If $u(x,y)$ Real Part given.

then $f(z) = 2u\left(\frac{z}{2}, \frac{z}{2i}\right) - u(0,0) + ci$

If Real part Given then asked directly $f(z)$ then how to get

Q. $u = x^2 - y^2$

$$u\left(\frac{z}{2}, \frac{z}{2i}\right) = \left(\frac{z}{2}\right)^2 - \left(\frac{z}{2i}\right)^2$$

$$= \frac{z^2}{4} + \frac{z^2}{4}$$

$$= \frac{z^2}{2}$$

$u(0,0) = 0 - 0 = 0.$

$$f(z) = 2 \times \frac{z^2}{2} - 0 + ci = z^2 + ci$$

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$$u = x^3 - 3xy^2 + 3x + 1.$$

$$u\left(\frac{z}{2}, \frac{z}{2i}\right) = \frac{z^3}{8} - \frac{3z}{2}\left(\frac{z^2}{4i^2}\right) + \frac{3z}{2} + 1$$

$$= \frac{z^3}{8} + \frac{3z^3}{8} + \frac{3z}{2} + 1$$

$$= \frac{z^3}{2} + \frac{3z}{2} + 1$$

$$u(0,0) = 0 - 0 + 0 + 1 = 1.$$

$$f(z) = 2\left(\frac{z^3}{8} + \frac{3z}{2} + 1\right) - 1 + ci$$

$$f(z) = z^3 + 3z + 1 + ci$$

→

Q. $f(z) = z^2$

$= (x+iy)^2$

$= x^2 + i^2y^2 + 2ixy.$

$= x^2 - y^2 + 2ixy.$

If real part given, how to get Imj part.

$$dv = \frac{\partial v}{\partial x}dx + \frac{\partial v}{\partial y}\cdot dy$$

$$\boxed{dv = -\frac{\partial u}{\partial y}dx + \frac{\partial u}{\partial x}dy}$$

which is exact.

Let $u = x^2 - y^2$.

$$\frac{\partial u}{\partial x} = 2x \quad , \quad \frac{\partial u}{\partial y} = -2y$$

$dv = 2y\,dx + 2x\,dy$

↳ exact. no need to check

Gen. Soln —

$$v = \int 2y\,dx + \int 0\,dy$$

$$v = 2xy + c$$

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Q. $u = x^3 - 3xy^2 + 3x + 1$

$\frac{\partial u}{\partial x} = 3x^2 - 3y^2 + 3$

$\frac{\partial u}{\partial y} = -6xy$

$dv = \int 6xy\,dx + \int (3x^2 - 3y^2 + 3)\,dy$

$=$

Q. $u = e^x \cos y$

$\frac{\partial u}{\partial x} = e^x \cos y$

$\frac{\partial u}{\partial y} = -e^x \sin y.$

$dv = -\frac{\partial u}{\partial y}dx + \frac{\partial u}{\partial x}dy$

$\int dv = \int e^x \sin y\,dx + \int e^x \cos y\,dy$

$\boxed{v = e^x \sin y.}$

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* If imj part given and asked real part then how to get?

$$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy.$$

or

$$\boxed{du = \frac{\partial v}{\partial y} dx - \frac{\partial v}{\partial x} dy}$$

which is exact..

ex:- ① $v = -xy$

$$\frac{\partial v}{\partial x} = -y, \quad \frac{\partial v}{\partial y} = -x.$$

$$\int du = \int -x\,dx + \int y\,dy.$$

$$u = -\frac{x^2}{2} + \frac{y^2}{2} = \frac{1}{2}(y^2 - x^2).$$

② Given $v = y^3 - 3x^2y$, find u. where $f(z)$ is harmonic

$$\frac{\partial v}{\partial x} = -6xy \qquad \frac{\partial v}{\partial y} = 3y^2 - 3x^2$$

$$\int du = \int (3y^2 - 3x^2)\,dx + \int 6xy\,dy$$

$$\boxed{u = 3xy^2 - x^3}$$

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Singular Points :- → or Pole

Singular points are those at which the given function $f(z)$ is not analytic.

These points are also called Singularities or poles.

Find Singular Points.

Q. ① $f(z) = \dfrac{1-2z}{z(z-1)(z-2)}$

$z = 0, 1, 2.$

Q. $f(z) = \dfrac{z}{z^2-z-2}$

$f(z) = \dfrac{z}{(z-2)(z+1)}$

$\Rightarrow z = 2, -1.$

Q. $f(z) = \dfrac{z}{z^4-1}$

$z = \pm 1, \pm i$

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Cauchy's Theorem :-

If $f(z)$ is an analytic fn and its derivative $f'(z)$ is continuous at all points inside and on a simple and closed curve C, then -

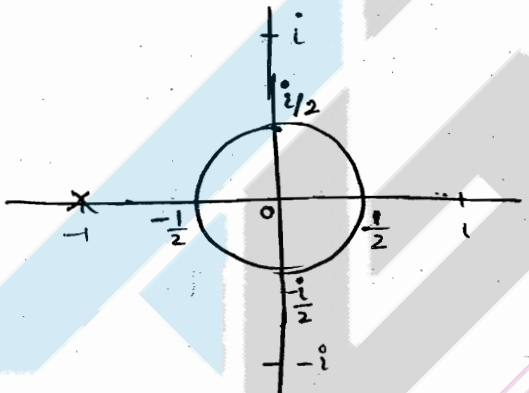
$$\oint_C f(z)\,dz = 0$$

means

if pole outside the limit, then $\int = 0$.

Q. $\oint_C \frac{3z^2+7z+1}{z+1}\,dz$, where C is $|z| = \frac{1}{2}$

→

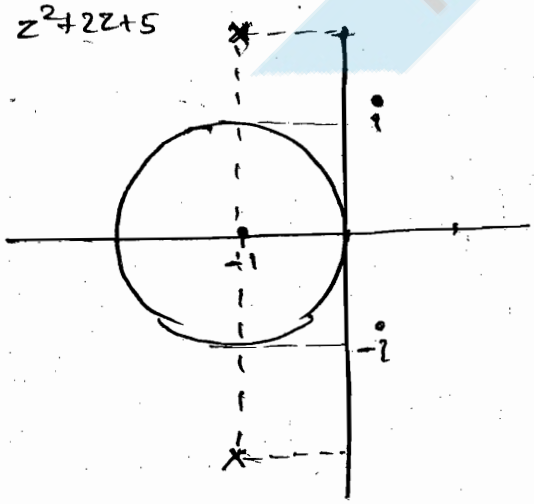


∵ pole is outside curve.

∴ Ans is 0.

Q. $\oint_C \frac{z+4}{z^2+2z+5}\,dz$ where C is $|z+1| = 1$

→



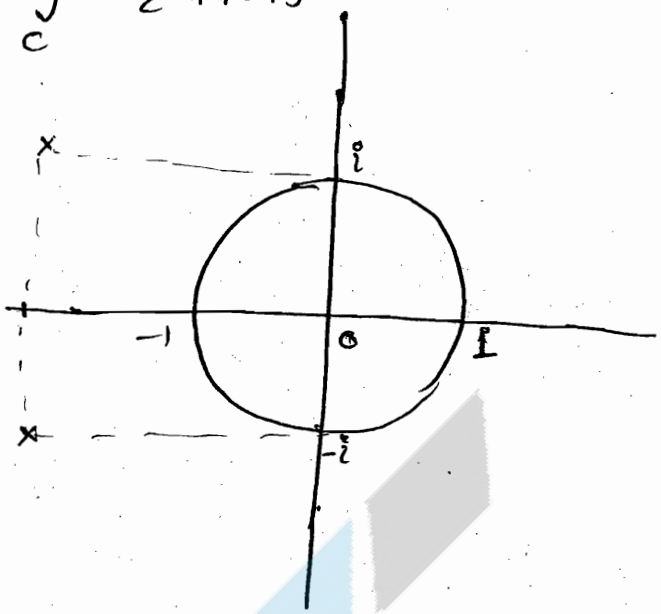
Pole $\Rightarrow z^2+2z+5 = 0$

$\Rightarrow \frac{-2 \pm \sqrt{4-20}}{2} = \frac{-2 \pm 4i}{2}$; $-1 \pm 2i$,

Pole = $-1+2i$, $-1-2i$ → outside the curve ⇒ ∴ Ans = 0

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Q. $\oint_C \frac{-3z+1}{z^2+4z+5}\,dz$ where C is $|z|=1$



$z^2+4z+5=0$

$\frac{-4 \pm \sqrt{16-20}}{2}$

$= \frac{-4 \pm 2i}{2}$

$= -2 \pm i$

↳ Pole outside the curve.

∴ Answer is '0'.

Cauchy's Integral formula

V.V.Imp 2 marks

If $f(z)$ is analytic fn inside and on a simple and closed curve 'C' and if a is any point inside C then,

$$\oint_C \frac{f(z)}{z-a}\,dz = 2\pi i f(a)$$

$$\oint_C \frac{f(z)}{(z-a)^2}\,dz = 2\pi i f'(a)$$

$$\oint_C \frac{f(z)}{(z-a)^3}\,dz = \frac{2\pi i}{2!} f''(a)$$

$$\oint_C \frac{f(z)}{(z-a)^{n+1}}\,dz = \frac{2\pi i}{n!} f^n(a)$$

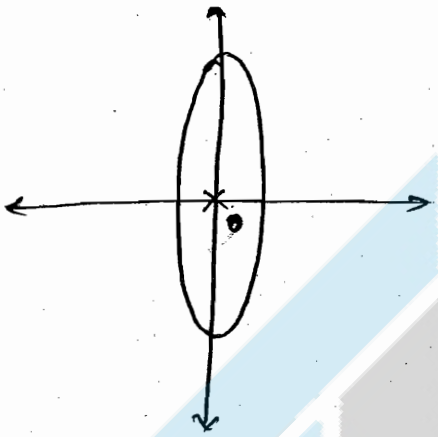
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Q. ① The value of $\oint_C \frac{\cos z}{z}\,dz$ where c is an ellipse

$9x^2+4y^2=1$

→ $9x^2+4y^2=1$

$\frac{x^2}{(\frac{1}{3})^2}+\frac{y^2}{(\frac{1}{2})^2}=1$



∵ Pole inside

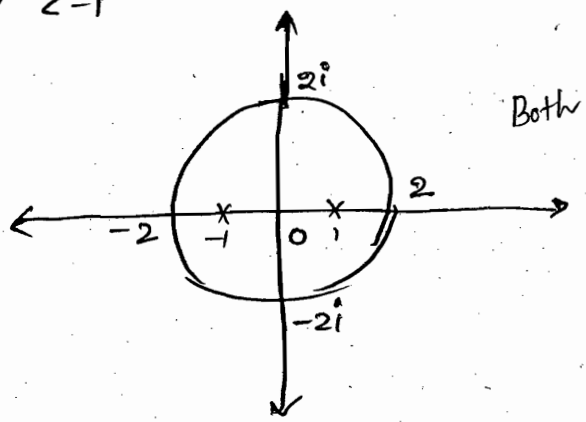
∵ Cauchy Integral formula applicable

$\therefore \oint \frac{\cos z}{z}\,dz = 2\pi i \cos z\Big|_{z=0}$

$= 2\pi i$ Ans.

Q. $\oint_C \frac{1}{z^2-1}\,dz$ where c is the circle $x^2+y^2=4$.

→



Both Pole inside

Partial fraction

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Method–1

Partial fraction —

$$\oint_C \frac{1}{(z-1)(z+1)}\,dz = \frac{1}{2}\oint\left[\frac{1}{z-1} - \frac{1}{z+1}\right]dz$$

$$= \frac{1}{2}\left(2\pi i(1) - 2\pi i(1)\right)$$

$$= 0$$

Observation :-

Sometimes even also pole inside then $\int = 0$

Method–2

Shortcut

$$\oint_C \frac{1}{z^2-1}\,dz = \oint_C \frac{1}{(z-1)(z+1)}\,dz$$

$$= \oint \frac{\left(\frac{1}{z+1}\right)}{(z-1)} + \frac{\left(\frac{1}{z-1}\right)}{(z+1)}\,dz$$

$$= 2\pi i\left(\frac{1}{z+1}\right)_{z=1} + 2\pi i\left(\frac{1}{z-1}\right)_{z=-1}$$

$$= 2\pi i\left(\frac{1}{2}\right) + 2\pi i\left(-\frac{1}{2}\right)$$

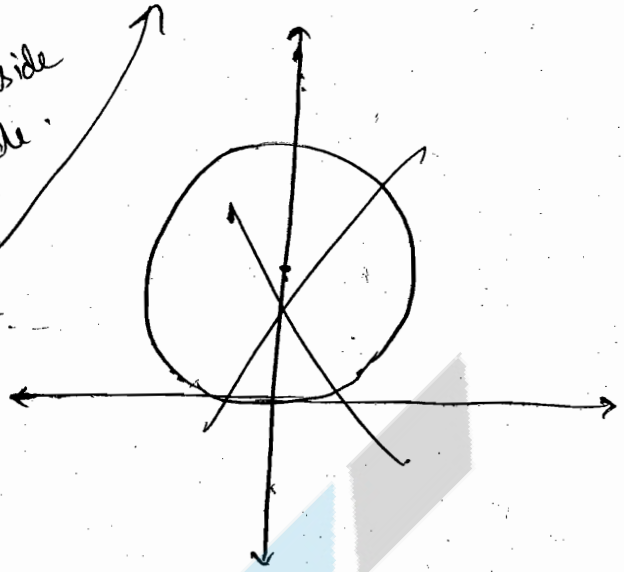
$$= 0$$

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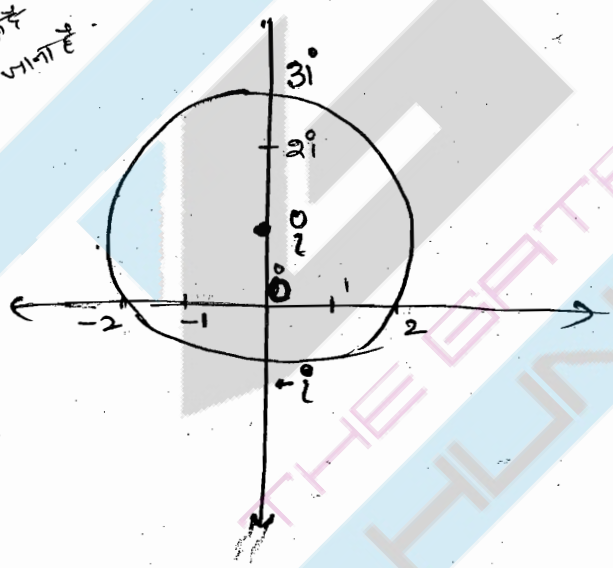
Q. Repeated 3 times.

$$\oint_C \frac{z-1}{(z+1)^2(z-2)}\,dz \quad \text{where } C \text{ is } |z-i| = 2$$

If one Pole inside another outside. then?



जो बाहर है उसको ऊपर से भगाते जो inside है उसको नीचे से भगाते।



$$\therefore \oint_C \frac{z-1}{(z+1)^2(z-2)}\,dz$$

$$= \oint_C \frac{\left(\frac{z-1}{z-2}\right)}{(z+1)^2}\,dz \qquad f(z)$$

$$f'(z) = \frac{-1}{(z-2)^2}$$

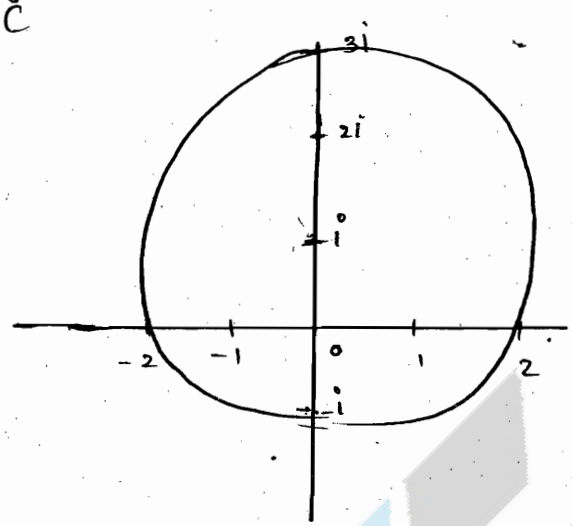
$$= 2\pi i \left[\frac{-1}{(z-2)^2}\right]_{z=-1}$$

$$= \frac{-2\pi i}{(-1-2)^2}$$

$$= \frac{-2\pi i}{9}$$

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Q. $\oint_C \frac{z^2-4}{z^2+4}\,dz$ where C is the circle $|z-i|=2$



$$\oint \frac{z^2-4}{z-2} \quad \frac{z^2-4}{z+2}$$
$$z+2 \qquad z-2$$

$= 2\pi i$ (4 -)

$$\oint \frac{z^2-4}{(z-2i)(z+2i)}\,dz$$
✓ ×

$$= \oint \frac{\frac{z^2-4}{z+2i}}{z-2i}\,dz$$

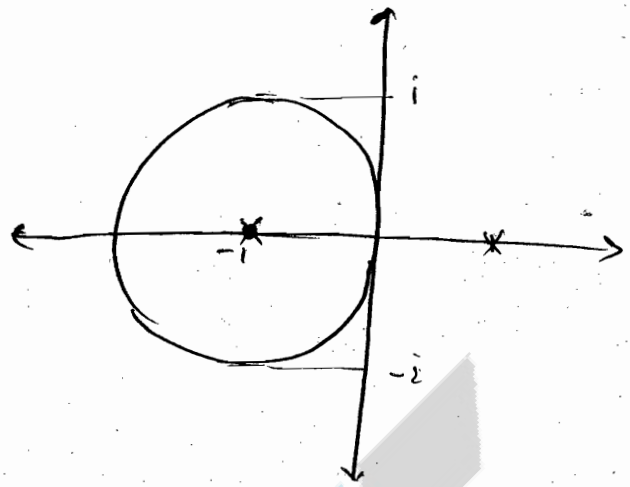
$$= 2\pi i \left(\frac{z^2-4}{z+2i}\right)_{z=2i}$$

$$= 2\pi i \left(\frac{-4-4}{2i+2i}\right) = 2\pi i \left(\frac{-8}{4i}\right)$$

$$= -4\pi.$$

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Q. $\oint_C \frac{z^2}{z^4+1}\,dz$ C is $|z+1|=1$.



$$\oint \frac{z^2}{(z^2-1)(z^2+1)} = \oint \frac{z^2}{(z-1)(z+1)(z-i)(z+i)}$$

× × × ×

outside pole

{ inside नीचे रखो
outside वाला ऊपर से उठाओ

$$\oint \frac{\frac{z^2}{(z-1)(z^2+1)}}{z+1}\,dz.$$

$$= 2\pi i \left(\frac{z^2}{(z-1)(z^2+1)}\right)_{z=-1}$$

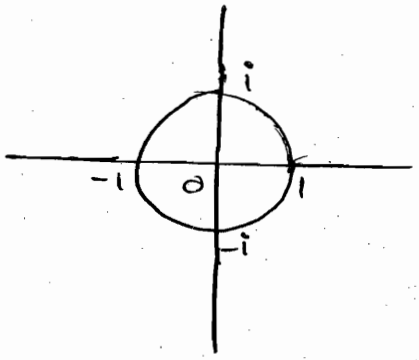
$$= 2\pi i \left(\frac{1}{-2\times 2}\right)$$

$$= -\frac{\pi i}{2}$$

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Q. $\oint \frac{5z^2+7z+3}{(z-1)^3}$ where C is $|z|=1$

→



$$\oint \frac{5z^2+7z+3}{(z-1)^3}$$

$f(z) = 5z^2+7z+3.$

$f'(z) = 10z+7.$

$f''(z) = 10.$

$$\therefore \oint \frac{5z^2+7z+3}{(z-1)^3} = \frac{2\pi i}{2!}(10)$$

$$= 10\pi i$$

***

Residue at $z=a$ where a is the pole of order n.

$$\lim_{z\to a} \frac{1}{(n-1)!} \frac{d^{n-1}}{dz^{n-1}} \left[(z-a)^n \cdot f(z) \right].$$

→ Q. find residue of fn $f(z) = \frac{1-2z}{z(z-1)(z-2)}$

at $z=0$, $\frac{1-0}{(0-1)(0-2)} = \frac{1}{2}$

at $z=1$, $\frac{1-2}{1\times 1} = -1$

at $z=2$, $\frac{1-4}{2\times 1} = -\frac{3}{2}$

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Q① find residue of for. $f(z) = \frac{z^2}{(z-1)(z+1)(z-2)}$

at $z=1$ $\quad \frac{1}{2\times -1} = -\frac{1}{2}$

$z=-1$ $\quad \frac{1}{-2\times -3} = \frac{1}{6}$.

$z=2$, $\quad \frac{4}{1\times 3} = \frac{4}{3}$

Q.② $f(z) = \frac{1}{z(z+2)^3}$

$$\frac{1}{z(z+2)^3} = \frac{A}{z} + \frac{B_0}{(z+2)^3} + \frac{B_1}{(z+2)^2} + \frac{B_2}{(z+2)}$$

for $(z+2)^3 \rightarrow$ $\lim_{z\to -2} \frac{1}{2!} \frac{d^2}{dz^2}\left[(z+2)^3 \cdot \frac{1}{z(z+2)^3}\right]$

$= \frac{1}{2} \frac{d^2}{dz}\left(\frac{1}{z}\right)$

$= \frac{1}{2} \times \frac{2}{z^3}$

$= \frac{1}{(-2)^3} = \frac{1}{8}$

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Workbook

(1) $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ or $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

$\Delta = 1$ $\quad$ $\Delta = -1$

$\Delta = \pm 1$

(3) $xy = y$

$yx = x$

$x^2 + y^2 = ?$

$= xx + yy$

~~$= (yx)x + (xy)$~~

$= (yx)(yx) + (xy)(xy)$

$= y(xy) + x(xy)$

$= yx + xy$

$= x + y$ .. Ans

(7) (c) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

Total 8

(8) (c) $A = [a_{ij}]_{m \times n}$.

$a_{ij} = i + j$

$A_{2\times 2} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix}$.

↳ 12 Check with option

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(9) 4^4

3×4^4

(10) wrong Q.

(11) 10

(12)

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$$

(13) (b) $5! = 120$

(14) a_{ij}

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -3 & -8 \\ 3 & 0 & -5 \\ 8 & 5 & 0 \end{bmatrix} \quad \because \Delta = 0$$

$\therefore$ inverse not possible.

(15)

(16) Correct $2x + 3y + z = \lambda z \longrightarrow 3^{rd}$

(17)

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(38) $\begin{bmatrix} \frac{1}{1\cdot 2} & 0 & \cdots \\ & & \\ & & \end{bmatrix}$

$$\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\frac{1}{3\cdot 4}+\cdots\frac{1}{n(n+1)}\cdots$$

$$S_n = \sum T_n$$

$$= \sum \frac{1}{n(n+1)}$$

$$= \sum \frac{1}{n}-\frac{1}{n+1}$$

$$= \frac{1}{1}-\frac{1}{2}+\frac{1}{2}-\frac{1}{3}+\frac{1}{3}-\frac{1}{4}+\cdots+\frac{1}{n}-\frac{1}{n+1}$$

$$= 1-\frac{1}{n+1}$$ ∴

(39) $$1\cdot\frac{1}{2}\cdot\frac{1}{3}\cdot\frac{1}{4}\cdots\frac{1}{n}$$

$$= \frac{1}{n!}$$ ∴

(40) $$A = \begin{bmatrix} 40 & -29 & -11 \\ -18 & 30 & -12 \\ 26 & 24 & -50 \end{bmatrix}$$

$\lambda_1, \lambda_2 = ?$

$\lambda_1+\lambda_2+\lambda_3 = 20$

$\lambda_3 = 0 \therefore D = 0$

$\lambda_2 = 20-\lambda$ ∴

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(45) $3, 2, -1$

$B = A^2 - A$.

$= 9 - 3 = 6$.

$= 2^2 - 2 = 2$

$= (-1)^2 - (-1) = 2$.

$|B| = 6 \times 2 \times 2 = 24$.

(46) $$A = \begin{bmatrix} 1 & 0 & 0 \\ i & \frac{-1+i\sqrt{3}}{2} & 0 \\ 0 & 1+2i & \frac{-1-i\sqrt{3}}{2} \end{bmatrix}$$

Trace ?

Trace is Σ principal diagonals

$$(A)^{102} = (1)^{102}, \left(\frac{-1+i\sqrt{3}}{2}\right)^{102}, \left(\frac{-1-i\sqrt{3}}{2}\right)^{102}$$

$$= 1, \left(e^{-i\frac{\pi}{3}}\right)^{102}, \left(e^{i\frac{\pi}{3}}\right)^{102}$$

$$= 1, e^{-34\pi i}, e^{34\pi i}$$

$$= 1 + 1 + 1 = 3$$

(48) $\lambda = -1, 1, 0$

$|A^{100} + I|$

$2, 2, 1$

$|A^{100} + I| = 2 \times 2 \times 1 = 4$

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(49) $\lambda_1 = 1$, $\lambda_2 = 4$

$X_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$. $X_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$.

(a) $\begin{bmatrix} -4 & -8 \\ 5 & 9 \end{bmatrix}$

(b) $\begin{bmatrix} 9 & -8 \\ 5 & -4 \end{bmatrix}$

(c) $\begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix}$

(d) $\begin{bmatrix} 2 & 2 \\ 1 & 3 \end{bmatrix}$

(51) $A = 1, -1, \lambda$

$A+3I = 1+3 , -1+3, \lambda+3.$

$A+3I = 4 , 2 , \lambda+3.$

$A^2+3I = 1(4), -1(2), \lambda(\lambda+3).$

$= 4 , -2, \lambda^2+3\lambda.$

18

$\lambda^2+3\lambda = 18$

$\hookrightarrow \lambda \neq 0$

$\therefore \Delta \neq 0$

$\therefore$ Inverse exists.

Similarly inverse exists for A

Correction → 2X2 not 3X3

(62) $A^{-1} = I - 2A.$

$I = A - 2A^2$

$1 = \lambda - 2\lambda^2$

$2\lambda^2 - \lambda + 1 = 0$

$\Delta = \lambda_1 \lambda_2$

$= \frac{1}{2}$

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(63) $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$a + d = ad - bc = 1$

Then $A^3 = ?$

→ $\lambda_1 + \lambda_2 = 1$

$\lambda_1 \lambda_2 = 1$

$\lambda_1 + \frac{1}{\lambda_1} = 1$

$\lambda_1^2 + 1 - \lambda_1 = 0$

$\lambda_1^2 - \lambda_1 + 1 = 0$

$\lambda_1 = ?$

$\lambda_1^2 + 1 = \lambda_1$

$A^2 = A - I$

$A^3 = A^2 - A$

$= (A - I)A$

$= A^2 - A$

$= A - I - A$

$= -I$

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Laplace Transform:

→ Let $f(t)$ be any fn. of t defined $\forall$ positive values of T, then the Laplace Transform of $f(t)$ is denoted by $\mathcal{L}[f(t)]$ and is defined as —

$$\mathcal{L}[f(t)] = \int_0^\infty e^{-st} \cdot f(t)\,dt = F(s)$$

$$\Rightarrow f(t) = \mathcal{L}^{-1}[F(s)]$$

is called the Inverse Laplace Transform of $F(s)$.

① $\mathcal{L}(t^n) = \int_0^\infty t^n e^{-st}\,dt$

Let $x = st \Rightarrow t = \frac{x}{s} \Rightarrow dt = \frac{dx}{s}$.

$t \to \infty \Rightarrow x \to \infty$

$t \to 0 \Rightarrow x \to 0$

$$= \int_0^\infty e^{-x} \left(\frac{x}{s}\right)^n \times \frac{dx}{s}$$

$$= \int_0^\infty e^{-x} \frac{x^n}{s^n} \cdot \frac{dx}{s}$$

$$= \frac{1}{s^{n+1}} \int_0^\infty e^{-x} x^n\,dx.$$

$$= \frac{1}{s^{n+1}} \int_0^\infty e^{-x} x^{\overline{n+1}-1}\,dx$$

$$\therefore \quad = \frac{\overline{n+1}}{s^{n+1}}.$$

$\therefore \mathcal{L}[t^n] = \frac{n!}{s^{n+1}}$, if $n \in N$.

$\mathcal{L}[t^n] = \frac{n\overline{n}}{s^{n+1}}$, if $n > 0$.

$\therefore L(1) = \frac{1}{s}$

$L(t) = \frac{1}{s^2}$

L

Improper Integral.

$$\overline{n} = \int_0^\infty e^{-x} x^{n-1}\,dx.$$

$\overline{n+1} = n\overline{n}$, if $n > 0$.

$\overline{n+1} = n!$, if $n \in N$

$\overline{1} = 1$

$\overline{\frac{1}{2}} = \sqrt{\pi}$

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$L(1) = \frac{1}{s}$

$L(t) = \frac{1}{s^2}$

$L(t^2) = \frac{2}{s^3}$

$L(e^{at}) = \frac{1}{s-a}$

$L(e^{-at}) = \frac{1}{s+a}$

$L(\sin at) = \frac{a}{s^2+a^2}$

$L(\cos at) = \frac{s}{s^2+a^2}$

$L(\sinh at) = \frac{a}{s^2-a^2}$

$L(\cosh at) = \frac{s}{s^2-a^2}$

Q. find $L(t^2+6t+8)$, $L(\sqrt{t})$, $L\left(\frac{1}{\sqrt{t}}\right)$, $L(\sin^2 t)$, $L(\sin^3 t)$, $L(\sin 2t \cdot \cos t)$

& $f(t) = \begin{cases} 0, & 0<t<2 \\ 3, & t \geq 2 \end{cases}$

find $L[f(t)]$.

→ ① $L(t^2+6t+8) = \frac{2}{s^3} + \frac{6}{s^2} + \frac{8}{s}$

② $L(t^{1/2}) = \frac{\Gamma(\frac{1}{2}+1)}{s^{\frac{1}{2}+1}} = \frac{\frac{1}{2}\Gamma\frac{1}{2}}{s^{3/2}} = \frac{\sqrt{\pi}}{2s^{3/2}}$ (+ve no.)

③ $L(t^{-\frac{1}{2}}) = \frac{\Gamma(-\frac{1}{2}+1)}{s^{-\frac{1}{2}+1}} = \frac{\Gamma\frac{1}{2}}{s^{\frac{1}{2}}} = \frac{\sqrt{\pi}}{\sqrt{s}} = \sqrt{\frac{\pi}{s}}$

④ $L(\sin^2 t) = L\left(\frac{1-\cos 2t}{2}\right) = \frac{1}{2s} - \frac{1}{2}\frac{s}{(s^2+4)} = \frac{1}{2s} - \frac{s}{2(s^2+4)}$

$\sin 3t = 3\sin t - 4\sin^3 t$

⑤ $L(\sin^3 t) = L\left(\frac{3\sin t - \sin 3t}{4}\right) = \frac{1}{4}\left[\frac{3}{s^2+1} - \frac{3}{s^2+9}\right] = \frac{3}{4}\left[\frac{8^2}{(s^2+1)(s^2+9)}\right]$

$= \frac{6}{(s^2+1)(s^2+9)}$

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⑥ $L(\sin 2t \cdot \cos t) = \frac{1}{2} L[\sin 3t + \sin t]$

$= \frac{1}{2}\left[\frac{3}{s^2+9} + \frac{1}{s^2+1}\right]$

⑦ $f(t) = \begin{cases} 0, & 0<t<2 \\ 3, & t \geq 2 \end{cases}$

$L[f(t)] = \int_0^\infty f(t) e^{-st} dt$

$= \int_0^2 0 \, dt + \int_2^\infty 3 e^{-st} dt$

$= \frac{3 e^{-st}}{-s} \Big|_2^\infty$

$= \frac{3}{-s}\left[0 - e^{-2s}\right]$

$= \frac{3e^{-2s}}{s}$

Shifting Theorem:-

If $L[f(t)] = \int_0^\infty e^{-st} f(t) dt = F(s)$

then $L[e^{at} f(t)] = F(s-a)$

L.H.S $= L[e^{at} f(t)] = \int_0^\infty f(t) e^{-t(s-a)} dt$

$= F(s-a)$

Formulae:-

$L(t^n) = \frac{n!}{s^{n+1}}$

$L(e^{at} t^n) = \frac{n!}{(s-a)^{n+1}}$

$L[e^{-at} t^n] = \frac{n!}{(s+a)^{n+1}}$

$L[e^{at} \sin bt] = \frac{b}{(s-a)^2+b^2}$

$L[e^{at} \cdot \cos bt] = \frac{(s-a)}{(s-a)^2+b^2}$

$L[e^{-at} \sin bt] = \frac{b}{(s+a)^2+b^2}$

$L[e^{-at} \cos bt] = \frac{s+a}{(s+a)^2+b^2}$

$L[e^{at} \sinh bt] = \frac{b}{(s-a)^2-b^2}$

$L[e^{-at} \sinh bt] = \frac{b}{(s+a)^2-b^2}$

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$L(t \sin at)$

$= I.P \text{ of } L(t e^{ait})$ $\qquad L(t) = \frac{1}{s^2}$

$= I.P \text{ of } \left[\frac{1}{(s-ai)^2} \times \frac{(s+ai)^2}{(s+ai)^2}\right]$

$= I.P \text{ of } \left[\frac{s^2-a^2+2ias}{(s^2+a^2)^2}\right]$

$L(t \sin at) = \frac{2as}{(s^2+a^2)^2}$ $\qquad L(t \sin t) = \frac{2s}{(s^2+1)^2}$

$L(t \cos at) = \frac{s^2-a^2}{(s^2+a^2)^2}$ $\qquad L(t \cos t) = \frac{s^2-1}{(s^2+1)^2}$

1 marks Q.

Inverse Laplace Transform :-

$L(t^n) = \frac{n!}{s^{n+1}}$

$L\left(\frac{t^n}{n!}\right) = \frac{1}{s^{n+1}}$

$L^{-1}\left(\frac{1}{s^{n+1}}\right) = \frac{t^n}{n!}$

$\therefore L^{-1}\left[\frac{1}{s}\right] = 1$

$L^{-1}\left[\frac{1}{s^2}\right] = t$

$L^{-1}\left[\frac{1}{s^3}\right] = \frac{t^2}{2}$

$L^{-1}\left(\frac{1}{s-a}\right) = e^{at}$

$L^{-1}\left(\frac{1}{s+a}\right) = e^{-at}$

$L^{-1}\left(\frac{1}{s^2+a^2}\right) = \frac{1}{a} \sin at$

$L^{-1}\left(\frac{s}{s^2+a^2}\right) = \cos at$

$L^{-1}\left(\frac{1}{s^2-a^2}\right) = \frac{1}{a} \sinh at$

$L^{-1}\left(\frac{s}{s^2-a^2}\right) = \cosh at$

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Q. Find $L^{-1}\left[\frac{1}{2s-5}\right]$, $L^{-1}\left[\frac{1}{s(s+1)}\right]$, $L^{-1}\left[\frac{1}{(s+1)^2}\right]$

$L^{-1}\left[\frac{s}{(s+a)^2}\right]$, $L^{-1}\left[\frac{s+23}{s^2+4s+13}\right]$, $L^{-1}\left[\frac{s-2}{s^2+2s+2}\right]$

→ (1) $L^{-1}\left[\frac{1}{2s-5}\right] = \frac{1}{2}L^{-1}\left[\frac{1}{s-\frac{5}{2}}\right] = \frac{1}{2}e^{\frac{5}{2}t}$

(2) $L^{-1}\left[\frac{1}{s(s+1)}\right] = L^{-1}\left[\frac{1}{s} - \frac{1}{s+1}\right] = 1 - e^{-t}$

(3) $L^{-1}\left[\frac{1}{(s+1)^2}\right] = te^{-t}$

(4) $L^{-1}\left(\frac{s}{(s+a)^2}\right) = L^{-1}\left[\frac{s+a-a}{(s+a)^2}\right] = e^{-at} - ate^{-at}$

$= e^{-at}(1-at)$

(5) $L^{-1}\left[\frac{s+23}{s^2-4s+13}\right]$

$= L^{-1}\left[\frac{s+23}{s^2-2\times 2s+4+13-4}\right]$

$= L^{-1}\left[\frac{s-2+25}{(s-2)^2+3^2}\right]$

$= e^{2t}\cos 3t + \frac{25}{3}e^{2t}\sin 3t$

(6) $L^{-1}\left[\frac{s-2}{s^2+2\times s\times 1+1+2-1}\right]$

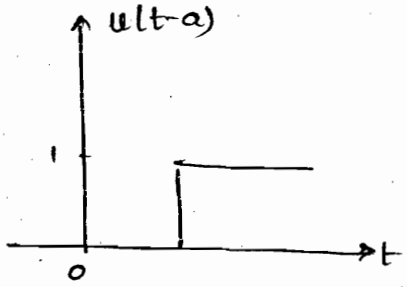
$= L^{-1}\left[\frac{s+1-3}{(s+1)^2+1}\right]$

$= e^{-t}\cos t - 3e^{-t}\sin t$

$= e^{-t}[\cos t - 3\sin t]$

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Unit Step function :-



The unit step function $u(t-a)$ is defined by —

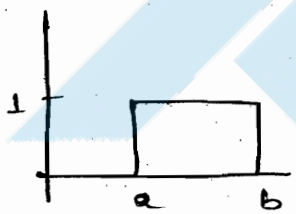
$$u(t-a) = \begin{cases} 0, & 0 < t < a \\ 1, & t \geq a \end{cases}$$

Then $\mathcal{L}[u(t-a)] = \int_0^\infty e^{-st} u(t-a)\, dt$

$$= \int_0^a 0\, dt + \int_a^\infty 1 \cdot e^{-st} dt$$

$$= \frac{-1}{s}\left[e^{-\infty s} - e^{-as}\right]$$

$$\boxed{\mathcal{L}[u(t-a)] = \frac{e^{-as}}{s}}$$

① 

$$= u(t-a) - u(t-b)$$

$$= \frac{e^{-as}}{s} - \frac{e^{-bs}}{s} = \frac{1}{s}\left[e^{-as} - e^{-bs}\right]$$

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Periodic function:-

→ A fn: $f(t)$ is said to be periodic with the period 'T', if $f(t+T) = f(t)$, where T is the least +ve number.

ex:- $f(t) = \sin t$

$f(t+2\pi) = \sin(t+2\pi)$

$= \sin t$

$f(t+2\pi) = f(t)$

Laplace Transform of Periodic function:-

Let $f(t)$ be a periodic fn with period T, then the laplace Transform of the periodic fn is defined by —

$$L[f(t)] = \frac{1}{1-e^{-sT}} \int_0^T e^{-st} \cdot f(t)\, dt$$

ex:- $f(t) = \begin{cases} 3, & 0<t<2 \\ 0, & 2<t<4 \end{cases}$

and $f(t+4) = f(t)$, then find $L[f(t)]$.

→ $$L[f(t)] = \frac{1}{1-e^{-4s}} \left[\int_0^2 e^{-st} \times 3\, dt + \int_3^\infty 0 \right]$$

$$= \frac{1}{1-e^{-4s}} \left[\frac{-3}{s} \left[e^{-2t} - 1 \right] \right]$$

$$= \frac{3\,(1-e^{-2s})}{s\,(1-e^{-4s})}$$

$$(1-e^{-2s})(1+e^{-2s})$$

$$= \frac{3}{s(1+e^{-2s})}$$

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Q. $f(t) = \begin{cases} 1, & 0<t<b \\ -1, & b<t<2b \end{cases}$, $f(t+2b) = f(t)$, find $L[f(t)]$

→ (b) $L[f(t)] = \frac{1}{1-e^{-2bs}} \left[\int_0^b e^{-st}\,dt + \int_b^{2b} e^{-st} \right]$

Imp Result

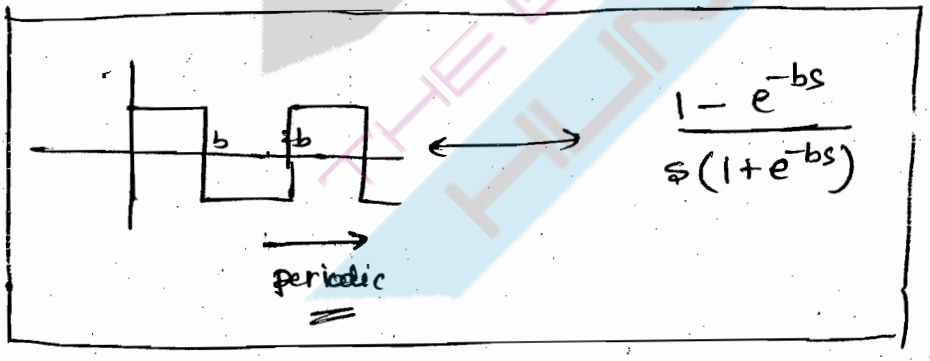
$$= \frac{1}{1-e^{-2bs}} \left[\frac{1-e^{-bs}}{s} + \frac{e^{-2bs} - e^{-bs}}{s} \right]$$

$$= \frac{1}{(1-e^{-2bs})} \cdot \frac{1 - e^{-bs} + e^{-2bs} - e^{-bs}}{s}$$

$$= \frac{1}{(1-e^{-bs})(1+e^{-bs})} \left[\frac{1 - 2e^{-bs} + e^{-2bs}}{s} \right] (1-e^{-bs})$$

$$= \frac{1-e^{-bs}}{s(1+e^{-bs})}$$

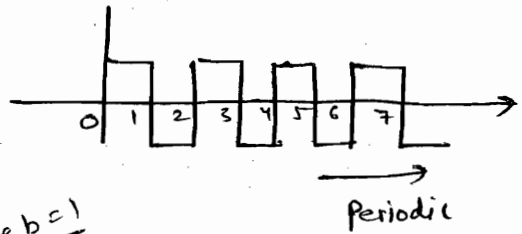
Remember i.e, Imp Result



$$\frac{1-e^{-bs}}{s(1+e^{-bs})}$$

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ex:-



$L[f(t)] = ?$

Here $b = 1$

$$L[f(t)] = \frac{1 - e^{-s}}{s(1+e^{-s})}$$

Q. ~~[illegible]~~

$$L[f(t)] = F(s) = \frac{5s^2+23s+6}{s(s^2+2s+2)}$$

then $\lim_{t\to\infty} f(t) =$ ____

(a) 3 (b) 5 (c) $\frac{17}{2}$ (d) ∞.

Q. $f(t) = L^{-1}\left[\frac{3s+1}{s^3+4s^2+(k-3)s}\right]$

$\lim_{t\to\infty} f(t) = 1$

then $k =$ ____

→. A/c to F.V.T — $\frac{1}{k-3} = 1 \Rightarrow k = 4$

Q. The Laplace Transform —

$$L[f(t)] = \frac{1}{s^2(s+1)}$$

$f(t) = ?$

→ $\frac{1}{s^2(s+1)} = \frac{a_0}{s^2} + \frac{a_1}{s} + \frac{A}{s+1}$

$\frac{1}{2} = 1 + a_1 + \frac{1}{2}$

$a_1 = -1$

$= t^2 - 1 + e^{-t}$

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Probability

→ If an experiment is conducted under essentially given conditions upto n times, Let m cases are favourable to an event E then the Probability of E is denoted by P(E) and is defined by –

$$\boxed{P(E) = \frac{m}{n}}$$

$$P(\bar{E}) = \frac{n-m}{n} = 1 - \frac{m}{n} = 1 - P(E).$$

$$\therefore \boxed{P(E) + P(\bar{E}) = 1}$$

Sum of Succes & failure is always 1.

Axioms of Probability :-

① Axiom of Positivity :- $0 \leq P(E) \leq 1$.

② Certainity :- $P(S) = 1$.

③ Union :- $P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$

Note :- If E_1 and E_2 are mutually Exclusive or Disjoint events

i.e, $E_1 \cap E_2 = \phi$

then $P(E_1 \cup E_2) = P(E_1) + P(E_2)$

Sample Space :-

The set of all possible outcomes in an experiment is said to be the sample space.

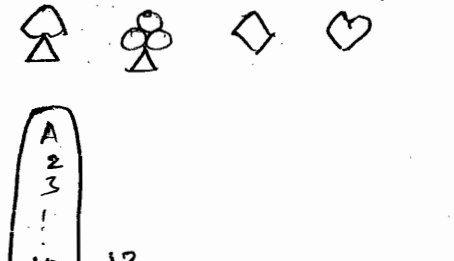
2 coins tossed $S = \{HH, HT, TH, TT\}$.

1 dice thrown $S = \{1, 2, 3, 4, 5, 6\}$.

2 dice thrown. $S = \{(1,1)(1,2)(1,3) \cdots (6,6)\}$ → total 36

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Spade club diamond Heart



13

Conditional Probability :-

Let S be the sample space, and Event $E_1, E_2 \subset S$ then the conditional Probability of E_1 after the occurance of E_2 is denoted by —

$$P\left(\frac{E_1}{E_2}\right) = \frac{P(E_1 \cap E_2)}{P(E_2)}, \quad (P(E_2) \neq 0)$$

$$P\left(\frac{E_2}{E_1}\right) = \frac{P(E_1 \cap E_2)}{P(E_1)}, \quad (P(E_1) \neq 0).$$

Random Variable :-

Mean :- $\sum x_i P(x = x_i)$ if exists, it is called the mean of R.V.

$$E = \mu = \sum x_i P(x = x_i).$$

Variance of R.V $\sum (x_i - \mu)^2 P(x = x_i)$ if exists, it is called the variance of R.V

$$\sigma^2 = \sum (x_i - \mu)^2 P(x = x_i)$$

$$= \sum (x_i^2 + \mu^2 - 2x_i\mu) P(x = x_i)$$

$$= \sum x_i^2 P(x = x_i) + \sum \mu^2 \cdot P(x = x_i) - \sum 2x_i \mu P(x = x_i)$$

$$= \sum x_i^2 P(x = x_i) + \mu^2 \sum P(x = x_i) - 2\mu \sum x_i P(x = x_i)$$

$$= \sum x_i^2 [P(x = x_i)] + \mu^2 - 2\mu^2$$

$$\boxed{\sigma^2 = \sum x_i^2 P(x = x_i) - \mu^2}$$

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③ Standard Deviation :-

The standard deviation is the sq. root of variance and is denoted by σ.

ABOVE WERE FOR GENERAL DISTRIBUTION :

Binomial Distribution :-

n = no. of trials

p = prob. of success.

q = prob of failures.

$$p+q=1$$

$$P(X=k) = {}^{n}C_k \, q^{n-k} \, p^k$$

If $p=q$

$$P(X=k) = {}^{n}C_k \, p^k$$

Mean = $n \cdot p$

Variance = npq

S.D = $\sqrt{npq}$

Poisson Distribution :-

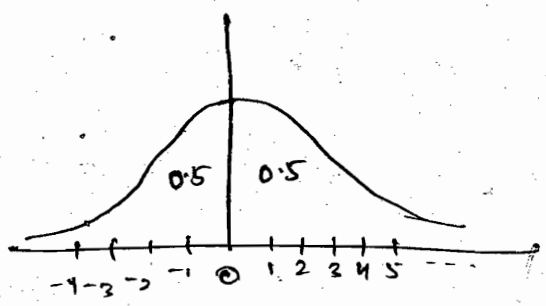
$$P(X=k) = \frac{\lambda^k}{k!} e^{-\lambda}$$, where λ is a parameter.

Mean = λ

Variance = λ

S.D = $\sqrt{\lambda}$

Normal Distribution :-



$$z = \frac{x-\mu}{\sigma}$$

μ = mean.

σ = S.D

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Q. What is the chance that a Leap year will have 53 Sundays.

→ $52 \times 7 = 364$

for rest 2 days —

S M →
M T
T W
W Th
Th F
F Sat
Sat Sun →

} $\frac{2}{7}$

for Non leap year — $P(53 \text{ Sunday}) = \frac{1}{7}$

Q. from a deck of playing Cards - 2 Cards are drawn at Random what will be the probab. that both Cards will be kings? if -

① the first Card is not replaced

② Replacement is allowed.

→ ① $\frac{{}^4C_1}{{}^{52}C_1} \times \frac{3}{51}$

$= \frac{\cancel{4}}{\cancel{52}_{13}} \times \frac{\cancel{3}}{\cancel{51}_{17}} = \frac{1}{221}$.

② $\frac{\cancel{4}}{\cancel{52}_{13}} \times \frac{\cancel{4}}{\cancel{52}_{13}} = \frac{1}{169}$

Q. A Bag Contain 3 red 6 white and 7 black balls. what is the Prob that 2 balls drawn are white and Black.

→ 6 3 7
(W R B)

$\frac{{}^6C_1 \times {}^7C_1}{{}^{16}C_2} = \frac{6 \times 7}{\frac{16 \times 15}{2}} = \frac{7}{20}$.

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Q. A problem in electronics is given to 3 students A, B, C whose chances of solving it are $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}$ respectively. What is the Prob. that the prob. will be solved.

→ $P(A) = \frac{1}{2}$
$P(B) = \frac{1}{3}$
$P(C) = \frac{1}{4}$

indivisual ↓ then

$1 - P(\bar{A} \cap \bar{B} \cap \bar{C})$ → $P(\bar{A}) \cdot P(\bar{B}) \cdot P(\bar{C})$

$= 1 - P(\bar{A})\, P(\bar{B})\, P(\bar{C})$

$= 1 - \frac{1}{2} \times \frac{1}{3} \times \frac{3}{4}$

$= \frac{3}{4}$

$= 75\%$ chances.

Q. A husband & wife appear in interview for 2 vacancies for the same post. The prob. of husband selection — $\frac{1}{7}$ & that of wife selection is $\frac{1}{5}$.
What is the Prob. that — ① both are selected.
② only one of them is selected
③ None of them is selected.

→ $P(A) = \frac{1}{7}$
$P(B) = \frac{1}{5}$

① $P(A \cap B) = P(A) \cdot P(B) = \frac{1}{7} \times \frac{1}{5} = \frac{1}{35}$.

②
$P(A \cap \bar{B}) + P(\bar{A} \cap B)$
$= \frac{1}{7} \times \frac{4}{5} + \frac{6}{7} \times \frac{1}{5}$
$= \frac{10}{35} = \frac{2}{7}$.

Don't use $1 - \frac{1}{35}$ Coz indivisual selection ≠ Either or

③ $P(\bar{A} \cap \bar{B}) = P(\bar{A}) \cdot P(\bar{B})$
$= \frac{6}{7} \times \frac{4}{5} = \frac{24}{35}$

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Q. An integer is selected at random from the first 200 digits what is the prob. that the integer choosen is devisible by either 6 or 8 ?

→, 1, 2, 3, ... 200

÷6 → 6, 12, 18 198

÷8 → 8, 16, 24 ..... 200.

÷6 & ÷8 → 24, 48 ..... 192.

no. of integers ÷6 = $\frac{198}{6}^{33}$ = 33 = $n(E_1)$

no. of integer ÷ 8 = $\frac{200}{8}^{25}$ = 25 = $n(E_2)$

no. of integers ÷6 & ÷8 = $\frac{192}{24}^{8}$ = 8 = $n(E_1 \cap E_2)$

$n(E_1 \cup E_2) = n(E_1) + n(F_2) - n(E_1 \cap E_2)$

$= 33 + 25 - 8$

$= 50.$

$\therefore P(E_1 \cup E_2) = \frac{50}{200_4} = \frac{1}{4}$ //

Q. A hits a target 4 times in 5 shots → B → 3 times in 4 shots

C → 2 in 3 shots.

The fire a volley. What is Prob. that 2 shots at least hit the target ?

(↓ means A, B, C then again ABC.. continuous)

→. A → 4 in 5

B → 3 in 4

C → 2 in 3

for at least 2 shots —

$P(A \cap B \cap \bar{C}) + P(A \cap \bar{B} \cap C) + P(\bar{A} \cap B \cap C) + P(A \cap B \cap C)$

$= \frac{4}{5} \times \frac{3}{4} \times \frac{1}{3} + \frac{4}{5} \times \frac{1}{4} \times \frac{2}{3} + \frac{1}{5} \times \frac{3}{4} \times \frac{2}{3} + \frac{4}{5} \times \frac{3}{4} \times \frac{2}{3}$

$= \frac{12}{60} + \frac{8}{60} + \frac{6}{60} + \frac{24}{60} = \frac{50}{60} = \frac{5}{6}$ //

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For at least 1 shot → $1 - P(\bar{A} \cap \bar{B} \cap \bar{C})$

Q On a college 25% of student fail in Math, 15% failed in Chemistry, 10% failed in both Maths and chemistry if a student is selected at random —

① - If he failed in chem. then what is the probablity that he fails in Math

② vice-versa

③ What is the Prob. that he failed neither Mathematics, nor chemistry?

→

→ $P(M) = 0.25$

$P(C) = 0.15$

$P(M \cap C) = 0.10$

① $P\left(\frac{M}{C}\right) = \frac{P(M \cap C)}{P(C)} = \frac{0.10}{0.15} = \frac{2}{3}$.

② $P\left(\frac{C}{M}\right) = \frac{P(C \cap M)}{P(M)} = \frac{0.10}{0.25} = \frac{2}{5}$

③ $P(\overline{M \cup C}) = 1 - P(M \cup C)$

$= 1 - [P(M) + P(C) - P(M \cap C)]$

$= 1 - [0.25 + 0.15 - 0.10]$

$= 1 - 0.3$

$= 0.7$.

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Q. A single dice is thrown twice, what is the probability that their sum is neither 8 nor nor 9.

→ $n(S) = 36$

$n(E_1) = \{(2,6)(3,5),(4,4)(5,3),(6,2)\}$

$\therefore P(E_1) = 5/36.$

$n(E_2) = \{(3,6),(6,3),(5,4)(4,5)\}$

$P(E_2) = \frac{4}{36}.$

∵ These are mutually Exclusive → (eg. 8 આવે તો 9 નહી આવવાના, 9 આવે તો 8 નહી આવવાના)

$\therefore P(\overline{E_1 \cup E_2}) = 1 - P(E_1 \cup E_2)$

$= 1 - [P(E_1) + P(E_2)]$

$= 1 - \left[\frac{5}{36} + \frac{4}{36}\right]$

$= \frac{\cancel{27}}{\cancel{36}}\,{}^{3}_{4} = \frac{3}{4}$ //.

Q. A pair of fair dice is thrown, if the two numbers appearing are different, find the Prob that —

① The sum is 6

② an ace appears

③ The sum is 4 or less

→ $n(S) = 36 - \cancel{6} = 30$

(Same no appearing.)

for Sum is 6 —

$n(E_1) = \{(1,5),(2,4),(5,1)(4,2)\} \Rightarrow P(E_1) = \frac{4^{2}}{\cancel{30}_{15}} = \frac{2}{15}$

for Ace Appears — $P(E) = 0$ if option available

But Here (Ace means 1)

$n(E_2) = \{(1,2),(1,3),(1,4),(1,5),(1,6), \times 2\} = 10$

$\therefore P(E_2) = \frac{\cancel{10}}{\cancel{30}} = \frac{1}{3}$

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③ The sum is 4 or less

$$n(E) = \{(1,2)(1,3),(2,1)(3,1)\} = 4$$

$$P(E) = \frac{4}{30} = \frac{2}{15}$$

Binomial Type-2

Q. Team A has probability $\frac{2}{3}$ of winning whenever it plays. Suppose A plays 4 games. find the prob. that A wins more than half of its games.

→ Team A → $\frac{2}{3}$.

$n = 4$

$p = \frac{2}{3}$

$q = \frac{1}{3}$

How to know Binomial?

n choose करने के बाद k

Given not imp

asked imp, a/c to that p is selected

$$P(X>2) = P(X=3) + P(X=4)$$

$$= {}^4C_k \, q^{4-k} \, p^k$$

$$= {}^4C_3 \left(\frac{1}{3}\right)^1 \left(\frac{2}{3}\right)^3 + {}^4C_4 \left(\frac{2}{3}\right)^4$$

$$= \frac{4 \times 8}{81} + \frac{1 \times 16}{81} = \frac{48}{81} = \frac{16}{27}.$$

Q. In a bank 60% of all customers applied for a loan are rejected. If 4 new loans applications received. find the prob that 3 of them are accepted.

→ $n = 4$

$P = 0.4$

$q = 0.6$

$$P(X=3) = {}^4C_3 (0.6)^1 (0.4)^3$$

$$= 4 (0.6)(0.4)^3$$

$$= 0.1536.$$

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Q. A player tosses 2 fair coins. He wins $2 if 2 heads occurs and wins $1 if 1 head occurs. On the other hand he looses $3, if no head occurs. find the expected value 'E' of the game.

→ $2 → 2H
$1 → 1H
− $3 → 0H

Fair game means — no loss no gain //

fairness of game ∝ mean value of game.

if mean = +ve → favourable to player
= -ve → Unfavourable to player
Mean = 0 → fair Game

$S = \{(H,H),(H,T),(T,H),(TT)\}$

x_i	$2	$1	− $3
P_i	$\frac{1}{4}$	$\frac{2}{4}$	$\frac{1}{4}$

$E(X) = \sum x_i P(X = x_i)$.

$= 2\left(\frac{1}{4}\right) + 1\times\left(\frac{2}{4}\right) - 3\left(\frac{1}{4}\right)$

$= \frac{1}{2} + \frac{1}{2} - \frac{3}{4}$

$= \frac{1}{4} > 0 \rightarrow$ favourable to player.

Q. 4 fair coins are tossed simultaneously. find the Prob. that at least one head and one tail turn up.

→ $n = 4$
$P = 1/2$

$P(X=1) + P(X=2) + P(X=3)$

$= \left({}^4C_1 + {}^4C_2 + {}^4C_3\right)\left(\frac{1}{2}\right)$

$= \frac{4+6+4}{16} = \frac{14}{16} = \frac{7}{8}$ //

$= 1 - [P(X=4) + P(X=0)]$

$= 1 - \left[\frac{1+1}{16}\right]$

$= 1 - \frac{1}{8} = \frac{7}{8}$

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Q. Five fair coins are tossed simultaneously. find the prob. that at least one head turn up.

→ $1 - P(X=0)$

$= 1 - {}^5C_0 \left(\frac{1}{2}\right)^5$

$= 1 - \frac{1}{32} = \frac{31}{32}$

Q. 4 fair coins are tossed simultaneously. find the Prob. of event the no. of times heads show up is more than the no. of times Tails show up.

→ $P(X=3) + P(X=4)$

$= {}^4C_3 + {}^4C_4 + \left(\frac{1}{2}\right)^4$

$= \frac{4+1}{16} = \frac{5}{16}$ //

Q. There are 5 duplicate and 10 original items in an automobile shop. 3 items are bought up by customer at random find the Prob that none of the item is duplicate.

→ means all original.

$= \frac{{}^{10}C_3}{{}^{15}C_3} = \frac{10 \times 9 \times 8}{15 \times 14 \times 13} = \frac{24}{91}$ //

Q. Out of 800 families with 4 children each how many families would be expected to have —

① 2 boys and 2 girls.
② At least one boy
③ At most 2 girls.

→ for 1 family —

$n = 4$

$p = q = \frac{1}{2}$

$\therefore P(X=2) = {}^4C_2 \left(\frac{1}{2}\right)^4 = 6 \times \frac{1}{16} \times 800 = 300$

Now for 800 family

Poisson distribution बहुत बड़ा distribution n is large (very large) But discrimination individual नहीं है ∴ Not Applicable

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② $P(X \geq 1) = 1 - P(X=0)$

$= 1 - {}^4C_0 \left(\frac{1}{2}\right)^4$

$= 1 - \frac{1}{16}$

$= \frac{15}{16} \times 800$

$= 750$

③ $P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$

$= \left({}^4C_0 + {}^4C_1 + {}^4C_2\right) \left(\frac{1}{2}\right)^4$

$= \frac{1+4+6}{16}$

$= \frac{11}{16} \times 800$

$= 550.$

Q. A book of 300 pages containing 30 pointing mistakes. Assuming that these errors are randomly distributed throughout the book and X is the no. of errors per page has a poission's distribution. What is the prob. that 20 pages selected at random will be free of errors.

→ error in page $P = \frac{30}{300} = \frac{1}{10}$

Given $n = 20$

$\lambda = np$

$= 20 \times \frac{1}{10}$

$\boxed{\lambda = 2}$

$P(X=0) = \frac{\lambda^0}{0!} e^{-2} = e^{-2} = 0.135$

Imp

When in Poission distribution λ is not mentioned, we will find it by Binomial help.

i.e $\boxed{\lambda = np}$

$= 20 \times \frac{1}{10} = 2$

$\therefore \boxed{\lambda = 2}$

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Q. 1000 Students had written an examination with the mean of test = 35 ——→ and S.D = 5. Assuming the distribution to be normal. find how many students marks—

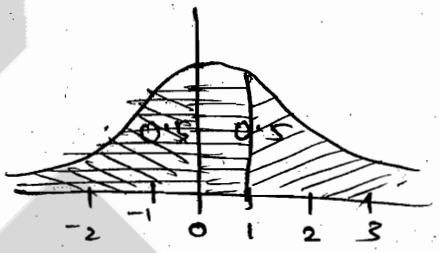
(a) lie between 25 & 40.
(b) How many got more than 40.
(c) How many get below 20.

Given $P(0 \le z \le 1) = 0.3415$
$P(0 \le z \le 2) = 0.4772$
$P(0 \le z \le 3) = 0.449$

→ $\mu = 35$
$\sigma = 5$.

$z = \frac{x - \mu}{\sigma}$

$P(25 \le x \le 40)$
$x_1 \quad x_2$



for $x_1 = 25$ — $z_1 = \frac{25-35}{5} = -2$

for $x_2 = 40$ — $z_2 = \frac{40-35}{5} = 1$.

$\therefore P(25 \le x \le 40) = P(-2 \le z \le 1)$

$= P(-2 \le z \le 0) + P(0 \le z \le 1)$

$= P(0 \le z \le 2) + P(0 \le z \le 1)$

$= 0.4772 + 0.3415$

$= 0.8187$

$= 0.819 \times 1000$ → Total Students

$= 819$.

$P(x \ge 40) = P(z \ge 1)$

$= 0.5 - P(0 \le z \le 1)$

$= 0.5 - 0.3415$

$= 0.1585$

$= 0.159 \times 1000$

$= 159$

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(c) $P(X \leq 20)$

$$z = \frac{20-35}{5} = \frac{-15}{5} = -3.$$

$P(z<3) = P(X<20) = 0.5 - P(0 \leq z \leq 3)$

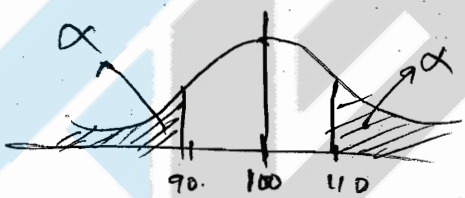
$= 0.5 - 0.499$

$= 0.001 \times 1000$

$= 1$ student.

Q. for a Random Variable — $(-\infty < x < \infty)$ following a normal distribution with the Mean is 100. If the prob. of $P(X \geq 110) = \alpha$ then $P(90 \leq x \leq 110) =$ ___

→



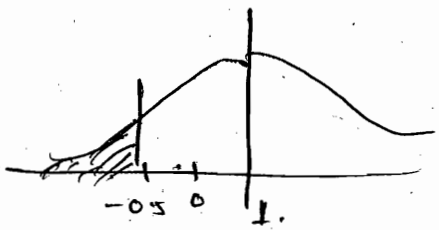
$P(X \geq 110) = \alpha$

$P(90 \leq x \leq 110) = 1 - 2\alpha.$

Q. Let x be the normal variant with Mean = 1 & $\sigma^2 = 4$ then the Prob. of $P(X<0)$ —

(a) 0.5 (b) >0 and <0.5 (c) >0.5 and <1 (d) 1.

→



$\mu = 1$

$\sigma^2 = 4$

$\sigma = 2$

$\therefore z = \frac{x-\mu}{\sigma} = \frac{-1}{2} = -0.5$

$P(X<0)$

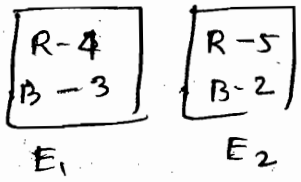
$= P(z < -0.5)$

$= 0.5 - P(0 \leq 0.5)$

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Q A box contain 3 blue and 4 red ball. Another identical Box contain 2 blue ball and 5 red ball. One ball is selected at Random from one of the two boxes and it is red. The Prob. that it come from the first box —

→



$P(E_1) = P(E_2) = \frac{1}{2}$

$P\left(\frac{A}{E_1}\right) = \frac{4}{7}$, $P\left(\frac{A}{E_2}\right) = \frac{5}{7}$

$$P\left(\frac{E_1}{A}\right) = \frac{P(E_1)\cdot P\left(\frac{A}{E_1}\right)}{P(E_1)\cdot P\left(\frac{A}{E_1}\right) + P(E_2)\cdot P\left(\frac{A}{E_2}\right)}$$

$$= \frac{\frac{1}{2}\left(\frac{4}{7}\right)}{\frac{1}{2}\left(\frac{4}{7}\right)+\frac{1}{2}\left(\frac{5}{7}\right)}$$

$$= \frac{\frac{4}{7}}{\frac{9}{7}} = \frac{4}{9}.$$

Q. Box P has 2 Red & 3 blue balls. Box Q has 3 Red & 1 Blue ball The Prob. of selecting boxes P and Q are $\frac{1}{3}$ and $\frac{2}{3}$ respectively One ball is selected from one of two boxes and it is red. the Prob that it came from box P.

→ $P(P) = \frac{1}{3}$, $P(Q) = \frac{2}{3}$, $P\left(\frac{A}{P}\right) = \frac{2}{5}$, $P\left(\frac{A}{Q}\right) = \frac{3}{4}$.

$$P\left(\frac{P}{A}\right) = \frac{P(P)\cdot P\left(\frac{A}{P}\right)}{P(P)\cdot P\left(\frac{A}{P}\right) + P(Q)\cdot P\left(\frac{A}{Q}\right)} = \frac{4}{19}$$

Q. The Probability density fn of the from $P(x) = k e^{-\alpha|x|}$, $x \in (-\infty, \infty)$, then the value of k —

→ $\int_{-\infty}^{\infty} P(x) = 1 \Rightarrow \int_{-\infty}^{\infty} k\, e^{-\alpha|x|} dx = 1 \Rightarrow 2\int_{0}^{\infty} k\, e^{-\alpha x} dx = 1.$

$\Rightarrow \frac{2k}{\alpha}\left[e^{-\infty} - e^{-0}\right] = 1$

$\Rightarrow -\frac{2k}{\alpha}(0-1) = 1$

$\Rightarrow k = \frac{\alpha}{2} = 0.5\alpha.$

Q. Two Bag Contain 10 coins each and the coins in each bag are numbered from 1 to 10. One coin is drawn at random from each bag. What is the prob that one of the coin has value (1, 2, 3, 4) while the other has value (7, 8, 9 or 10)..

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Differential Eq^n :-

- Formation of Diff^n eq^n → diff upto no. of constants.
- Solution of Diff eq^n
  - Variable - Seperable
    - normal
    - reducable

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